

BOSCH REXROTH INDRADRIIVE PROJECT PLANNING MANUAL



Quick Links

[Mains Filter Nfd / Nfe](#)

Table of Contents

Table of Contents

Documentation

About this Documentation

Documentations - Overview

General Information on How to Install

Appropriate Use

Introduction

Areas of Use and Application

Inappropriate Use

Introduction

Explanations

Hazards by Improper Use

General Information

Protection Against Contact with Electrical Parts

Protection Against Electric Shock by Protective Low Voltage (PELV)

Protection Against Dangerous Movements

Protection Against Magnetic and Electromagnetic Fields During
Operation and Mounting

Protection Against Contact with Hot Parts

3.10 Protection During Handling and Mounting

3.11 Battery Safety

3.12 Protection Against Pressurized Systems

Autotransformers

Type Code

Selection

Technical Data

Type Code NFE / NFD

Technical Data NFE / NFD

Dimensions NFE / NFD

Mains Filter HNF01.1

Type Code HNF01.1

Data HNF01.1

Dimensional Drawings HNF01.1

Mains Filter (-Combination) HNK01.1

Type Code HNK01.1

Type Plate HNK01.1

Mechanical Data HNK01.1

Technical Data HNK01.1

Arrangement of Components

Identification

Type Code

Type Plate

Mains Choke HNL01.1E (feeding)

HNL01.1E-0170-N0146-A-500-NNNN, HNL01.1E-0100-N0202-A-480-NNNN

Mains Choke HNL01.1R (regenerating)

HNL01.1R-0980-C0026-A-480-NNNN, HNL01.1R-0590-C0065-A-480-NNNN,
HNL01.1R-0540-C0094-A-480-NNNN

Mains Chokes HNL01.1E-****-S (current compensated)

Application Instructions

Identification

Types

Mounting

Details

Mechanical Data

Installation

Electrical Data

Connections

Description

Identification

Type Code HLB01.1

Mounting HLB01.1

Mechanical Data HLB01.1

Installation HLB01.1

Electrical Data HLB01.1

Connections

Diagnostic LEDs and Reset-Pushbutton

General Information

Identification

Type Code HLR01.1

Type Plate HLR01.1

Technical Characteristics of the Standard Braking Resistors HLR01.1

Mechanical Data of the Standard Braking Resistors HLR01.1

Dimensioned Drawings

Arrangement of the Components Device / Braking Resistor HLR01.1N-0300-N17R- A-007-NNNN

Technical Data of the Standard Braking Resistors HLR01.1

Technical Features of the Reinforced Braking Resistors HLR01.1

Technical Data of the Reinforced Braking Resistors HLR01.1

Braking Resistors for HCS03.1E-W0070

Braking Resistors for HCS03.1E-W0100

Braking Resistors for HCS03.1E-W0150

Braking Resistors for HCS03.1E-W0210

Line Cross Sections for Reinforced Braking Resistors HLR01.1

Requirements for Line Routing

Mechanical Data of the Reinforced Braking Resistors HLR01.1

Fixed Resistors IP 20 Type A

Steel Grid Fixed Resistors IP 20 Type B

Steel Grid Fixed Resistors IP 20 Type C

10.1 Type Code

10.2 Type Plate

10.3 Electrical Data

10.4 Mechanical Data

Dimensional Drawings

Arrangement of the Components HCS03.1 / Motor Filter HMF

Arrangement of the Components HCS03.1 / Motor Filter and Mains Filter

11.1 Helpdesk

11.2 Service-Hotline

11.3 Internet

Vor der Kontaktaufnahme

11.5 Kundenbetreuungsstellen

Sales & Service Facilities

Other ManualsLib Projects

Rexroth IndraDrive Additional Components

R911306140
Edition 02

Project Planning Manual



Title	Rexroth IndraDrive Additional Components
Type of Documentation	Project Planning Manual
Document Typecode	DOK-INDRV*-ADDCOMP****-PR02-EN
Internal File Reference	Document number 120-2400-B315-02

Record of Revisions

Description	Release Date	Notes
DOK-INDRV*-ADDCOMP****-PR01-EN-P	11.04	first edition
DOK-INDRV*-ADDCOMP****-PR02-EN-P	03.05	motor filter HMF01.1 added

Copyright © Bosch Rexroth AG 2004 2005

Copying this document, giving it to others and the use or communication of the contents thereof without express authority, are forbidden. Offenders are liable for the payment of damages. All rights are reserved in the event of the grant of a patent or the registration of a utility model or design (DIN 34-1).

Validity The specified data only serve to describe the product. No statements concerning a certain condition or suitability for a certain application can be derived from our information. The given information does not release the user from the obligation of own judgement and verification. It must be remembered that our products are subject to a natural process of wear and aging.

Published by Bosch Rexroth AG
Bgm.-Dr.-Nebel-Str. 2 • D-97816 Lohr a. Main
Telefon +49 (0)93 52 / 40-0 • Tx 68 94 21 • Fax +49 (0)93 52 / 40-48 85
<http://www.boschrexroth.de/>
Abt. EDY1

Note This document has been printed on chlorine-free bleached paper.

Contents

1	Introduction	1-1
1.1	Documentation.....	1-1
	About this Documentation	1-1
	Documentations - Overview	1-2
1.2	General Information on How to Install	1-3
2	Important Directions for Use	2-1
2.1	Appropriate Use.....	2-1
	Introduction	2-1
	Areas of Use and Application	2-2
2.2	Inappropriate Use	2-2
3	Safety Instructions for Electric Drives and Controls	3-1
3.1	Introduction	3-1
3.2	Explanations	3-1
3.3	Hazards by Improper Use.....	3-2
3.4	General Information	3-3
3.5	Protection Against Contact with Electrical Parts.....	3-5
3.6	Protection Against Electric Shock by Protective Low Voltage (PELV)	3-6
3.7	Protection Against Dangerous Movements	3-7
3.8	Protection Against Magnetic and Electromagnetic Fields During Operation and Mounting	3-9
3.9	Protection Against Contact with Hot Parts.....	3-10
3.10	Protection During Handling and Mounting.....	3-10
3.11	Battery Safety	3-11
3.12	Protection Against Pressurized Systems.....	3-11
4	Transformers	4-1
4.1	Autotransformers	4-2
	Type Code	4-2
	Selection	4-3
	Technical Data.....	4-4
5	Mains Filter	5-1
5.1	Mains Filter NFD / NFE.....	5-1
	Type Code NFE / NFD	5-1
	Technical Data NFE / NFD	5-4
	Dimensions NFE / NFD	5-5
5.2	Mains Filter HNF01.1	5-8

Type Code HNF01.1.....	5-8
Data HNF01.1.....	5-9
Dimensional Drawings HNF01.1	5-10
5.3 Mains Filter (-Combination) HNK01.1.....	5-12
Type Code HNK01.1	5-12
Type Plate HNK01.1.....	5-13
Mechanical Data HNK01.1	5-14
Technical Data HNK01.1	5-18
Arrangement of Components	5-19
6 Mains Choke	6-1
6.1 Identification.....	6-1
Type Code	6-1
Type Plate.....	6-2
6.2 Mains Choke HNL01.1E (feeding).....	6-3
HNL01.1E-1000-N0012-A-500-NNNN, HNL01.1E-1000-N0020-A-500-NNNN, HNL01.1E-0600-N0032-A-500-NNNN, HNL01.1E-0571-N0050-A-500-NNNN, HNL01.1E-0400-N0051-A-480-NNNN, HNL01.1E-0362-N0080-A-500-NNNN, HNL01.1E-0240-N0106-A-500-NNNN, HNL01.1E-0200-N0125-A-480-NNNN, HNL01.1E-0170-N0146-A-500-NNNN, HNL01.1E-0100-N0202-A-480-NNNN	6-3
6.3 Mains Choke HNL01.1R (regenerating)	6-6
HNL01.1R-0980-C0026-A-480-NNNN, HNL01.1R-0590-C0065-A-480-NNNN, HNL01.1R-0540-C0094-A-480-NNNN	6-6
6.4 Mains Chokes HNL01.1E-****-S (current compensated)	6-7
HNL01.1E-5700-S0051-A-480-NNNN, HNL01.1E-2800-S0125-A-480-NNNN, HNL01.1E-3400-S0202-A-480-NNNN, HNL01.1R-4200-S0026-A-480-NNNN, HNL01.1R-6300-S0065-A-480-NNNN, HNL01.1R-3000-S0094-A-480-NNNN	6-7
7 DC Bus Capacitor Unit HLC01.1	7-1
7.1 Application Instructions.....	7-1
7.2 Identification.....	7-2
Types	7-2
7.3 Mounting	7-3
Details.....	7-3
Mechanical Data.....	7-4
7.4 Installation.....	7-6
Electrical Data.....	7-6
Connections.....	7-6
8 DC Bus Resistor Unit HLB01.1C and HLB01.1D	8-1
8.1 Description	8-1
8.2 Identification.....	8-1
Type Code HLB01.1	8-1
8.3 Mounting HLB01.1.....	8-2
Mechanical Data HLB01.1	8-2
8.4 Installation HLB01.1.....	8-5
Electrical Data HLB01.1	8-5
Connections.....	8-6

8.5	Diagnostic LEDs and Reset-Pushbutton	8-15
9	Braking Resistors HLR01.1	9-1
9.1	General Information	9-1
9.2	Identification.....	9-3
	Type Code HLR01.1.....	9-3
	Type Plate HLR01.1	9-4
9.3	Technical Characteristics of the Standard Braking Resistors HLR01.1	9-4
9.4	Mechanical Data of the Standard Braking Resistors HLR01.1.....	9-5
	Dimensioned Drawings.....	9-5
	Arrangement of the Components Device / Braking Resistor HLR01.1N-0300-N17R-A-007-NNNN.....	9-7
9.5	Technical Data of the Standard Braking Resistors HLR01.1	9-10
9.6	Technical Features of the Reinforced Braking Resistors HLR01.1	9-10
9.7	Technical Data of the Reinforced Braking Resistors HLR01.1.....	9-11
	Braking Resistors for HCS03.1E-W0070	9-11
	Braking Resistors for HCS03.1E-W0100	9-13
	Braking Resistors for HCS03.1E-W0150	9-15
	Braking Resistors for HCS03.1E-W0210	9-17
9.8	Line Cross Sections for Reinforced Braking Resistors HLR01.1	9-18
	Requirements for Line Routing.....	9-18
9.9	Mechanical Data of the Reinforced Braking Resistors HLR01.1.....	9-19
	Fixed Resistors IP 20 Type A.....	9-19
	Steel Grid Fixed Resistors IP 20 Type B.....	9-20
	Steel Grid Fixed Resistors IP 20 Type C.....	9-21
10	Motor Filter HMF01.1	10-1
10.1	Type Code	10-1
10.2	Type Plate.....	10-2
10.3	Electrical Data.....	10-2
10.4	Mechanical Data	10-3
	Dimensional Drawings.....	10-3
10.5	Arrangement of the Components HCS03.1 / Motor Filter HMF.....	10-5
10.6	Arrangement of the Components HCS03.1 / Motor Filter and Mains Filter.....	10-7
11	Service & Support	11-1
11.1	Helpdesk.....	11-1
11.2	Service-Hotline	11-1
11.3	Internet.....	11-1
11.4	Vor der Kontaktaufnahme... - Before contacting us... ..	11-1
11.5	Kundenbetreuungsstellen - Sales & Service Facilities	11-2
12	Index	12-1

1 Introduction

1.1 Documentation

About this Documentation

This documentation contains the technical data of the additional components of Rexroth IndraDrive drive systems:

- Transformers
- Mains filters
- Mains chokes
- DC bus capacitor units
- DC bus resistor units
- Braking resistors
- Motor filter



WARNING

Personal injury and property damage caused by incorrect project planning for applications, machines and installations!

⇒ Take the content of the Project Planning Manual "Rexroth IndraDrive Drive System" (DOK-INDRV*-SYSTEM*****-PRxx-EN-P; part no. R911309636) into account.

For complete project planning of a Rexroth IndraDrive drive system you need, in any case, the Project Planning Manual "Rexroth IndraDrive Drive System" (DOK-INDRV*-SYSTEM*****-PRxx-EN-P; part no. R911309636). This Project Planning Manual, among other things, contains:

- specifications for the components of the drive system
- configuration of the drive system components
- arrangement of the components in the control cabinet
- electromagnetic compatibility (EMC)
- types of mains connection
- requirements to the mains connection
- control circuits for the mains connection
- connections of the components in the drive system
- fusing and selecting the mains contactor
- accessories in the drive system
- calculations (determining appropriate drive controller; mains connection; leakage capacitance; operating data of mains filters; selecting the 24V supply; braking behavior when using a DC bus resistor unit)
- Replacing devices

Documentations - Overview

For project planning of the drive system the following documentations are available:

Title	Kind of Documentation	Document Typecode ¹⁾
Rexroth IndraDrive Drive System	Project Planning Manual	DOK-INDRV*-SYSTEM****-PRxx-EN-P
Rexroth IndraDrive Drive Controllers Control Sections CSx	Project Planning Manual	DOK-INDRV*-CSH*****-PRxx-EN-P
Rexroth IndraDrive M Drive Controllers Power Sections HMx	Project Planning Manual	DOK-INDRV*-HMS+HMD****-PRxx-EN-P
Rexroth IndraDrive C Drive Controllers Power Sections HCS02.1	Project Planning Manual	DOK-INDRV*-HCS02.1****-PRxx-EN-P
Rexroth IndraDrive C Drive Controllers Power Sections HCS03.1	Project Planning Manual	DOK-INDRV*-HCS03.1****-PRxx-EN-P
Rexroth IndraDrive Supply Units	Project Planning Manual	DOK-INDRV*-HMV*****-PRxx-EN-P
Electromagnetic Compatibility (EMC) in Drive and Control Systems	Project Planning Manual	DOK-GENERAL-EMV*****-PRxx-EN-P
Rexroth IndraDrive Integrated Safety Technology	Functional and Application Description	DOK-INDRV*-SI*-*VRS*-FKxx-EN-P
Rexroth Connection Cables	Selection Data	DOK-CONNEX-CABLE*STAND-AUxx-EN-P
Safety Instructions for Electrical Drives	Safety Guidelines	DOK-GENERAL-DRIVE-*****-SVSx-MS-P
Rexroth IndraDrive Additional Components	Project Planning Manual	DOK-INDRV*-ADDCOMP****-PRxx-EN-P
Rexroth IndraDrive Firmware for Drive Controllers	Functional Description	DOK-INDRV*-MP*-02VRS*-FKxx-EN-P
Rexroth IndraDrive Firmware for Drive Controllers	Parameter Description	DOK-INDRV*-GEN-*VRS*-PAXx-EN-P
Rexroth IndraDrive Firmware for Drive Controllers	Troubleshooting Guide	DOK-INDRV*-GEN-*VRS*-WAXx-EN-P
Rexroth IndraDyn S Synchronous Motors MSK	Project Planning Manual	DOK-MOTOR*-MSK*****-PRxx-EN-P
Rexroth IndraDyn A Asynchronous Motors MAD/MAF	Project Planning Manual	DOK-MOTOR*-MAD/MAF****-PRxx-EN-P
Rexroth IndraDyn T Synchronous Torque Motors MBT	Project Planning Manual	DOK-MOTOR*-MBT*****-PRxx-EN-P
Rexroth IndraDyn H Synchronous Kit – Spindle Motors MBS-H	Project Planning Manual	DOK-MOTOR*-MBS-H*****-PRxx-EN-P
Rexroth IndraDyn L Synchronous Linear Motors MLF	Project Planning Manual	DOK-MOTOR*-MLF*****-PRxx-EN-P
Third Party Motors	Project Planning Manual and Commissioning	DOK-DRIVE*-3RDPART*MOT-AWxx-EN-P

1) in the document typecodes "xx" is a wild card for the current edition of the documentation (example: "PR01" is the first edition of a Project Planning Manual)

Fig. 1-1: Documentations - overview

1.2 General Information on How to Install

Damage can be caused to the drive controller or circuit boards if electrostatic charging present in people and/or tools is discharged across them. Therefore, please note the following information:



CAUTION

Electrostatic charges can cause damage to electronic components and interfere with their operational safety!

⇒ Exposed conductive parts coming into contact with components and circuit boards must be discharged by means of grounding. Otherwise errors may occur when triggering motors and moving elements.

Such exposed conductive parts include:

- the copper bit when soldering
- the human body (ground connection caused by touching a conductive, grounded item)
- parts and tools (place them on a conductive support)

Endangered components may only be stored or dispatched in conductive packaging.

Note: Rexroth connection diagrams are only to be used for producing installation connection diagrams. The machine manufacturer's installation connection diagrams must be used for wiring the installation!

- Lay signal lines separately from the load resistance lines because of the occurrence of interference.
- Transmit analog signals (e.g. command values, actual values) via shielded lines.
- Do not connect mains, DC bus or power leads to low voltages or allow them to come into contact with these.
- When carrying out a high voltage test or an applied-overvoltage withstand test on the machine's electrical equipment, disconnect all connections to the devices. This protects the electronic components (allowed in accordance with EN 60204-1). During their routine testing, Rexroth drive components are tested for high voltage and insulation in accordance with EN 50178.



CAUTION

Risk of damage to the drive controller by connecting and disconnecting live connections!

⇒ Do not connect and disconnect live connections.

Notes

2 Important Directions for Use

2.1 Appropriate Use

Introduction

Rexroth products represent state-of-the-art developments and manufacturing. They are tested prior to delivery to ensure operating safety and reliability.

The products may only be used in the manner that is defined as appropriate. If they are used in an inappropriate manner, then situations can develop that may lead to property damage or injury to personnel.

Note: Rexroth, as manufacturer, is not liable for any damages resulting from inappropriate use. In such cases, the guarantee and the right to payment of damages resulting from inappropriate use are forfeited. The user alone carries all responsibility of the risks.

Before using Rexroth products, make sure that all the pre-requisites for an appropriate use of the products are satisfied:

- Personnel that in any way, shape or form uses our products must first read and understand the relevant safety instructions and be familiar with appropriate use.
- If the product takes the form of hardware, then they must remain in their original state, in other words, no structural changes are permitted. It is not permitted to decompile software products or alter source codes.
- Do not mount damaged or faulty products or use them in operation.
- Make sure that the products have been installed in the manner described in the relevant documentation.

Areas of Use and Application

Drive controllers made by Rexroth are designed to control electrical motors and monitor their operation.

Control and monitoring of the motors may require additional sensors and actors.

Note: The drive controllers may only be used with the accessories and parts specified in this document. If a component has not been specifically named, then it may not be either mounted or connected. The same applies to cables and lines.

Operation is only permitted in the specified configurations and combinations of components using the software and firmware as specified in the relevant function descriptions.

Every drive controller has to be programmed before starting it up, making it possible for the motor to execute the specific functions of an application.

The drive controllers are designed for use in single or multiple-axis drive and control applications.

To ensure an application-specific use, the drive controllers are available with differing drive power and different interfaces.

Typical applications of drive controllers are:

- handling and mounting systems,
- packaging and foodstuff machines,
- printing and paper processing machines and
- machine tools.

The drive controllers may only be operated under the assembly, installation and ambient conditions as described here (temperature, system of protection, humidity, EMC requirements, etc.) and in the position specified.

2.2 Inappropriate Use

Using the drive controllers outside of the above-referenced areas of application or under operating conditions other than described in the document and the technical data specified is defined as "inappropriate use".

Drive controllers may not be used if

- they are subject to operating conditions that do not meet the above specified ambient conditions. This includes, for example, operation under water, in the case of extreme temperature fluctuations or extremely high maximum temperatures or if
- Rexroth has not specifically released them for that intended purpose. Please note the specifications outlined in the general safety instructions!

3 Safety Instructions for Electric Drives and Controls

3.1 Introduction

Read these instructions before the initial startup of the equipment in order to eliminate the risk of bodily harm or material damage. Follow these safety instructions at all times.

Do not attempt to install or start up this equipment without first reading all documentation provided with the product. Read and understand these safety instructions and all user documentation of the equipment prior to working with the equipment at any time. If you do not have the user documentation for your equipment, contact your local Rexroth representative to send this documentation immediately to the person or persons responsible for the safe operation of this equipment.

If the equipment is resold, rented or transferred or passed on to others, then these safety instructions must be delivered with the equipment.



Improper use of this equipment, failure to follow the safety instructions in this document or tampering with the product, including disabling of safety devices, may result in material damage, bodily harm, electric shock or even death!

3.2 Explanations

The safety instructions describe the following degrees of hazard seriousness in compliance with ANSI Z535. The degree of hazard seriousness informs about the consequences resulting from non-compliance with the safety instructions.

Warning symbol with signal word	Degree of hazard seriousness according to ANSI
	Death or severe bodily harm will occur.
	Death or severe bodily harm may occur.
	Bodily harm or material damage may occur.

Fig. 3-1: Hazard classification (according to ANSI Z535)

3.3 Hazards by Improper Use



DANGER

**High voltage and high discharge current!
Danger to life or severe bodily harm by electric shock!**



DANGER

Dangerous movements! Danger to life, severe bodily harm or material damage by unintentional motor movements!



WARNING

High electrical voltage due to wrong connections! Danger to life or bodily harm by electric shock!



WARNING

Health hazard for persons with heart pacemakers, metal implants and hearing aids in proximity to electrical equipment!



CAUTION

Surface of machine housing could be extremely hot! Danger of injury! Danger of burns!



CAUTION

Risk of injury due to improper handling! Bodily harm caused by crushing, shearing, cutting and mechanical shock or incorrect handling of pressurized systems!



CAUTION

Risk of injury due to incorrect handling of batteries!

3.4 General Information

- Bosch Rexroth AG is not liable for damages resulting from failure to observe the warnings provided in this documentation.
- Read the operating, maintenance and safety instructions in your language before starting up the machine. If you find that you cannot completely understand the documentation for your product, please ask your supplier to clarify.
- Proper and correct transport, storage, assembly and installation as well as care in operation and maintenance are prerequisites for optimal and safe operation of this equipment.
- Only persons who are trained and qualified for the use and operation of the equipment may work on this equipment or within its proximity.
 - The persons are qualified if they have sufficient knowledge of the assembly, installation and operation of the equipment as well as an understanding of all warnings and precautionary measures noted in these instructions.
 - Furthermore, they must be trained, instructed and qualified to switch electrical circuits and equipment on and off in accordance with technical safety regulations, to ground them and to mark them according to the requirements of safe work practices. They must have adequate safety equipment and be trained in first aid.
- Only use spare parts and accessories approved by the manufacturer.
- Follow all safety regulations and requirements for the specific application as practiced in the country of use.
- The equipment is designed for installation in industrial machinery.
- The ambient conditions given in the product documentation must be observed.
- Use only safety features and applications that are clearly and explicitly approved in the Project Planning Manual.

For example, the following areas of use are not permitted: construction cranes, elevators used for people or freight, devices and vehicles to transport people, medical applications, refinery plants, transport of hazardous goods, nuclear applications, applications sensitive to high frequency, mining, food processing, control of protection equipment (also in a machine).
- The information given in the documentation of the product with regard to the use of the delivered components contains only examples of applications and suggestions.

The machine and installation manufacturer must

 - make sure that the delivered components are suited for his individual application and check the information given in this documentation with regard to the use of the components,
 - make sure that his application complies with the applicable safety regulations and standards and carry out the required measures, modifications and complements.
- Startup of the delivered components is only permitted once it is sure that the machine or installation in which they are installed complies with the national regulations, safety specifications and standards of the application.

- Operation is only permitted if the national EMC regulations for the application are met.
The instructions for installation in accordance with EMC requirements can be found in the documentation "EMC in Drive and Control Systems".
The machine or installation manufacturer is responsible for compliance with the limiting values as prescribed in the national regulations.
- Technical data, connections and operational conditions are specified in the product documentation and must be followed at all times.

3.5 Protection Against Contact with Electrical Parts

Note: This section refers to equipment and drive components with voltages above 50 Volts.

Touching live parts with voltages of 50 Volts and more with bare hands or conductive tools or touching ungrounded housings can be dangerous and cause electric shock. In order to operate electrical equipment, certain parts must unavoidably have dangerous voltages applied to them.



DANGER

High electrical voltage! Danger to life, severe bodily harm by electric shock!

- ⇒ Only those trained and qualified to work with or on electrical equipment are permitted to operate, maintain or repair this equipment.
 - ⇒ Follow general construction and safety regulations when working on high voltage installations.
 - ⇒ Before switching on power the ground wire must be permanently connected to all electrical units according to the connection diagram.
 - ⇒ Do not operate electrical equipment at any time, even for brief measurements or tests, if the ground wire is not permanently connected to the points of the components provided for this purpose.
 - ⇒ Before working with electrical parts with voltage higher than 50 V, the equipment must be disconnected from the mains voltage or power supply. Make sure the equipment cannot be switched on again unintended.
 - ⇒ The following should be observed with electrical drive and filter components:
 - ⇒ Wait thirty (30) minutes after switching off power to allow capacitors to discharge before beginning to work. Measure the voltage on the capacitors before beginning to work to make sure that the equipment is safe to touch.
 - ⇒ Never touch the electrical connection points of a component while power is turned on.
 - ⇒ Install the covers and guards provided with the equipment properly before switching the equipment on. Prevent contact with live parts at any time.
 - ⇒ A residual-current-operated protective device (RCD) must not be used on electric drives! Indirect contact must be prevented by other means, for example, by an overcurrent protective device.
 - ⇒ Electrical components with exposed live parts and uncovered high voltage terminals must be installed in a protective housing, for example, in a control cabinet.
-

To be observed with electrical drive and filter components:



DANGER

**High electrical voltage on the housing!
High leakage current! Danger to life, danger of
injury by electric shock!**

- ⇒ Connect the electrical equipment, the housings of all electrical units and motors permanently with the safety conductor at the ground points before power is switched on. Look at the connection diagram. This is even necessary for brief tests.
- ⇒ Connect the safety conductor of the electrical equipment always permanently and firmly to the supply mains. Leakage current exceeds 3.5 mA in normal operation.
- ⇒ Use a copper conductor with at least 10 mm² cross section over its entire course for this safety conductor connection! The cross section must not be smaller than the cross section of a phase of the mains supply wire.
- ⇒ Prior to startups, even for brief tests, always connect the protective conductor or connect with ground wire. Otherwise, high voltages can occur on the housing that lead to electric shock.

3.6 Protection Against Electric Shock by Protective Low Voltage (PELV)

All connections and terminals with voltages between 0 and 50 Volts on Rexroth products are protective low voltages designed in accordance with international standards on electrical safety.



WARNING

**High electrical voltage due to wrong
connections! Danger to life, bodily harm by
electric shock!**

- ⇒ Only connect equipment, electrical components and cables of the protective low voltage type (PELV = Protective Extra Low Voltage) to all terminals and clamps with voltages of 0 to 50 Volts.
- ⇒ Only electrical circuits may be connected which are safely isolated against high voltage circuits. Safe isolation is achieved, for example, with an isolating transformer, an opto-electronic coupler or when battery-operated.

3.7 Protection Against Dangerous Movements

Dangerous movements can be caused by faulty control of the connected motors. Some common examples are:

- improper or wrong wiring of cable connections
- incorrect operation of the equipment components
- wrong input of parameters before operation
- malfunction of sensors, encoders and monitoring devices
- defective components
- software or firmware errors

Dangerous movements can occur immediately after equipment is switched on or even after an unspecified time of trouble-free operation.

The monitoring in the drive components will normally be sufficient to avoid faulty operation in the connected drives. Regarding personal safety, especially the danger of bodily injury and material damage, this alone cannot be relied upon to ensure complete safety. Until the integrated monitoring functions become effective, it must be assumed in any case that faulty drive movements will occur. The extent of faulty drive movements depends upon the type of control and the state of operation.

**DANGER****Dangerous movements! Danger to life, risk of injury, severe bodily harm or material damage!**

- ⇒ Ensure personal safety by means of qualified and tested higher-level monitoring devices or measures integrated in the installation. Unintended machine motion is possible if monitoring devices are disabled, bypassed or not activated.
- ⇒ Pay attention to unintended machine motion or other malfunction in any mode of operation.
- ⇒ Keep free and clear of the machine's range of motion and moving parts. Possible measures to prevent people from accidentally entering the machine's range of motion:
 - use safety fences
 - use safety guards
 - use protective coverings
 - install light curtains or light barriers
- ⇒ Fences and coverings must be strong enough to resist maximum possible momentum, especially if there is a possibility of loose parts flying off.
- ⇒ Mount the emergency stop switch in the immediate reach of the operator. Verify that the emergency stop works before startup. Don't operate the machine if the emergency stop is not working.
- ⇒ Isolate the drive power connection by means of an emergency stop circuit or use a starting lockout to prevent unintentional start.
- ⇒ Make sure that the drives are brought to a safe standstill before accessing or entering the danger zone. Safe standstill can be achieved by switching off the power supply contactor or by safe mechanical locking of moving parts.
- ⇒ Secure vertical axes against falling or dropping after switching off the motor power by, for example:
 - mechanically securing the vertical axes
 - adding an external braking/ arrester/ clamping mechanism
 - ensuring sufficient equilibration of the vertical axes

The standard equipment motor brake or an external brake controlled directly by the drive controller are not sufficient to guarantee personal safety!

- ⇒ Disconnect electrical power to the equipment using a master switch and secure the switch against reconnection for:
 - maintenance and repair work
 - cleaning of equipment
 - long periods of discontinued equipment use
 - ⇒ Prevent the operation of high-frequency, remote control and radio equipment near electronics circuits and supply leads. If the use of such equipment cannot be avoided, verify the system and the installation for possible malfunctions in all possible positions of normal use before initial startup. If necessary, perform a special electromagnetic compatibility (EMC) test on the installation.
-

3.8 Protection Against Magnetic and Electromagnetic Fields During Operation and Mounting

Magnetic and electromagnetic fields generated near current-carrying conductors and permanent magnets in motors represent a serious health hazard to persons with heart pacemakers, metal implants and hearing aids.



WARNING

Health hazard for persons with heart pacemakers, metal implants and hearing aids in proximity to electrical equipment!

- ⇒ Persons with heart pacemakers, hearing aids and metal implants are not permitted to enter the following areas:
 - Areas in which electrical equipment and parts are mounted, being operated or started up.
 - Areas in which parts of motors with permanent magnets are being stored, operated, repaired or mounted.
 - ⇒ If it is necessary for a person with a heart pacemaker to enter such an area, then a doctor must be consulted prior to doing so. Heart pacemakers that are already implanted or will be implanted in the future, have a considerable variation in their electrical noise immunity. Therefore there are no rules with general validity.
 - ⇒ Persons with hearing aids, metal implants or metal pieces must consult a doctor before they enter the areas described above. Otherwise, health hazards will occur.
-

3.9 Protection Against Contact with Hot Parts



CAUTION

**Housing surfaces could be extremely hot!
Danger of injury! Danger of burns!**

- ⇒ Do not touch housing surfaces near sources of heat! Danger of burns!
- ⇒ After switching the equipment off, wait at least ten (10) minutes to allow it to cool down before touching it.
- ⇒ Do not touch hot parts of the equipment, such as housings with integrated heat sinks and resistors. Danger of burns!

3.10 Protection During Handling and Mounting

Under certain conditions, incorrect handling and mounting of parts and components may cause injuries.



CAUTION

Risk of injury by incorrect handling! Bodily harm caused by crushing, shearing, cutting and mechanical shock!

- ⇒ Observe general installation and safety instructions with regard to handling and mounting.
- ⇒ Use appropriate mounting and transport equipment.
- ⇒ Take precautions to avoid pinching and crushing.
- ⇒ Use only appropriate tools. If specified by the product documentation, special tools must be used.
- ⇒ Use lifting devices and tools correctly and safely.
- ⇒ For safe protection wear appropriate protective clothing, e.g. safety glasses, safety shoes and safety gloves.
- ⇒ Never stand under suspended loads.
- ⇒ Clean up liquids from the floor immediately to prevent slipping.

3.11 Battery Safety

Batteries contain reactive chemicals in a solid housing. Inappropriate handling may result in injuries or material damage.



CAUTION

Risk of injury by incorrect handling!

- ⇒ Do not attempt to reactivate discharged batteries by heating or other methods (danger of explosion and cauterization).
- ⇒ Never charge non-chargeable batteries (danger of leakage and explosion).
- ⇒ Never throw batteries into a fire.
- ⇒ Do not dismantle batteries.
- ⇒ Do not damage electrical components installed in the equipment.

Note: Be aware of environmental protection and disposal! The batteries contained in the product should be considered as hazardous material for land, air and sea transport in the sense of the legal requirements (danger of explosion). Dispose batteries separately from other waste. Observe the legal requirements in the country of installation.

3.12 Protection Against Pressurized Systems

Certain motors and drive controllers, corresponding to the information in the respective Project Planning Manual, must be provided with pressurized media, such as compressed air, hydraulic oil, cooling fluid and cooling lubricant supplied by external systems. Incorrect handling of the supply and connections of pressurized systems can lead to injuries or accidents. In these cases, improper handling of external supply systems, supply lines or connections can cause injuries or material damage.



CAUTION

Danger of injury by incorrect handling of pressurized systems!

- ⇒ Do not attempt to disassemble, to open or to cut a pressurized system (danger of explosion).
- ⇒ Observe the operation instructions of the respective manufacturer.
- ⇒ Before disassembling pressurized systems, release pressure and drain off the fluid or gas.
- ⇒ Use suitable protective clothing (for example safety glasses, safety shoes and safety gloves)
- ⇒ Remove any fluid that has leaked out onto the floor immediately.

Note: Environmental protection and disposal! The media used in the operation of the pressurized system equipment may not be environmentally compatible. Media that are damaging the environment must be disposed separately from normal waste. Observe the legal requirements in the country of installation.

Notes

4 Transformers

Transformers are only needed if the systems voltage exceeds the rated voltage of the drive controller.

Grounded power supply lines

For grounded power supply lines, the line voltage is matched to the rated voltage of the units using **autotransformers** which are suited to **one specific output voltage range**.

Ungrounded power supply lines

To match the voltage for ungrounded power supply lines, always connect **isolating transformers** to prevent excess phase to ground voltages.

4.1 Autotransformers

Type Code

Abbrev. Column		1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	2	1	2	3	4	5	6	7	8	9	0	3	1	2	3	4	5	6	7	8	9	0	4		
Example:		T	R	A	F	O						D	S	T		•	4	,	0	0	/	L	/	3	8	0	,	4	1	5	,	4	4	0	-	2	2	0					1	0	M	M

1.

Object

1.1

Transformer = TRAFO

2.

Product

2.1

DST. = DST

3.

Nominal power

3.1

2.0 kVA = •2,00

3.2

2.5 kVA = •2,50

3.3

4.0 kVA = •4,00

3.4

5.0 kVA = •5,00

3.5

7.5 kVA = •7,50

3.6

10.0 kVA = 10,00

3.7

12.5 kVA = 12,50

3.8

15.0 kVA = 15,00

3.9

18.0 kVA = 18,00

3.10

20.0 kVA = 20,00

3.11

25.0 kVA (for vertical mounting **only**) = 25,00

3.12

35.0 kVA (for vertical mounting **only**) = 35,00

3.13

50.0 kVA (for vertical mounting **only**) = 50,00

4.

Construction (Design)

4.1

suitable for mounting into IP55 housing = G

4.2

horizontal mounting = L

4.3

vertical mounting = S

5.

Nominal input voltage (Phase-Phase)

5.1

e.g., AC 380 V, AC 415 V, AC 440 V = 380, 415, 440

6.

Nominal output voltage (Phase-Phase)

6.1

e.g., AC 230 V. = 220

7.

Special design

7.1

Does not apply to standard transformers.

7.2

Frequency: e.g., 100 Hz = 100HZ

7.3

max. line diameter: e.g., 10 mm² = 10MM

7.4

Protecting style: e.g., IP23, in protecting housing ST0 = IP23

7.5

Nema-type = NEMA

7.6

UL standard = UL-N

8.

Standard reference

Standard

DIN EN 60529

Title

Degrees of protection provided by enclosures (IP code)
(IEC 60529:1989 + A1:1999); German version
EN 60529:1991 + A1:2000

Edition

2000-09

Note:

• = Field does not apply

Selection

Select an autotransformer suited to both the line voltage and the power requirements of the system.

Proceed with the selection as follows:

- ⇒ Determine the rating group using the required rated line voltage range in the diagram "Classification of the three-phase alternating current autotransformers and then locate the transformer ratio "i".
- ⇒ Calculate the actual transformer output voltage using the rated line voltage and the transformation ratio "i".
- ⇒ Check the drive data. The output voltage of the transformer has an effect on the drive data.
- ⇒ Select a three-phase autotransformer corresponding to the required connected load.

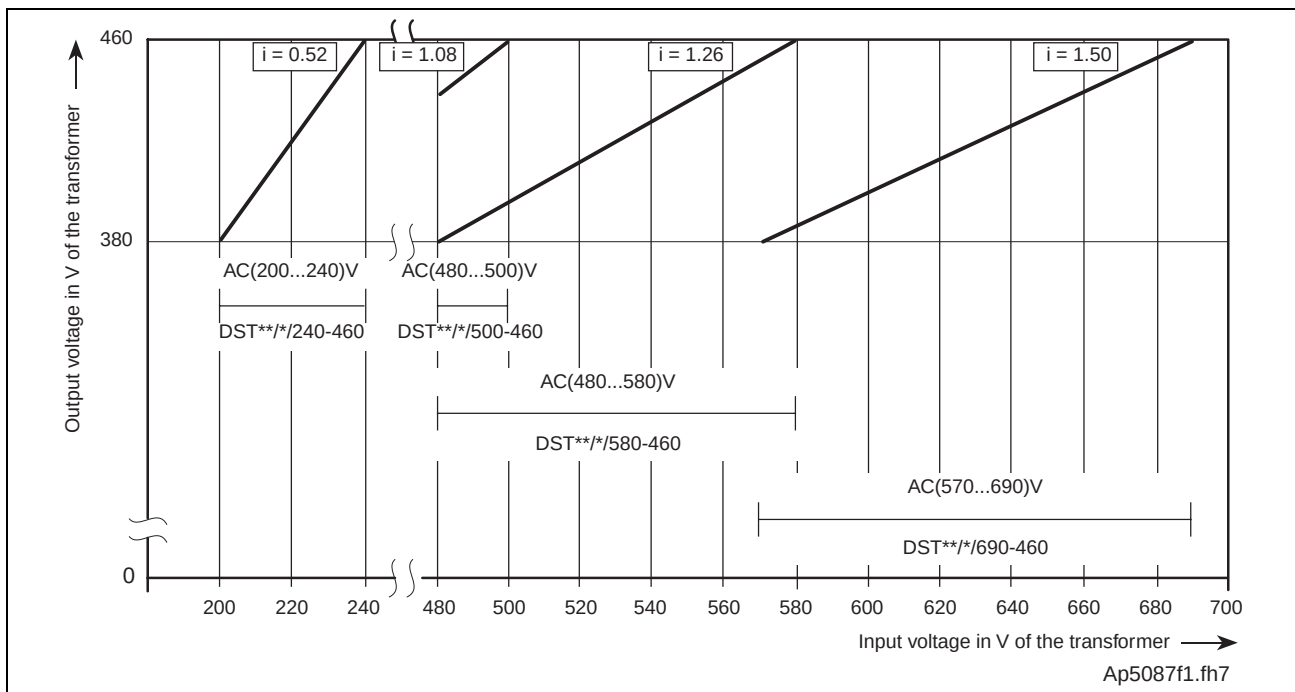
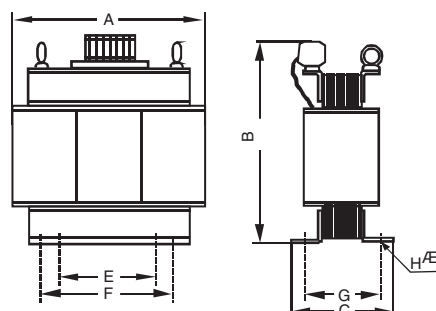


Fig. 4-2: Classification of three-phase autotransformers in rating groups

Technical Data

DST autotransformer with secondary or output voltage AC (380 to 460) V

Standing version for mounting with base: DST.../S



Rating plate (example)

Type	DST 20/S/580 - 480	Bj.	2004
Prim.	480 580 V		
Sec.	380 460 V	30 25 A	
S	20 kVA	YNa0	T 40/B f 50/60 Hz

Block diagram



1) Temperature switch max. load:
DC 24V/1A; AC 230V/1A

Type designation DST...	Conn. output in KVA	Trans- mission ratio	Dimensions in mm							Power loss in W	max. cross section in mm ²	Weight in kg
			A	C	B	F	E	G	HØ			
Input voltage: AC (200...240) V ±10%												
4/S/240-460	4		240	150	260	170	110	120	11	120	10	24.5
7,5/S/240-460	7.5		335	175	365	230	160	145	11	225	10	55
12,5/S/240-460	12.5	0.52	360	190	395	250	170	160	11	310	10	70
25/S/240-460	25		480	195	500	356	-----	158	13	500	35	135
50/S/240-460	50		580	265	540	400	270	215	18	750	70	195
Input voltage: AC (480...500) V ±10%												
4/S/500-460	4		180	105	190	125	80	75	7	160	4	8.5
7,5/S/500-460	7.5		205	130	210	145	95	95	7	260	4	13
12,5/S/500-460	12.5	1.08	240	140	260	170	110	110	11	440	10	22
25/S/500-460	25		300	155	325	210	140	125	11	750	16	36
50/S/500-460	50		335	175	365	230	160	145	11	1050	35	53
Input voltage: AC (480...580) V ±10%												
4/S/580-460	4		240	130	260	170	110	100	11	140	4	18
7,5/S/580-460	7.5		240	140	260	170	110	110	11	260	4	22
12,5/S/580-460	12.5	1.26	300	155	325	210	140	125	11	375	10	37
25/S/580-460	25		360	190	395	250	170	160	11	625	10	72
50/S/580-460	50		420	215	450	280	190	155	14	1000	35	95
Input voltage: AC (570...690) V ±10%												
4/S/690-460	4		240	140	260	170	110	110	11	140	10	22
7,5/S/690-460	7.5		300	155	325	210	140	125	11	225	10	37
12,5/S/690-460	12.5	1.5	335	175	365	230	160	145	11	375	10	57
25/S/690-460	25		420	205	450	280	190	145	14	500	16	88
50/S/690-460	50		480	222	500	356	-----	185	13	750	35	178

MB5007f1.fh7

Fig. 4-3: Technical data

5 Mains Filter

5.1 Mains Filter NFD / NFE

Type Code NFE / NFD

NFE01.1

[illegible]

- | | | |
|-----|-----------------|-------|
| 1. | Product group | |
| 1.1 | NFE | = NFE |
| 2. | Line | |
| 2.1 | 1 | = 01 |
| 3. | Design | |
| 3.1 | 1 | = 1 |
| 4. | Nominal voltage | |
| 4.1 | AC 250 V | = 250 |
| 5. | Nominal current | |
| 5.1 | 6,0 A | = 006 |

Illustration example: NFE01.1

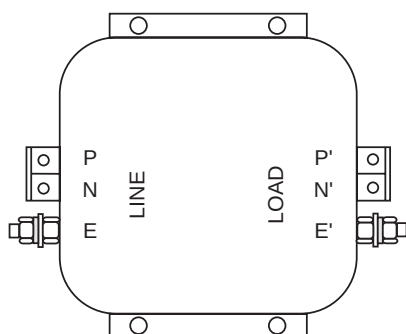


Fig. 5-1: Type code NFE01.1

NFE02.1

Abbrev.
Column

→

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	2	1	2	3	4	5	6	7	8	9	0	3	1	2	3	4	5	6	7	8	9	0	4
Example: N F E 0 2 . 1 - 2 3 0 - 0 0 8																																										

1. Product group

1.1 NFE = NFE

2. Line

2.1 2 = 02

3. Design

3.1 1 = 1

4. Nominal voltage

4.1 AC 230 V = 230

5. Nominal current

5.1 8,0 A = 008

Illustration example: NFE02.1

Fig. 5-2: Type code NFE02.1

NFD03.1

[illegible]

- | | | |
|-----|------------------------|-------|
| 1. | Product group | |
| 1.1 | NFD..... | = NFD |
| 2. | Line | |
| 2.1 | 3..... | = 03 |
| 3. | Design | |
| 3.1 | 1..... | = 1 |
| 4. | Nominal voltage | |
| 4.1 | AC 115 to 480 V | = 480 |
| 5. | Rated current | |
| 5.1 | 7 A | = 007 |
| 5.2 | 16 A | = 016 |
| 5.3 | 30 A | = 030 |
| 5.4 | 55 A | = 055 |
| 5.5 | 75 A | = 075 |
| 5.6 | 130 A | = 130 |
| 5.7 | 180 A | = 180 |

Illustration example: NFD03.1



Fig. 5-3: Type code NFD03.1

Technical Data NFE / NFD

Maximum connecting voltage of mains 50..60 Hz U_N in V	Nominal mains current I_{Netz} (1) in A	No. of phases	Type of mains filter	Terminals (3)			Power dissipation approx W	Weight kg	Type
				flexibel mm ²	starr mm ²	AWG			
AC 480V+10%	7	3	NFD 03.1-480-007	4 (3)	6 (3)	AWG 12	3,9	0,7	vertical
AC 480V+10%	16	3	NFD 03.1-480-016	4 (3)	6 (3)	AWG 12	6,4	1,0	vertical
AC 480V+10%	30	3	NFD 03.1-480-030	10	16	AWG 6	11,9	1,4	vertical
AC 480V+10%	55	3	NFD 03.1-480-055	16	25	AWG 4	25,9	2,0	vertical
AC 480V+10%	75	3	NFD 03.1-480-075	25	35	AWG 3	30,4	3,5	vertical
AC 480V+10%	130	3	NFD 03.1-480-130	50	50	AWG 1/0	38	4,7	vertical
AC 480V+10%	180	3	NFD 03.1-480-180	95	95	AWG 4/0	61	10	vertical
AC 230V+10%	7,5	1	NFE 02.1-230-008	4 (3)	6 (3)	AWG 10	7,2	1,1	vertical
AC 230 V+10%	4,7	1	NFE 01.1-250-006 (2)	tab connectors 6,3 x 0,8 mm			4	0,245	horizontal

NFD = three-phase filter, NFE = single-phase filter

(1) = maximum continuous current on mains side at 45°C ambient temperature
 (2) = to be used only for interference suppression of the NTM power supply unit
 (3) = a conductor cross section of 10 mm² is to be connected for the ground wire by means of a plug connector or ring cable lug

Fig. 5-4: Technical Data

Operating frequency	From 0 to 60 Hz at 45 °C
Power dissipation	measured at $2 \text{ or } 3 \times R I_{\text{Nenn DC}}^2$
Temperature range	-25 ... + 85 °C
Overload	1.5 I_{Nenn} 1 minute per hour or 4 I_{Nenn} für 10 s
Effective attenuation	frequency range 150 kHz – 30 MHz
Saturation	reduction of filter attenuation by 6 dB with 2.5-to 3-fold nominal current
Test voltage	L/N -> PE or L -> PE: 2700 V _{DC} 2 s at 25 °C N -> L 2100 V _{DC} 2 s at 25 °C
Current reduced with overtemperature	see formula for calculating the reduction
Leakage current at 50 Hz	symmetrical three-phase operation: typ. 30 mA, single-phase operation or in the case of fuse failure of a phase: typ. 175 ... 190mA
Protection category	IP20, except for NFE01.1-250-006: IP 10

Fig. 5-5: Technical Data

Dimensions NFE / NFD

Mains filter type	A	B	C	D	E	F	G	H	J	K	L	M	M _{AE}	M _{AKI}
NFD 03.1-480-007	190	90	50	160	180	20	5,4	---	---	190	---	M5	2,2	0,8
NFD 03.1-480-016	250	90	55	220	235	25	5,4	---	---	250	---	M5	2,2	0,8
NFD 03.1-480-030	270	100	60	240	255	30	5,4	---	---	270	---	M5	2,2	2
NFD 03.1-480-055	250	105	90	220	235	60	5,4	---	---	260	---	M6	4	2,2
NFD 03.1-480-075	270	145	90	240	255	60	6,5	---	---	280	---	M6	4	4,5
NFD 03.1-480-130	270	160	100	240	255	65	6,5	---	---	330	---	M10	18	8
NFD 03.1-480-180	380	180	130	350	365	102	6,5	---	---	455	---	M10	18	20
NFE 02.1-230-008	90	210	60	60	80	40	5,3	40	0,75	---	15	10	0,8	0,8
NFE 01.1-250-006	see figure													

M_{AE} = maximum tightening torque of the ground stud in Nm

M_{AKI} = maximum tightening torque of the terminal connector in Nm

Fig. 5-6: Dimensions of the mains filters NFD/NFE

NFD03.1

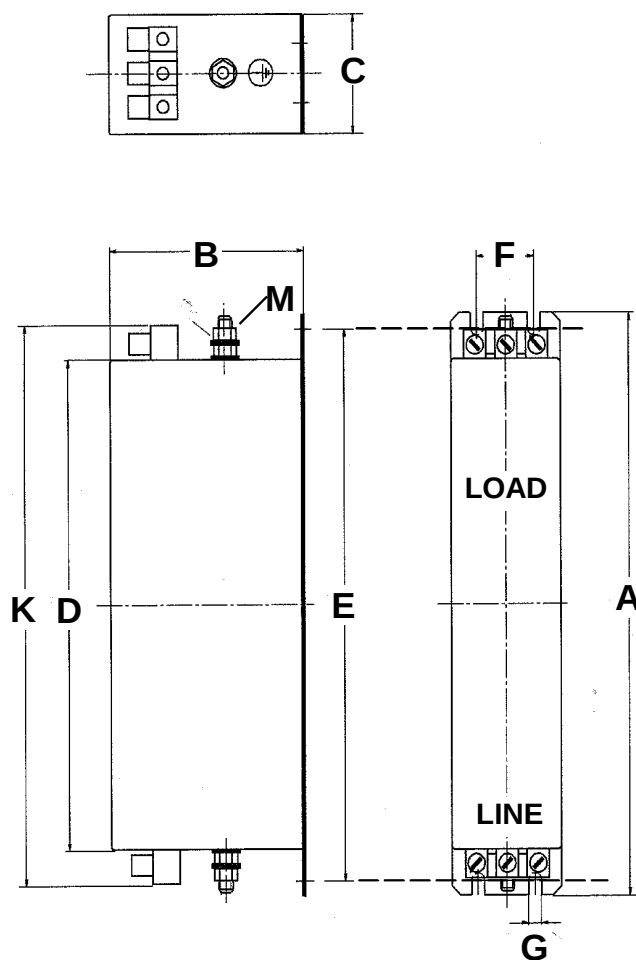
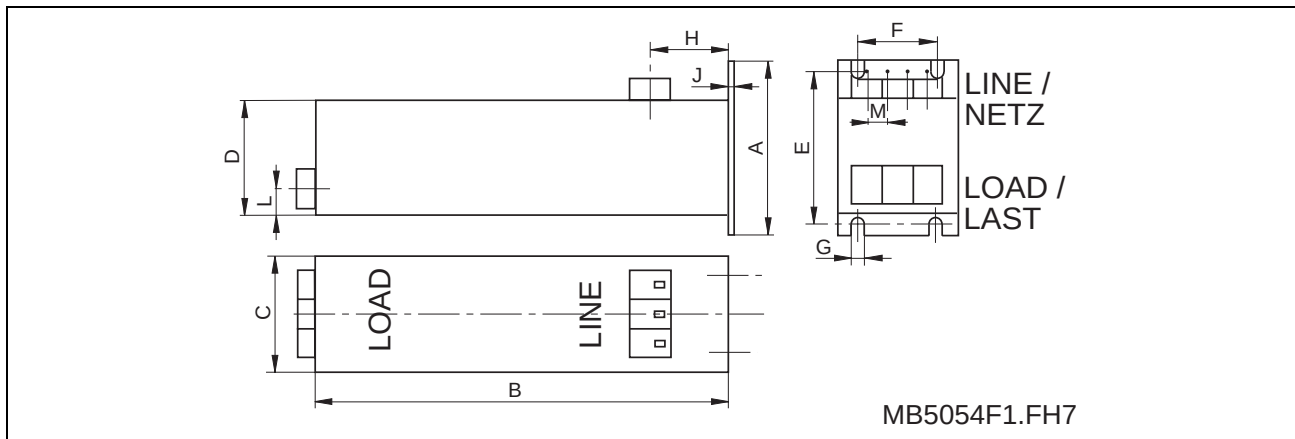


Fig. 5-7: AC filter NFD03.1 for drives

Tolerance limits for NFD03.1:

- Dimensions B, C, D, K are maximum values. They can be reduced up to 15 mm.
- The ground studs M can also stand vertically, seen from the front side, instead of horizontally.

NFE02.1

NFE02.1-230-008 (with 3 terminal connectors)

Fig. 5-8: Single-phase filter NFE02.1 for drives

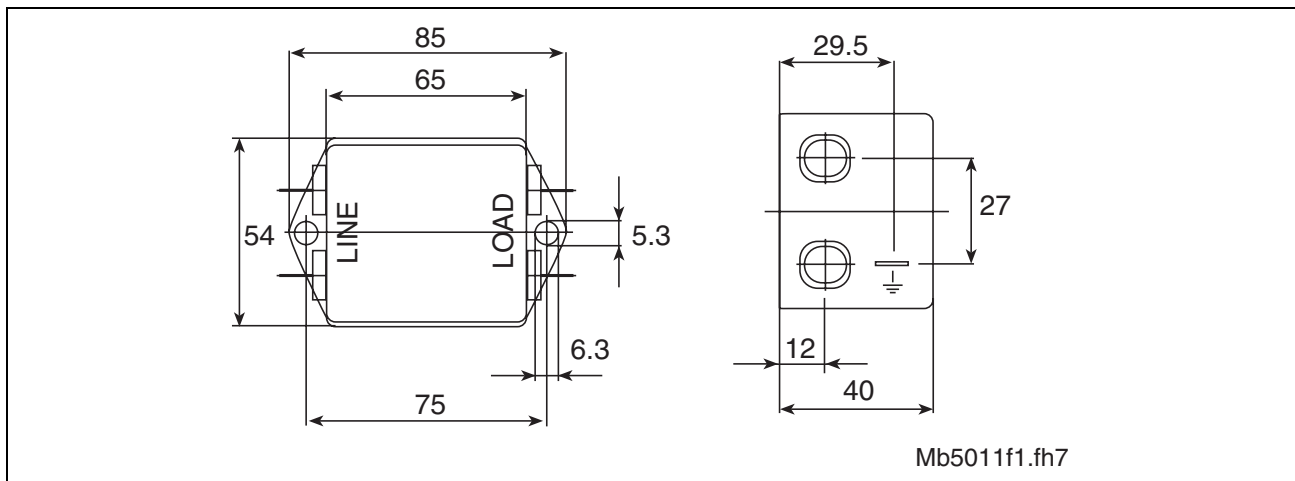
NFE01.1

Fig. 5-9: Single-phase filter NFE01.1-250-006 for interference suppression of the power supply unit NTM

Note: The mains filter is connected by means of tab receptacles (b = 6.3 mm, d = 1 mm).

5.2 Mains Filter HNF01.1

Type Code HNF01.1

Abbrev. Column		1	2	3	4	5	6	7	8	9	1	0	1	2	3	4	5	6	7	8	9	2	0	1	2	3	4	5	6	7	8	9	3	0	1	2	3	4	5	6	7	8	9	4	0
Example:		H	N	F	0	1	.	1	A	-	F	2	4	0	-	R	0	0	9	4	-	A	-	4	8	0	-	N	N	N	N														

1. Product

1.1 HNF = HNF

2. Line

2.1 1..... = 01

3. Design

3.1 1 = 1

4. EMC area per DIN EN 61800-3

4.1 industrial area = A

4.2 residential area = B

5. Applications

5.1 6 axes, cable length max. 240 m = F240

5.2 15 axes, cable length max. 900 m = M900

6. Supply system

6.1 only for feeding units = E

6.2 only for regenerative units = R

7. Nominal current

7.1 e.g., 94 A..... = 0094

8. Degree of protection

8.1 IP20. = A

9. Mains connecting voltage

9.1 3 AC 400 to 480V -15+10%, 50/60 Hz = 480

9.2 3 AC 400 to 500V -15+10%, 50/60 Hz = 500

10. Other design

10.1 none = NNNN

11. Standard reference

Standard

Title

Edition

DIN EN 60529

Degrees of protection provided by enclosures (IP-Code)

2000-09

DIN EN 61800-3

Adjustable speed electrical power drive systems - Part 3:
EMC product standard including specific test methods
(IEC 61800-3:1996); German version EN 61800:1996 + A11:2000

2001-02

Fig. 5-10: Type code

Data HNF01.1

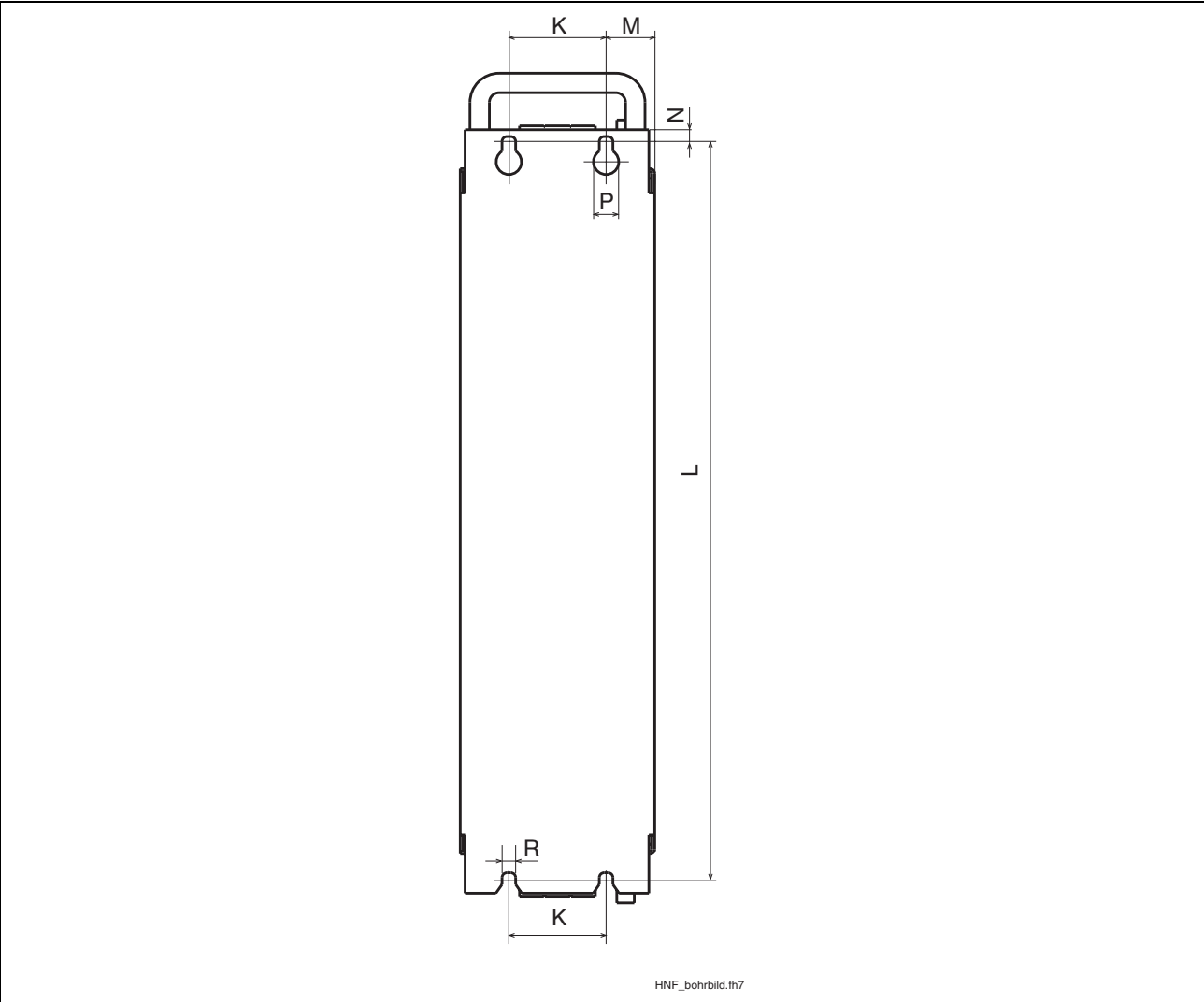
	Unit	HNF01.1A- ... -A-480-NNNN											
		F240-E0051	M900-E0051	F240-E0125	M900-E0125	F240-E0202	M900-E0202	F240-R0026	M900-E0026	F240-R0065	M900-E0065	F240-R0094	M900-E0094
Weight	kg	15	15	18	30	29	37	14	17	25	26	28	29
Terminal L1, L2, L3 resp. L1.1, L2.1, L3.1:													
Cross section flexible min.	mm ²	0,5	0,5	16	16	50	50	0,5	0,5	10	10	16	16
Cross section flexible max.	mm ²	16	16	50	50	150	150	10	10	25	25	50	50
Cross section rigid min.	mm ²	0,5	0,5	16	16	35	35	0,5	0,5	6	6	16	16
Cross section rigid max.	mm ²	25	25	50	50	150	150	16	16	35	35	50	50
Cross section AWG min.	--	20	20	6	6	2	2	20	20	10	10	6	6
Cross section AWG max.	--	4	4	0	0	00/3	00/3	6	6	2	2	0	0
Tightening torque	Nm	4,8 ±0,5	4,8 ±0,5	< 25	< 25	< 25	< 25	4,8 ±0,5	4,8 ±0,5	4,8 ±0,5	4,8 ±0,5	< 25	< 25
Power dissipation	W	< 89	< 91	< 127	< 174	< 238	< 373	< 73	< 77	< 163	< 157	< 135	< 146
Phase current (continuous current)	A	51	51	125	125	202	202	26	26	65	65	94	94
Phase current (peak current ¹⁾)	A	77	77	188	188	303	303	65	65	163	163	235	235
Leakage current at filters input (line side)	A	< 2	< 2	< 2,2	< 2,2	< 2,5	< 2,5	< 2	< 2	< 2	< 2	< 2,2	< 2,2
Leakage current at filters output (load side)	A	< 2	< 2	< 2,2	< 2,2	< 2,5	< 2,5	< 2	< 2	< 2	< 2	< 2,2	< 2,2
allowed THD (Total Harmonic Distortion)	%	see Project Planning Manual "Rexroth IndraDrive Drive System"											
Reduction of allowed operating data due to harmonic content		see Project Planning Manual "Rexroth IndraDrive Drive System"											
Input supply voltage	V	3 x AC 380...480V ±10%, 50/60Hz ±2%											
Sum of leakage currents at filters input with power supply unit switched off: • 1 phase failed • 2 phases failed	A	< 0,83 < 1,4											
Leakage resistance (Phase – Ground)	MOhm	> 15											
Protection category	-	IP20											
Materials	-	free of asbestos and silicone											

1) duration: 300 ms ;
cycle duration:
for feeding units: 0,67 s;
for regenerating units: 1,42 s;
base load: 60% (continuous current)

Fig. 5-11: Data

Dimensional Drawings HNF01.1

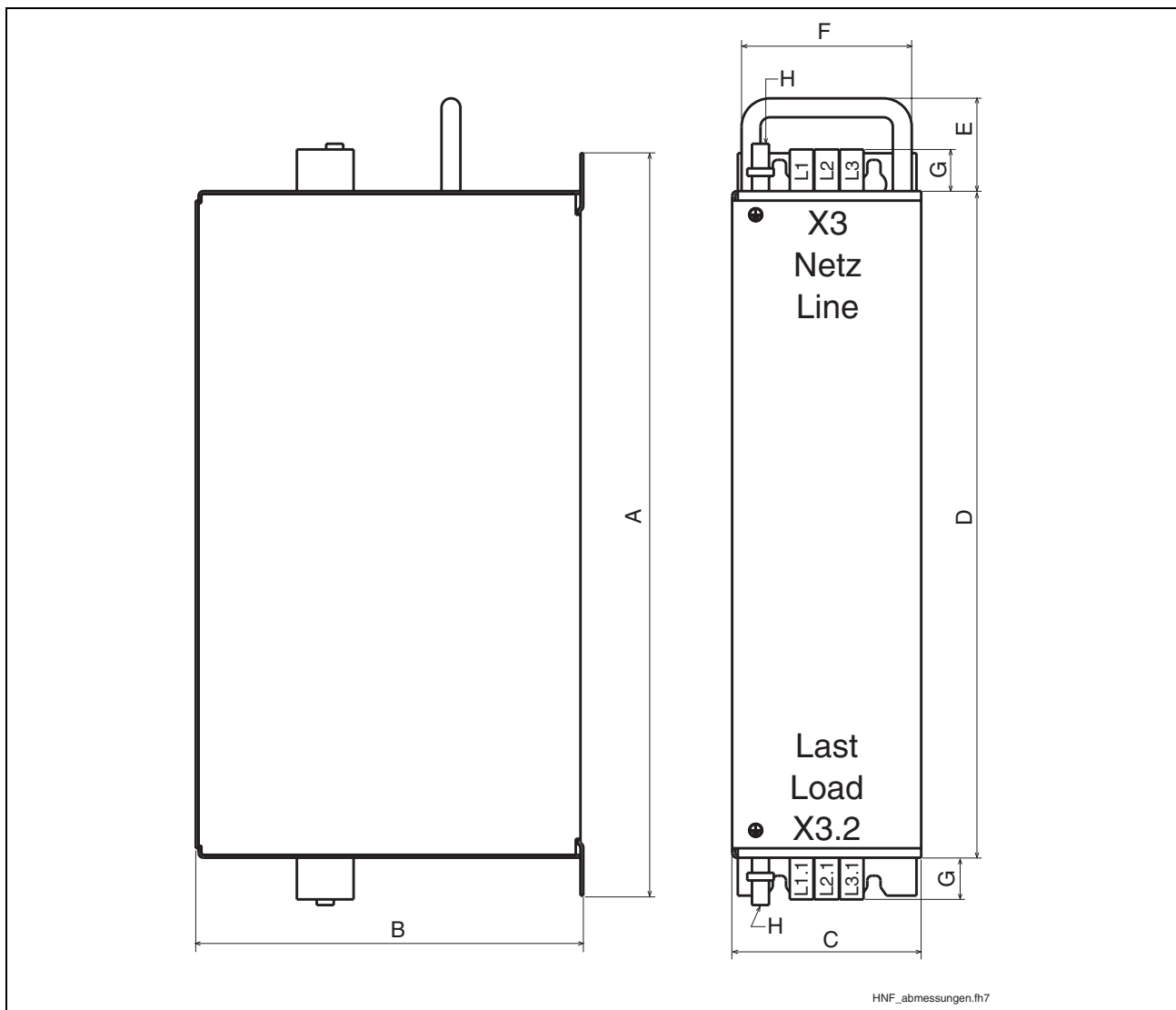
Drilling Pattern



Mains filter	K [mm]	L [mm]	M [mm]	N [mm]	P [mm]	R [mm]
HNF01.1A-F240-E0051-A-480-NNNN HNF01.1A-M900-E0051-A-480-NNNN	50	466	25	7	13	6,5
HNF01.1A-F240-E0125-A-480-NNNN HNF01.1A-M900-E0125-A-480-NNNN	125	466	12,5	7	13	6,5
HNF01.1A-F240-E0202-A-480-NNNN	125	466	12,5	7	13	6,5
HNF01.1A-M900-E0202-A-480-NNNN	150	466	50	7	13	6,5
HNF01.1A-F240-R0026-A-480-NNNN	50	466	25	7	13	6,5
HNF01.1A-M900-R0026-A-480-NNNN	100	466	25	7	13	6,5
HNF01.1A-F240-R0065-A-480-NNNN HNF01.1A-M900-R0065-A-480-NNNN	100	466	25	7	13	6,5
HNF01.1A-F240-R0094-A-480-NNNN HNF01.1A-M900-R0094-A-480-NNNN	125	466	12,5	7	13	6,5

Fig. 5-12: Drilling pattern

Dímenions



Mains filter	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	F [mm]	G [mm]	H [mm]
HNF01.1A-F240-R0026-A-480-NNNN	480	262	100	440	-	-	26	M6x20 8.6 Nm
HNF01.1A-F240-E0051-A-480-NNNN HNF01.1A-M900-E0051-A-480-NNNN	480	262	100	440	-	-	33	M6x16 2.3 Nm
HNF01.1A-M900-R0026-A-480-NNNN	480	262	150	440	-	-	26	M6x20 8.6 Nm
HNF01.1A-F240-R0065-A-480-NNNN HNF01.1A-M900-R0065-A-480-NNNN	480	262	150	440	40	110	40	M6x16 4.8 Nm
HNF01.1A-F240-R0094-A-480-NNNN HNF01.1A-M900-R0094-A-480-NNNN	480	262	150	440	40	110	45	M10x30 25 Nm
HNF01.1A-F240-E0125-A-480-NNNN HNF01.1A-M900-E0125-A-480-NNNN	480	262	150	440	40	110	45	M10x30 25 Nm
HNF01.1A-F240-E0202-A-480-NNNN	480	262	150	440	40	110	63.5	M10x30 25 Nm
HNF01.1A-M900-E0202-A-480-NNNN	480	262	250	440	40	110	63.5	M10x30 25 Nm

Fig. 5-13: Dimensions

5.3 Mains Filter (-Combination) HNK01.1

Type Code HNK01.1

[illegible]

Fig. 5-14: Type code HNK01.1

Type Plate HNK01.1

Each mains filter is identified by a type designation. A type plate is attached to all units.

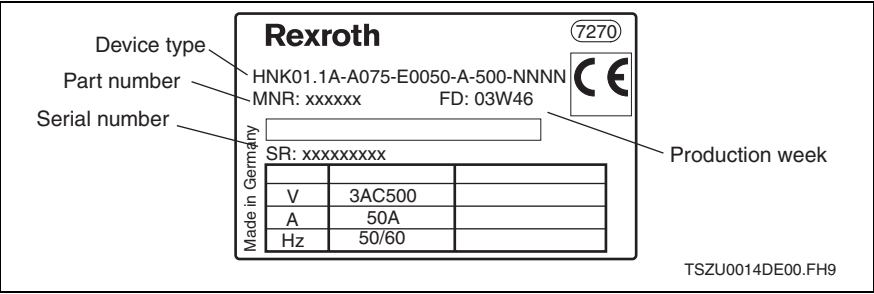


Fig. 5-15: Example of a type plate HNK01.1

Type plate arrangement:

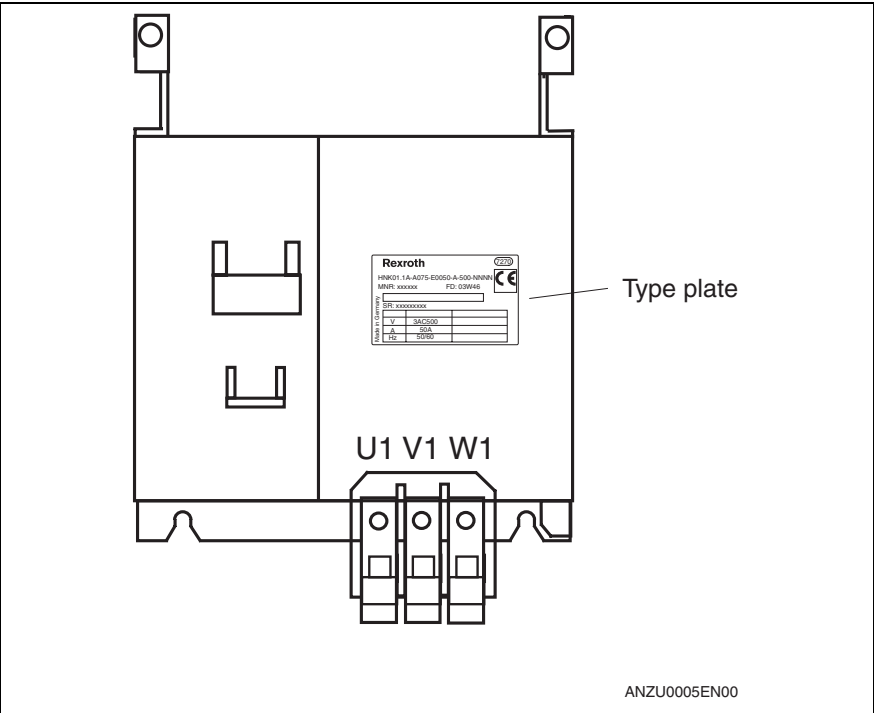
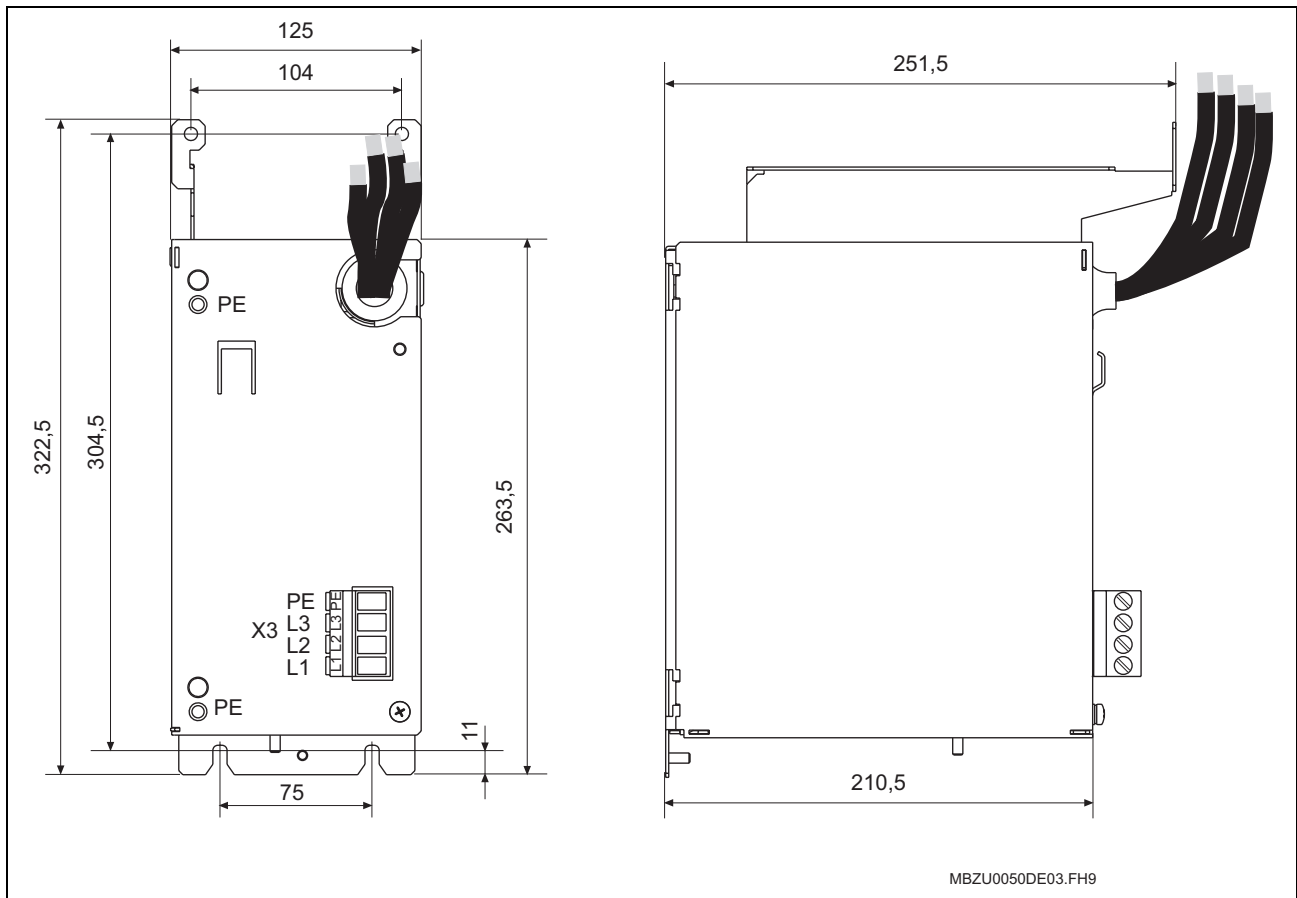
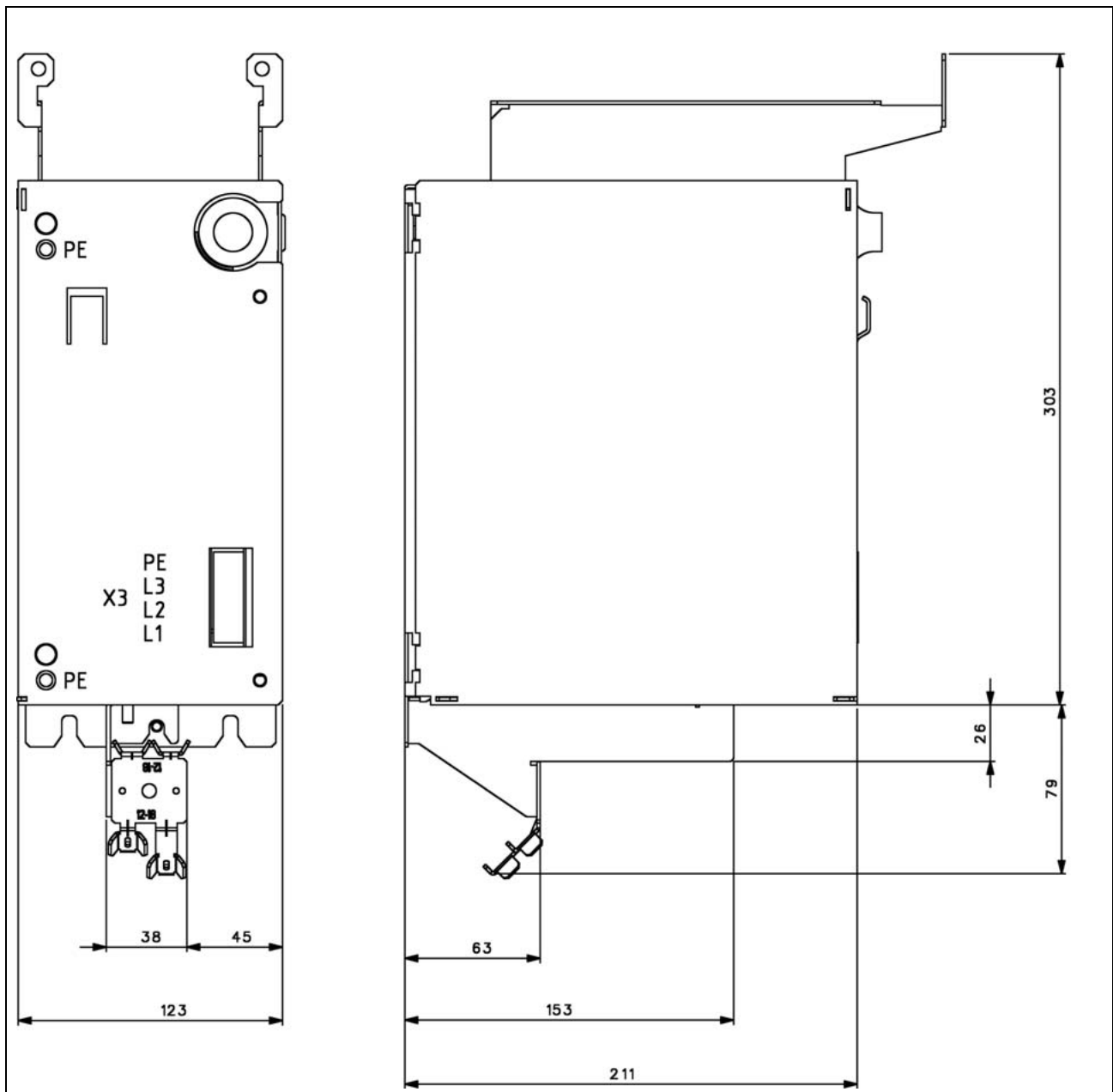


Fig. 5-16: Type plate arrangement

Mechanical Data HNK01.1

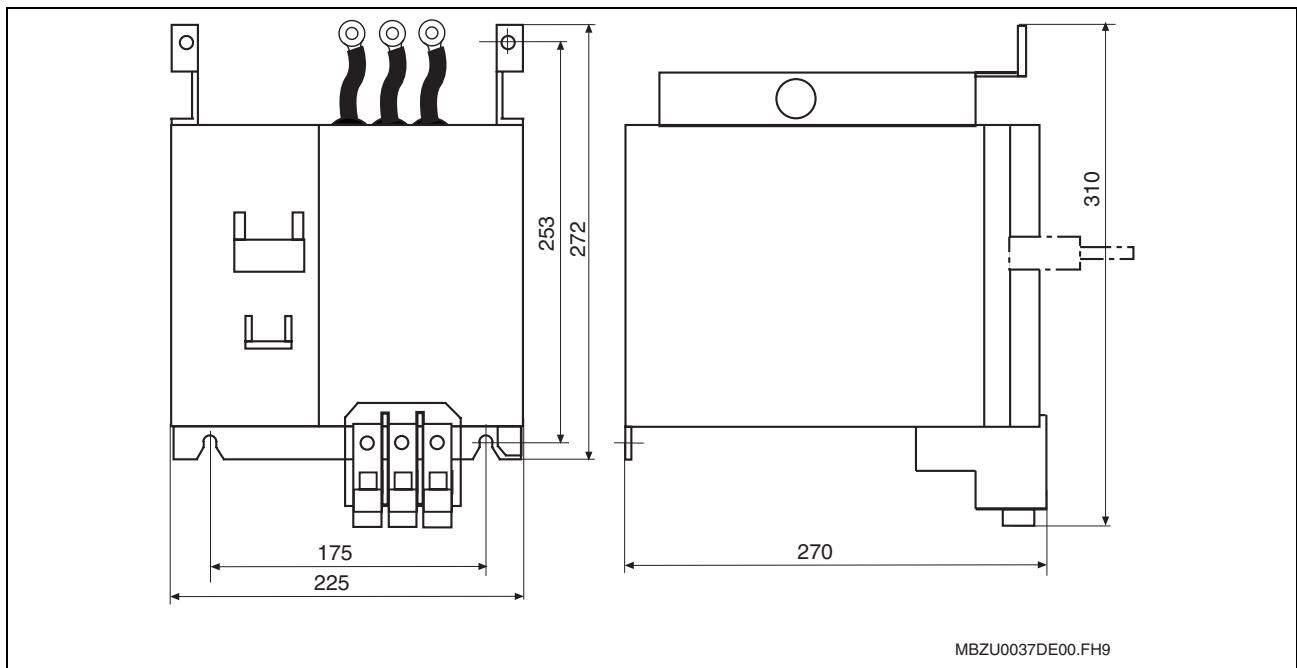
Dimensional Drawings





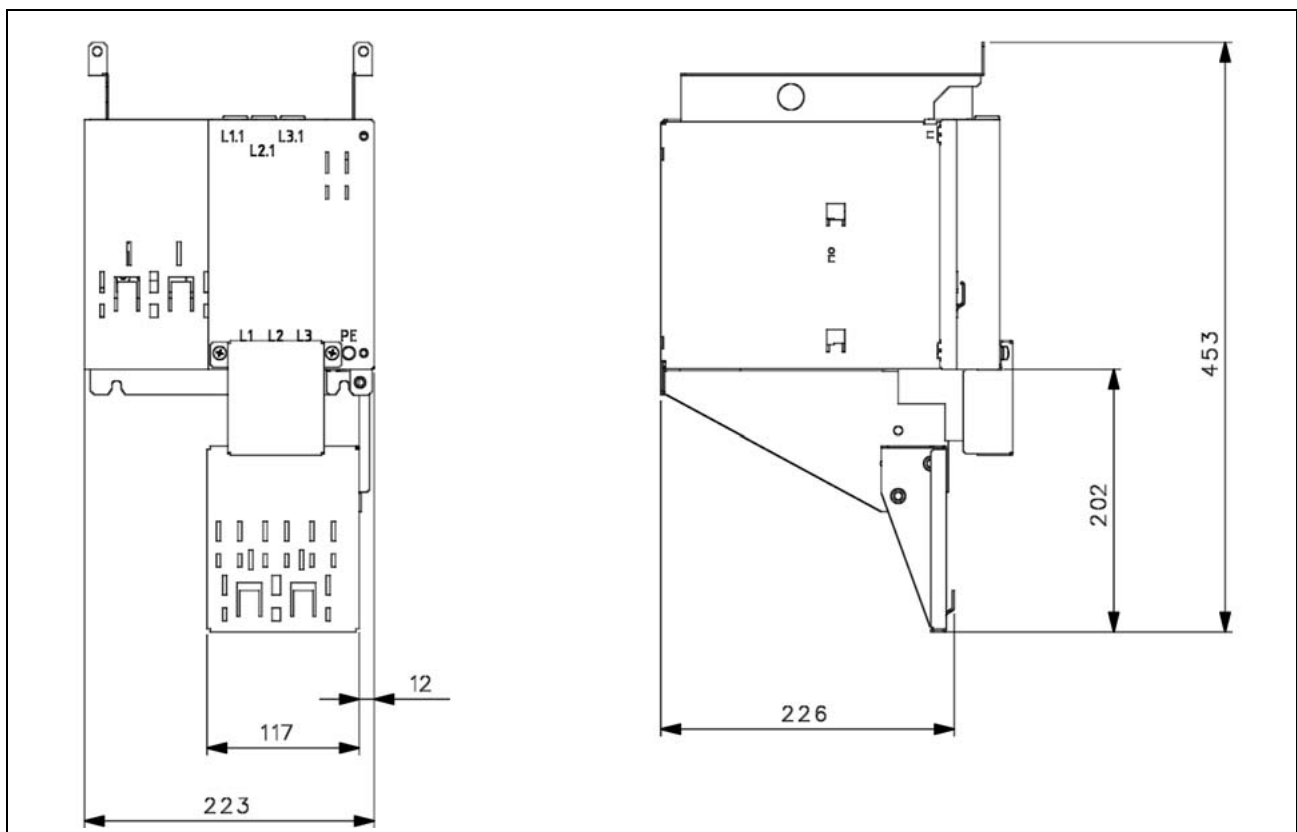
dimensions in mm

Fig. 5-18: Dimensional drawing mains filter HNK01.1A-A075-E0050 with HAS02



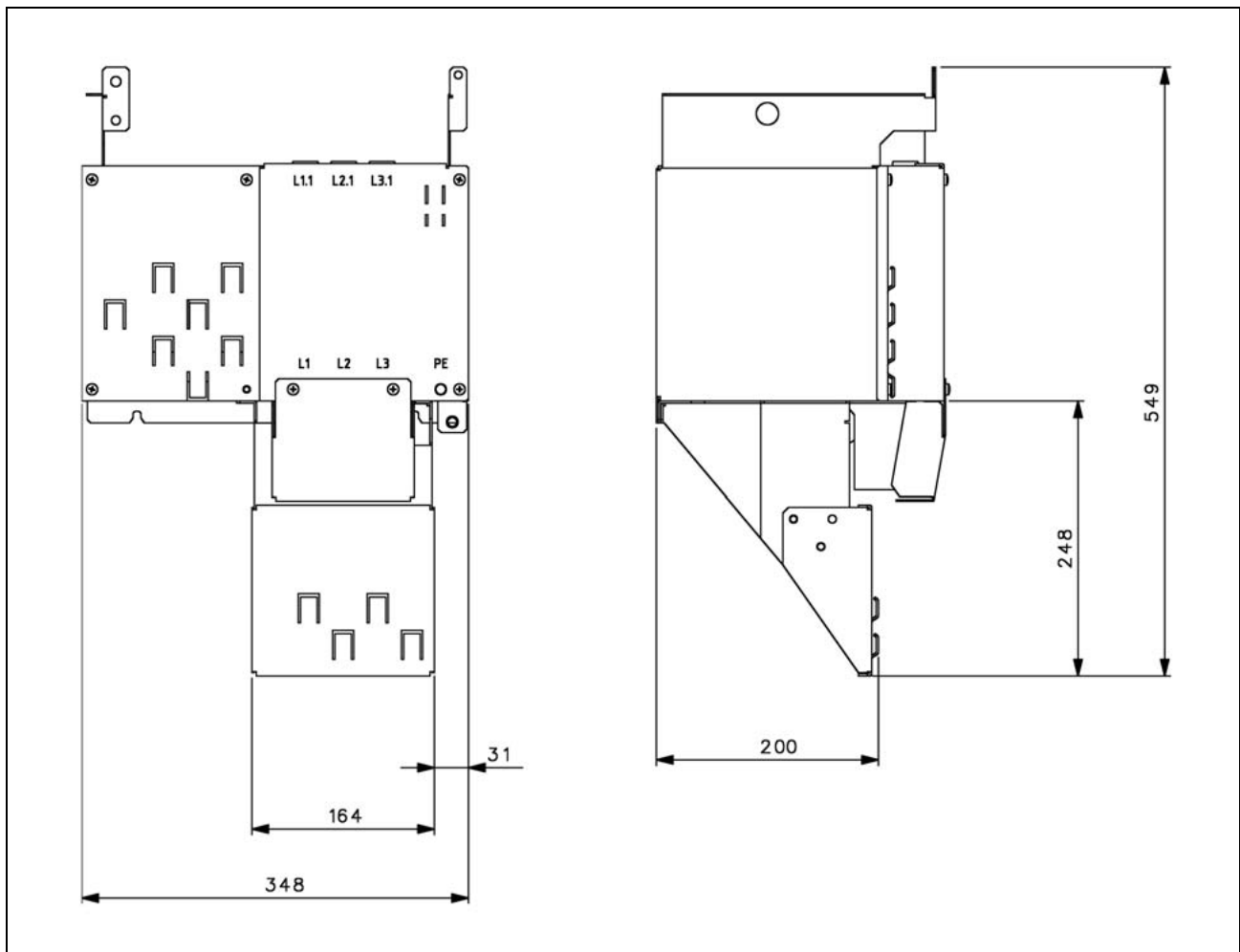
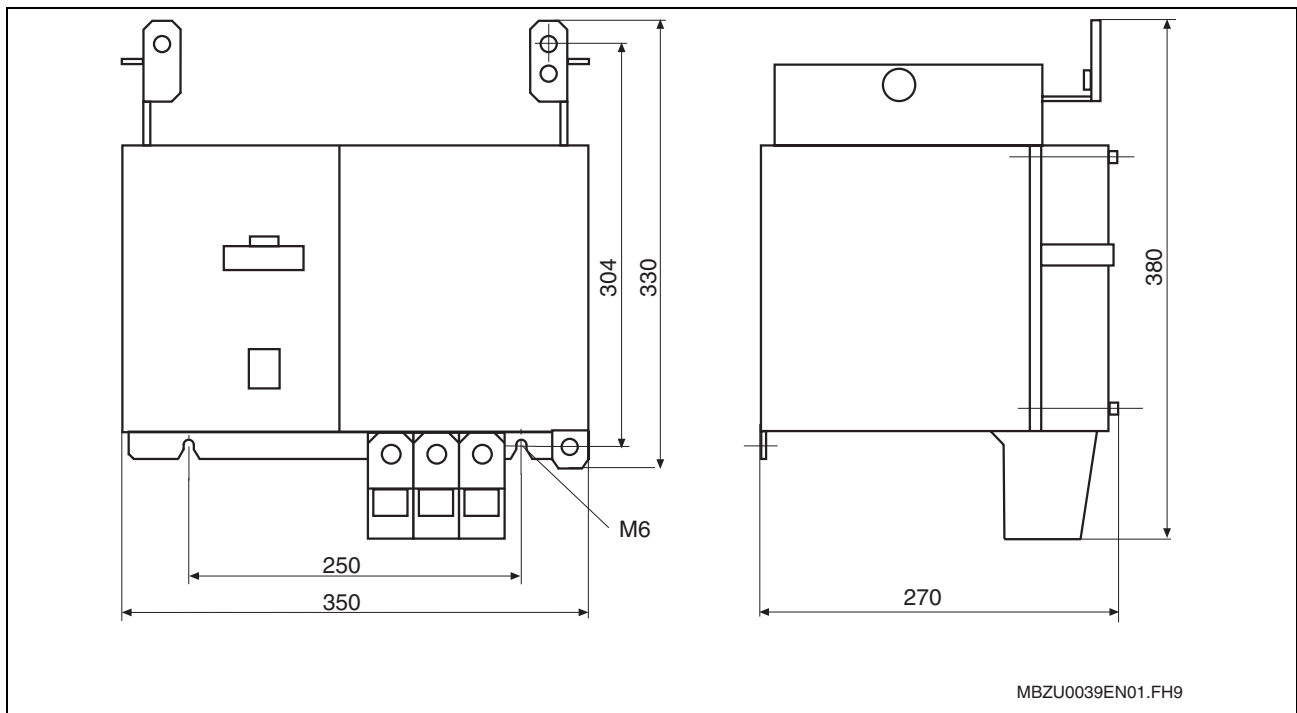
dimensions in mm

Fig. 5-19: Dimensional drawing mains filter HNK01.1A-A075-E0080/0106



dimensions in mm

Fig. 5-20: Dimensional drawing mains filter HNK01.1A-A075-E0080/0106 with HAS02



Technical Data HNK01.1

Mains filter HNK01.1A-A075	Unit	E0050	E0080	E0106	E0146
Nominal voltage	V	3*AC400...500			
Nominal current	A	50	80	106	146
Inductance	μH	571	362	240	170
Connection cross section	mm²	16 (AWG 6)	25 (AWG 3)	25 (AWG 3)	50 (AWG 1/0)
Protection category	-	IP20			
Weight	kg	15	20	20	28

Fig. 5-23: Technical Data HNK01.1

Allocation of Mains Filter HNK01 to HCS03 drive controllers

Mains filter HNK01	Drive controller
HNK01.1A-A075-E0050	HCS03.1E-W0070-
HNK01.1A-A075-E0080	HCS03.1E-W0100-
HNK01.1A-A075-E0106	HCS03.1E-W0150-
HNK01.1A-A075-E0146	HCS03.1E-W0210-

Fig. 5-24: Allocation of HNK01 to HCS03

Arrangement of Components

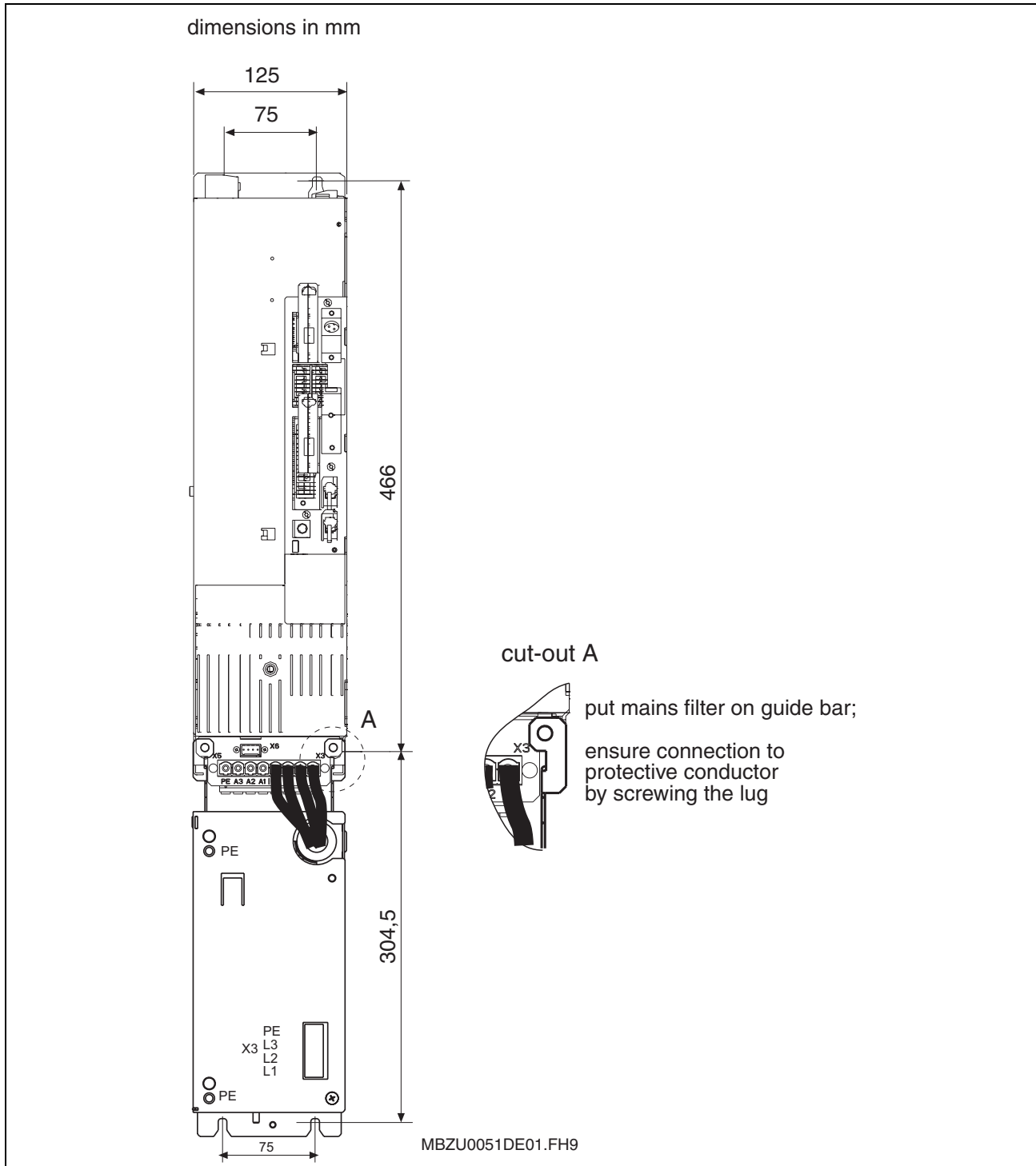


Fig. 5-25: Arrangement of drive controller and mains filter E0050

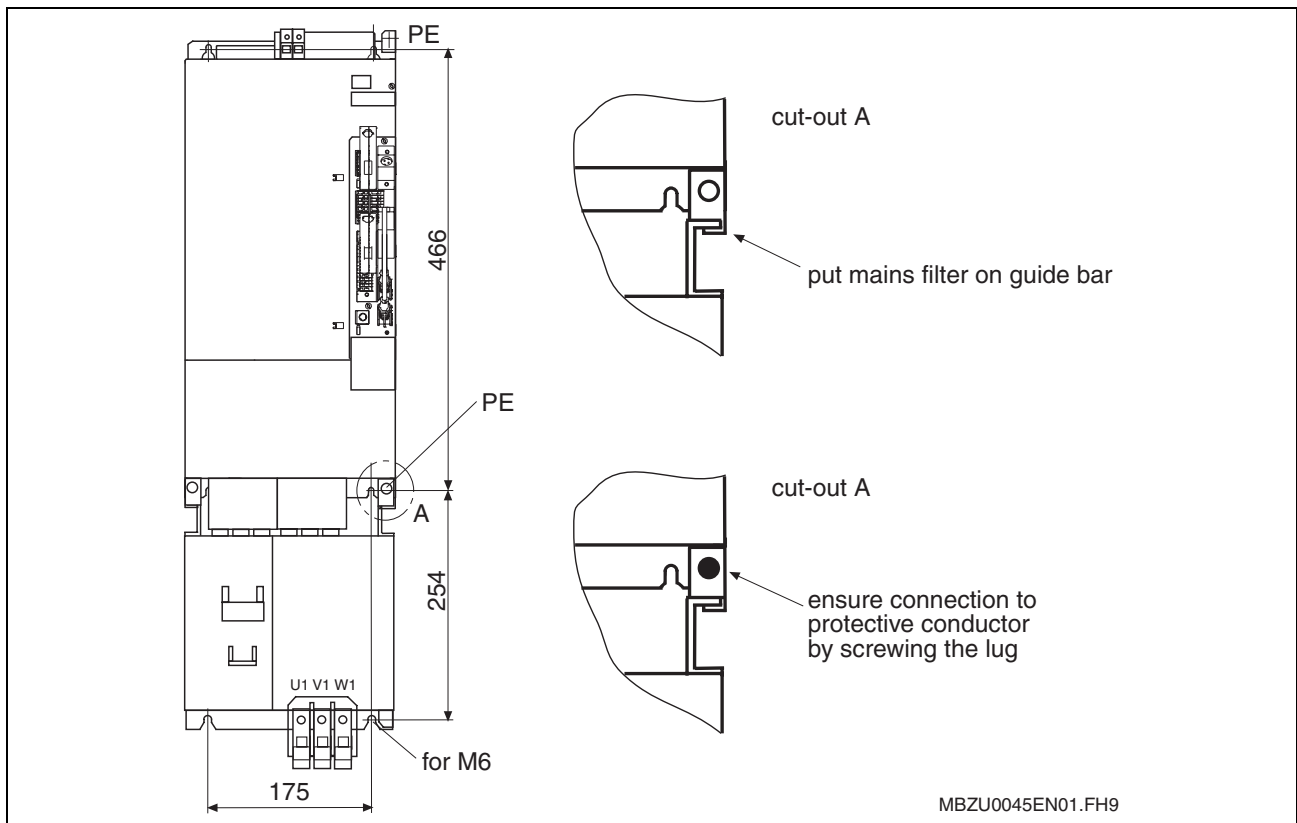


Fig. 5-26: Arrangement of drive controller and mains filter E0080 and E0106

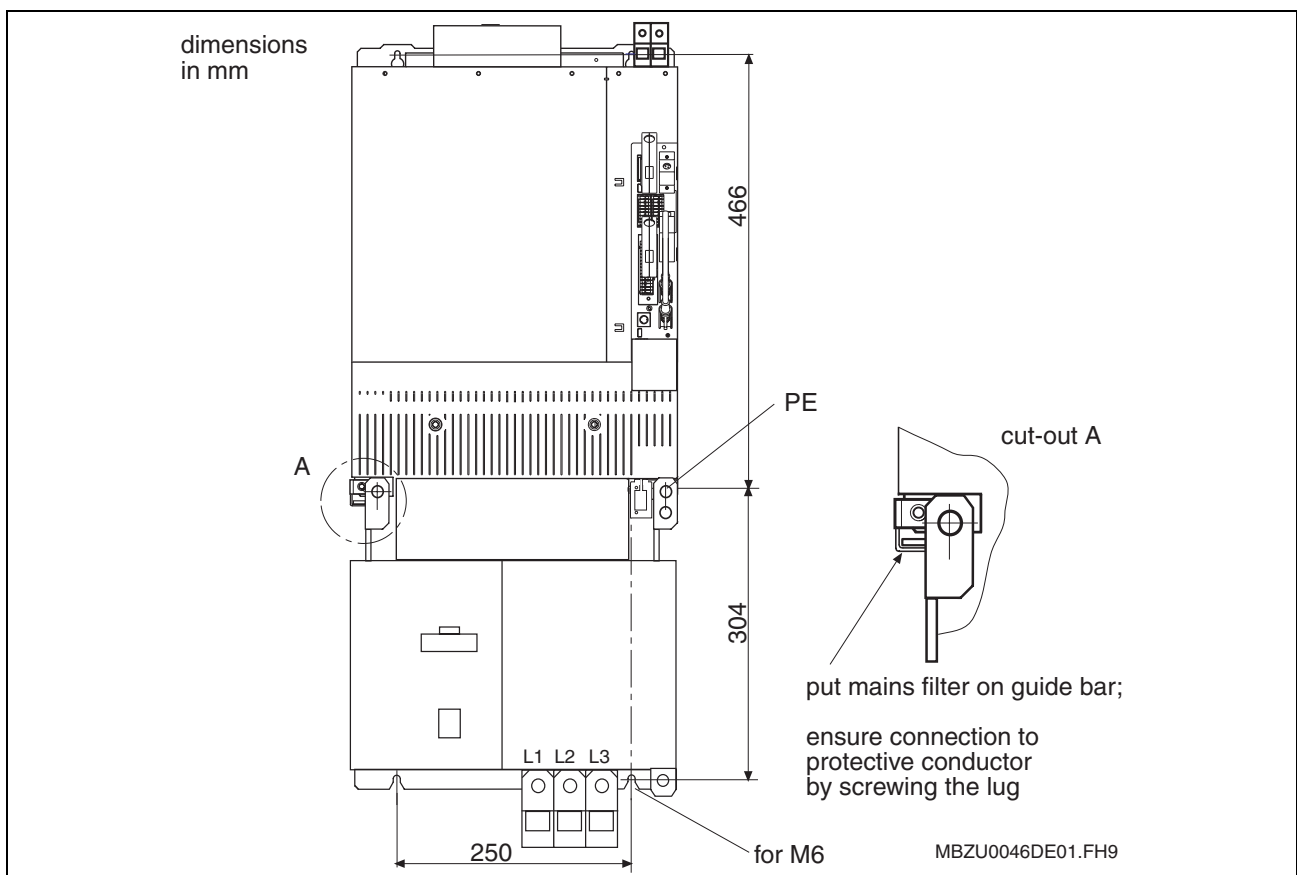


Fig. 5-27: Arrangement of drive controller and mains filter E0146

6 Mains Choke

6.1 Identification

Type Code

Abbrev.	Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40		
Example:		H	N	L	0	1	.	1	E	-	0	9	8	0	-	N	0	0	2	6	-	A	-	4	8	0	-	N	N	N	N												

1. Product

1.1 HNL = HNL

2. Line

2.1 1..... = 01

3. Design

3.1 1..... = 1

4. Supply system

4.1 only for feeding units = E

4.2 only for regenerative units = R

5. Nominal inductance

5.1 e.g., 980 µH..... = 0980

6. Additionl option

6.1 with capacitors = C

6.2 current-compensated = S

6.3 none = N

7. Nominal current

7.1 e.g., 26 A..... = 0026

8. Degree of protection

8.1 IP20 = A

9. Mains connecting voltage

9.1 3 AC 400 to 480V -15+10%, 50/60 Hz. = 480

9.2 3 AC 400 to 500V -15+10%, 50/60 Hz. = 500

10. Other design

10.1 none = NNNN

11. Standard reference

Standard

DIN EN 60529

Title

Degrees of protection provided by enclosures (IP-Code)

Edition

2000-09

typenschl_hnl_de.EPS

Fig. 6-1: Type code

Type Plate

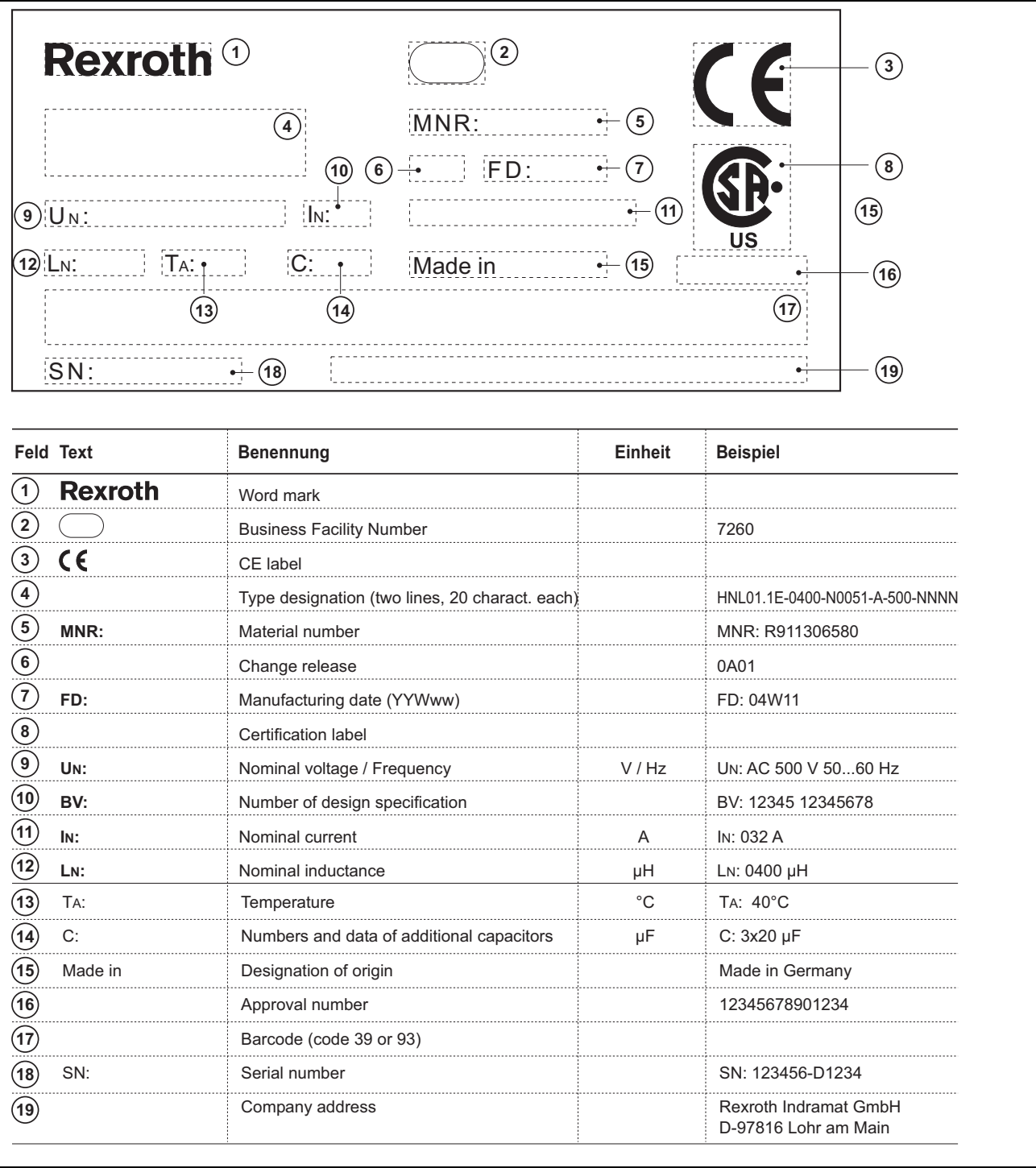


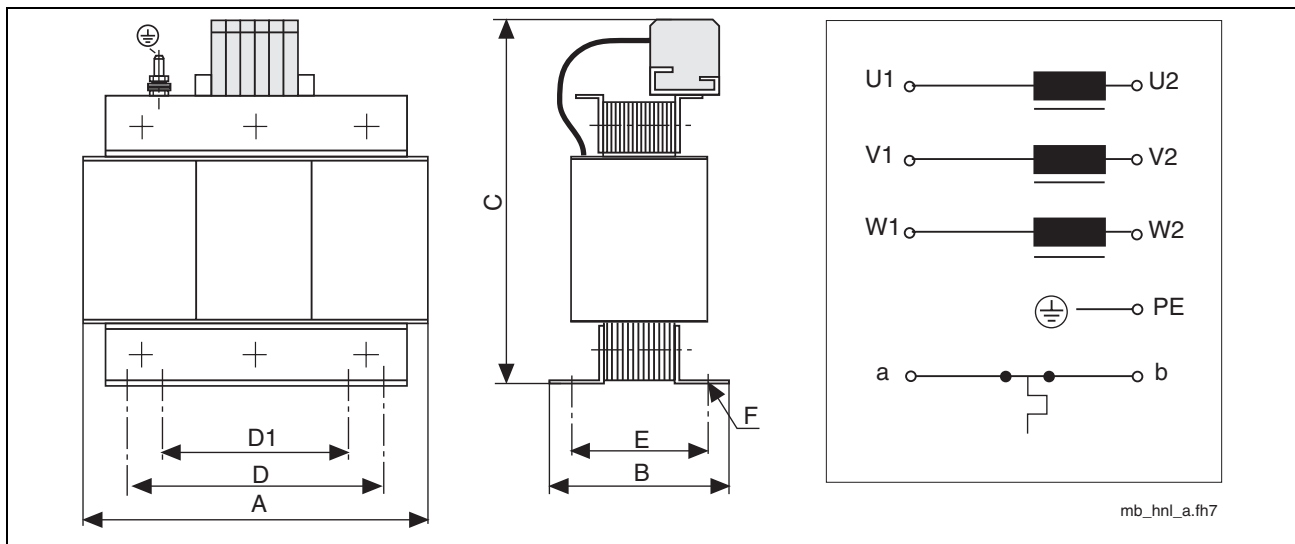
Fig. 6-2: Type Plate

6.2 Mains Choke HNL01.1E (feeding)

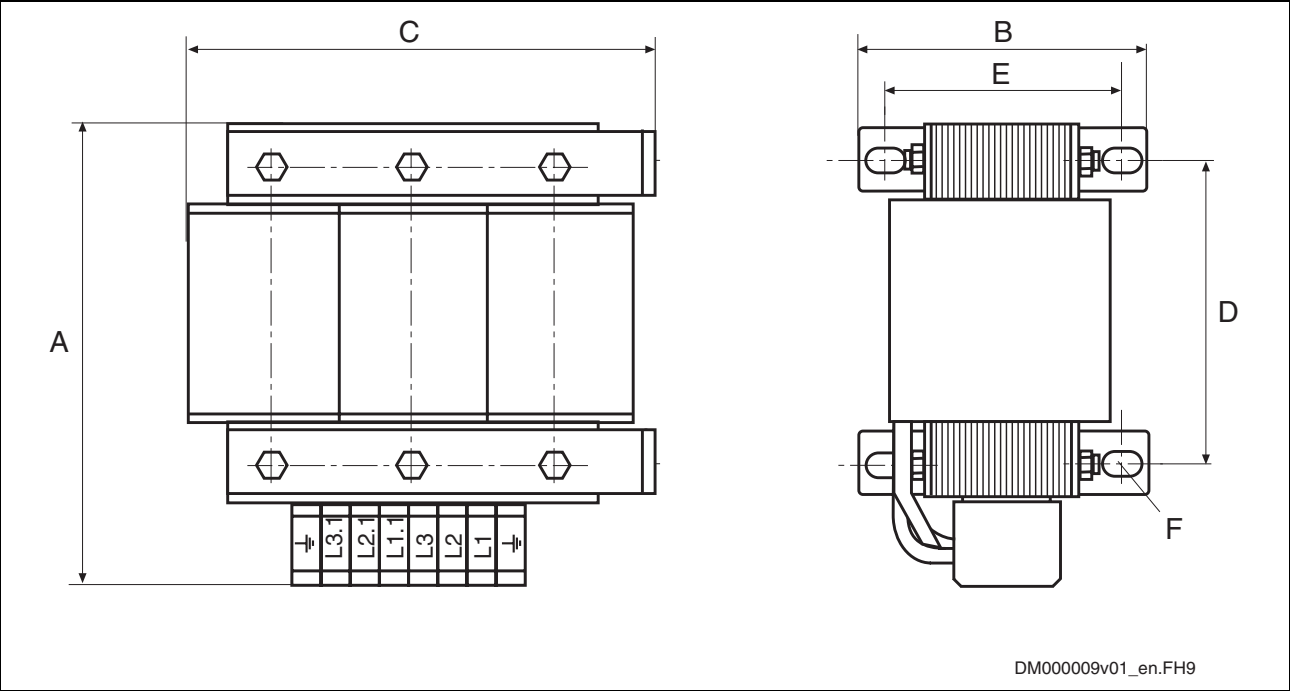
HNL01.1E-1000-N0012-A-500-NNNN,
 HNL01.1E-1000-N0020-A-500-NNNN,
 HNL01.1E-0600-N0032-A-500-NNNN,
 HNL01.1E-0571-N0050-A-500-NNNN,
 HNL01.1E-0400-N0051-A-480-NNNN,
 HNL01.1E-0362-N0080-A-500-NNNN,
 HNL01.1E-0240-N0106-A-500-NNNN,
 HNL01.1E-0200-N0125-A-480-NNNN,
 HNL01.1E-0170-N0146-A-500-NNNN,
 HNL01.1E-0100-N0202-A-480-NNNN

Technical Data

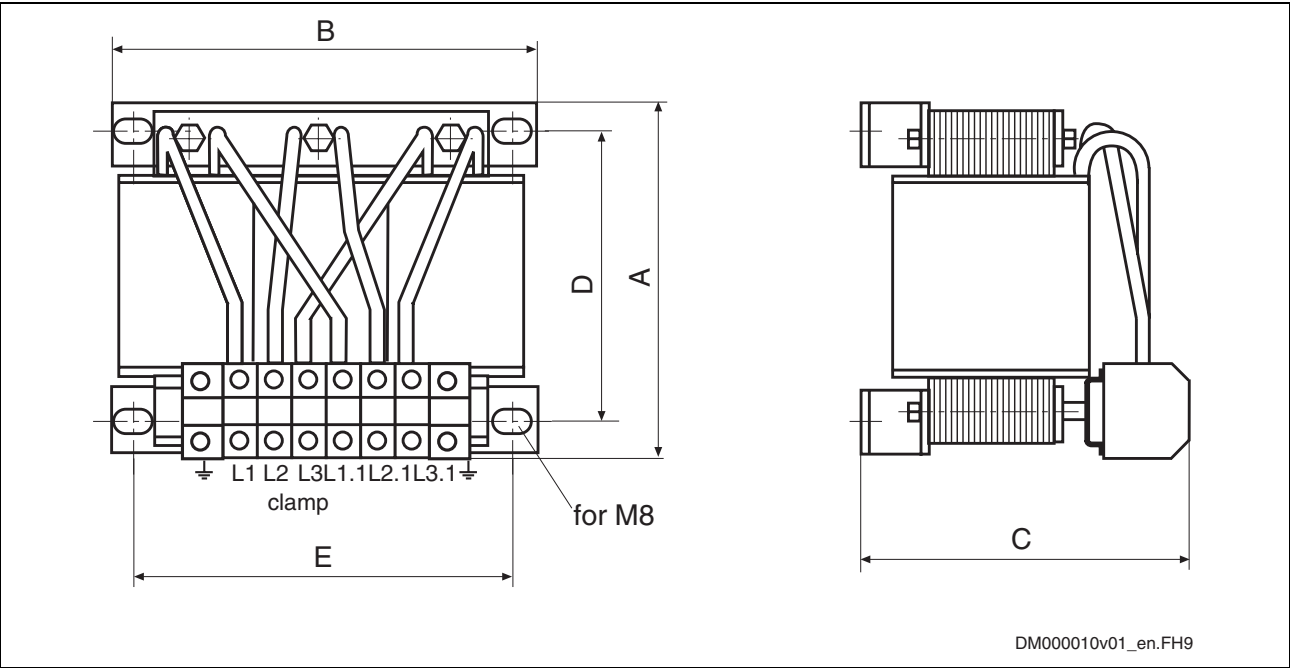
Dimensions Type 1:



Dimensions Type 2:



Dimensions Type 3:



Mains choke	Type	Dimensions [mm]	A	B	C	D	D1	E	F ¹⁾	Weight [kg]
HNL01.1E-1000-N0012-A-500-NNNN	1	120	61	164	81	-	44	6,4 x 11	2,7	
HNL01.1E-1000-N0020-A-500-NNNN	1	150	66,5	184	113	-	49,5	6,4 x 11	3,8	
HNL01.1E-0600-N0032-A-500-NNNN	1	150	66,5	185	113	-	49,5	6,4 x 11	4,5	
HNL01.1E-0571-N0050-A-500-NNNN	2	208	100	183	127	-	80	8,5 x 12	13	
HNL01.1E-0400-N0051-A-480-NNNN	1	180	112	225	125	80	87	7 x 15	13,5	
HNL01.1E-0362-N0080-A-500-NNNN	3	175	205	180	145	-	185	8,5 x 12	17	
HNL01.1E-0240-N0106-A-500-NNNN	3	193	205	210	145	-	185	8,5 x 12	17	
HNL01.1E-0200-N0125-A-480-NNNN	1	230	148	295	180	-	122	8 x 12	24	
HNL01.1E-0170-N0146-A-500-NNNN	3	205	250	230	152	-	230	8,5 x 12	23	
HNL01.1E-0100-N0202-A-480-NNNN	1	265	152	350	215	-	126	15 x 11	33	

1) long hole in "B" direction

Fig. 6-3: Mechanical data HNL

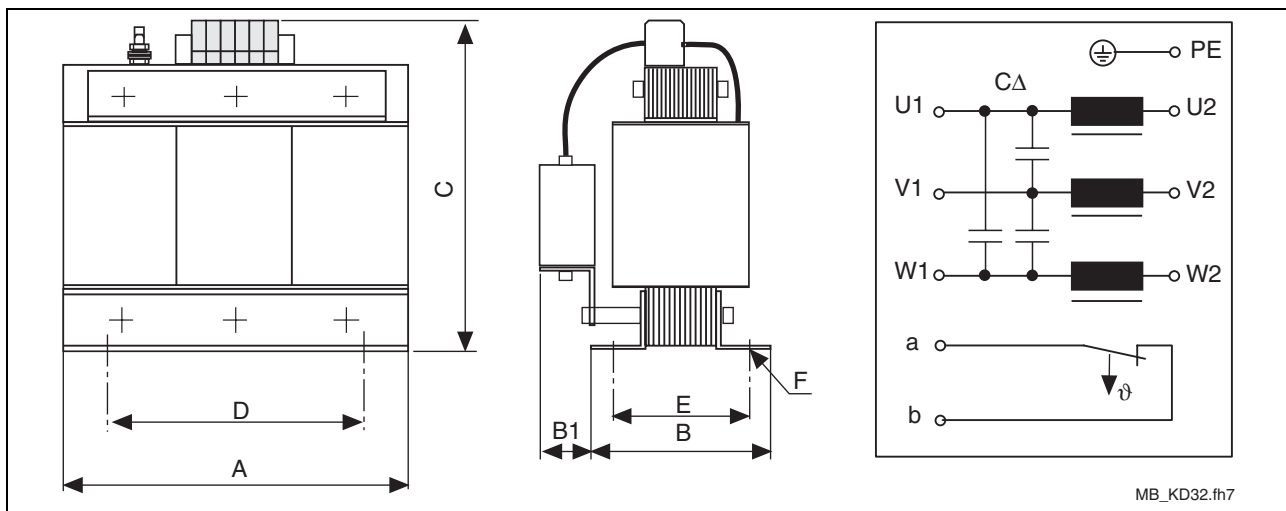
Mains choke	U _N [V]	I _N [A]	L _N [μH]	P _V [W]	I _{max} [A]	L _{min} at I _{max}	Cross section [mm ²]
HNL01.1E-1000-N0012-A-500-NNNN	500	12	3 x 1000	40	25	50% von L _N	4
HNL01.1E-1000-N0020-A-500-NNNN	500	20	3 x 1000	60	50	50% von L _N	6 (a, b: 4)
HNL01.1E-0600-N0032-A-500-NNNN	500	32	3 x 600	75	80	50% von L _N	10 (a, b: 4)
HNL01.1E-0571-N0050-A-500-NNNN	500	50	3 x 571	50	100	50% von L _N	16 (a, b: -)
HNL01.1E-0400-N0051-A-480-NNNN	480	51	3 x 400	165	77	50% von L _N	16 (a, b: 2,5)
HNL01.1E-0362-N0080-A-500-NNNN	500	80	3 x 362	80	160	50% von L _N	35 (a, b: -)
HNL01.1E-0240-N0106-A-500-NNNN	500	106	3 x 240	100	212	50% von L _N	50 (a, b: -)
HNL01.1E-0200-N0125-A-480-NNNN	480	125	3 x 200	170	188	50% von L _N	70 (a, b: 2,5)
HNL01.1E-0170-N0146-A-500-NNNN	500	146	3 x 170	130	292	50% von L _N	95 (a, b: -)
HNL01.1E-0100-N0202-A-480-NNNN	480	202	3 x 100	200	303	50% von L _N	150 (a, b: 2,5)

Fig. 6-4: Electrical data HNL

Switching Capacity 1A/250V AC; 24V DC
Switching Temperature 125 °C

6.3 Mains Choke HNL01.1R (regenerating)

HNL01.1R-0980-C0026-A-480-NNNN,
HNL01.1R-0590-C0065-A-480-NNNN,
HNL01.1R-0540-C0094-A-480-NNNN



Mains choke	Dimensions [mm]							Weight [kg]
	A	B	B1	C	D	E	F ¹⁾	
HNL01.1R-0980-C0026-A-480-NNNN	210	108	64	245	175	85	8 x 12	16
HNL01.1R-0590-C0065-A-480-NNNN	300	155	50	360	240	123	11 x 15	45
HNL01.1R-0540-C0094-A-480-NNNN	340	174	55	385	290	132	11 x 15	65

1) long hole in "B" direction
Fig. 6-5: Mechanical data HNL..C

Mains choke	U_N [V]	I_N [A]	P_V [W]	L_N [μH]	$I_{max}^{1)}$ [A]	L_{min} at I_{max}	$C\Delta$ [μF]	Cross section [mm²]
HNL01.1R-0980-C0026-A-480-NNNN	480	26	225	3 x 980	65	80% of L_N	3 x 10	16 (a, b: 2,5)
HNL01.1R-0590-C0065-A-480-NNNN	480	65	310	3 x 590	163	80% of L_N	3 x 20	35 (a, b: 2,5)
HNL01.1R-0540-C0094-A-480-NNNN	480	94	420	3 x 540	235	80% of L_N	3 x 20	70 (C: 16; a, b: 2,5)

1) for 300 ms at 1,42 s cycle duration and 60 % base load
Fig. 6-6: Electrical data HNL..C

Switching Capacity 1A/250V AC; 24V DC

Switching Temperature 125 °C

6.4 Mains Chokes HNL01.1E-****-S (current compensated)

HNL01.1E-5700-S0051-A-480-NNNN,
HNL01.1E-2800-S0125-A-480-NNNN,
HNL01.1E-3400-S0202-A-480-NNNN,
HNL01.1R-4200-S0026-A-480-NNNN,
HNL01.1R-6300-S0065-A-480-NNNN,
HNL01.1R-3000-S0094-A-480-NNNN

General Data

Input supply voltage	3 x AC 380...480V ±10%, 50/60Hz ±2%
Minimum inductance $L_{\min}$	$0,8 \cdot L_N$ at $I_{\max}$
Protection category	IP20
Materials	free of asbestos and silicone
Certification	according to UL 508 and CSA C22.2 No. 14-M91

Fig. 6-7: General data HNL..S

Specific Data

	Unit	HNL01.1 ... -A-480-NNNN					
		E-5700-S0051	E-2800-S0125	E-3400-S0202	R-4200-S0026	R-6300-S0065	R-3000-S0094
Weight	kg	11	26	30	12	14	24
Inductance L_N	mH	3 x 5,7	3 x 2,8	3 x 3,4	3 x 4,2	3 x 6,3	3 x 3,0
Nominal current	A	51	125	202	26	65	94
Peak current ¹⁾	A	77	188	303	65	163	235
Total leakage current at power terminal	A	< 2	< 2,2	< 2,2	< 2	< 2	< 2
Allowed power dissipation at nominal current and maximum leakage current	W	83	179	320	80	138	142

1) duration: 300 ms ;
cycle duration:
for feeding units: 0,67 s;
for regenerating units: 1,42 s;
base load: 60%

Fig. 6-8: Specific Data HNL..S

Dimensional drawing

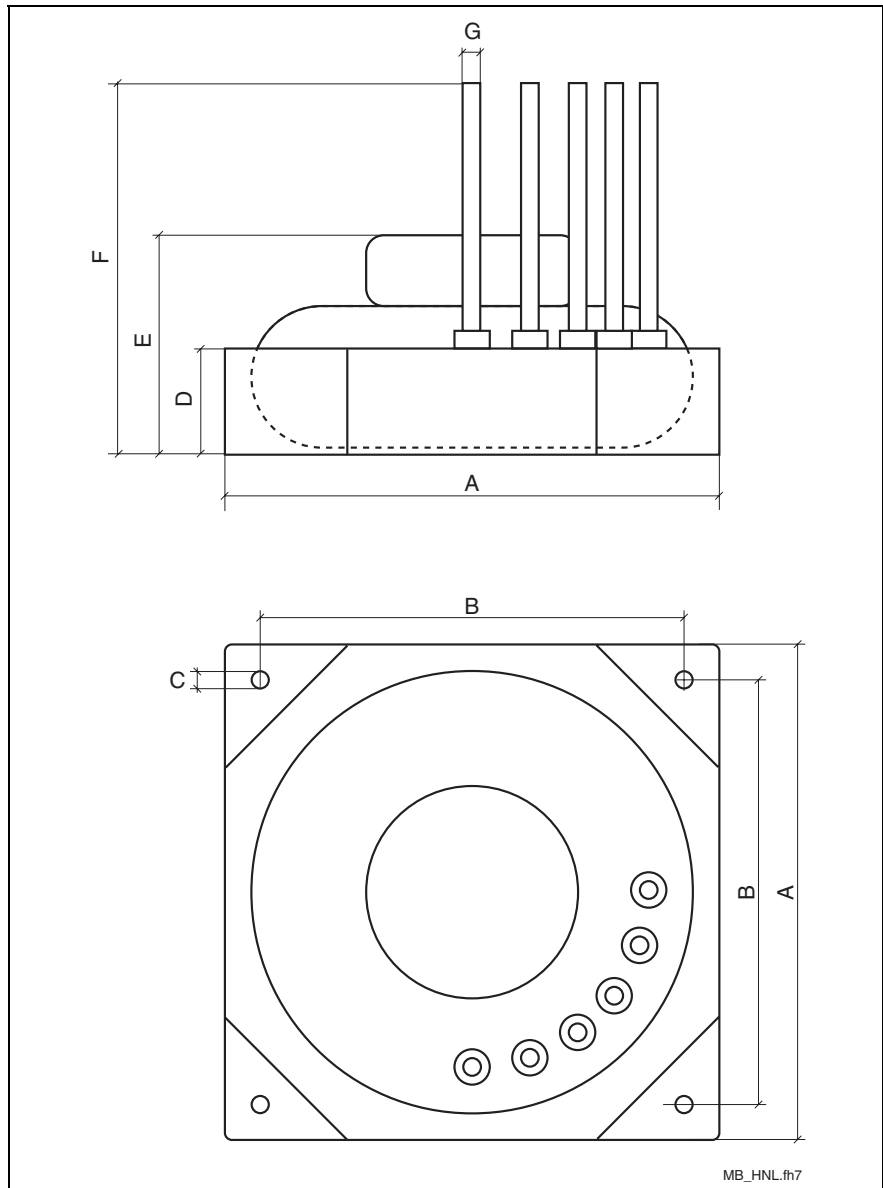


Fig. 6-9: Dimensional drawing

Mains choke	A	B	C	D	E	F	G
HNL01.1E-5700-S0051-A-480-NNNN	230	200	6.5	30	105	600	25 ¹⁾
HNL01.1E-2800-S0125-A-480-NNNN	340	280	8.4	83	150	600	1/0 ²⁾
HNL01.1E-3400-S0202-A-480-NNNN	340	280	8.4	83	185	600	5/0 ²⁾
HNL01.1R-4200-S0026-A-480-NNNN	230	200	6.5	30	125	600	16 ¹⁾
HNL01.1R-6300-S0065-A-480-NNNN	250	200	6.5	55	155	600	25 ¹⁾
HNL01.1R-3000-S0094-A-480-NNNN	340	280	8.4	83	170	600	50 ¹⁾

1)	mm ²
2)	AWG

Fig. 6-10: Dimensions HNL..S

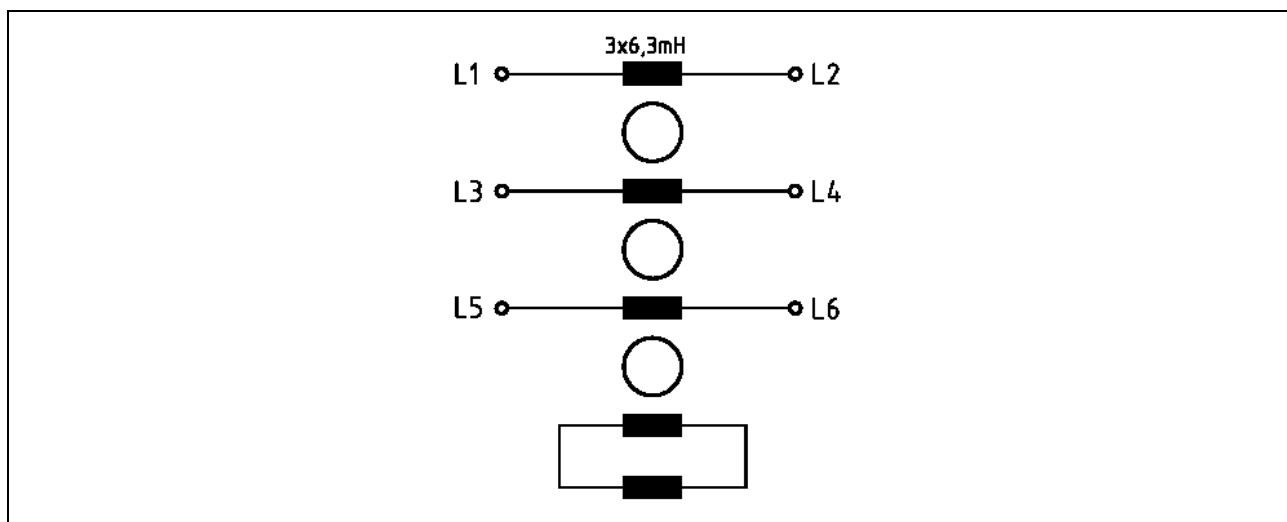
Circuit Diagram (Example HNL01.1R-6300-S0065-A-480-NNNN)

Fig. 6-11: Circuit diagram HNL...S

Notes

7 DC Bus Capacitor Unit HLC01.1

7.1 Application Instructions



CAUTION

Possible damage to the drive controller!

In operating cycles of more than 30 cycles per minute the DC bus capacitors **within the drive controller** can be overloaded when the mains supply voltage decreases.

⇒ Use DC bus capacitor units.

Using a DC bus capacitor unit you can increase the storable DC bus energy. This makes sense...

- for applications with short cycle times (> 30 cycles per minute). The continuous power of braking resistor will be reduced and with it the lost heat
- for applications, which need also in case of mains supply failure a certain amount of energy (e.g. for return motions)
- to back the DC bus for drive systems with DC bus connections > 2 m between the drive controllers

7.2 Identification

Types

Abbrev.																																	
Column	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9	2	0	1	2	3	4	5	6	7	8	9	3	0	1	2
Example:	H	L	C	0	1	.	1	D	-	0	5	M	0	-	A	-	0	0	7	-	N	N	N	N									

1. Product

1.1 HLC..... = HLC

2. Line

2.1 1 = 01

3. Design

3.1 1 = 1

4. Mounting depth^①

4.1 300 mm..... = C

4.2 400 mm..... = D

5. Nominal capacitance

5.1 5,0 mF..... = 05M0

5.2 2,4 mF..... = 02M4

5.3 1,0 mF..... = 01M0

6. Degree of protection

6.1 IP 20..... = A

7. DC bus nominal voltage

7.1 DC 700 V = 007

8. Other design

8.1 none = NNNN

9. Standard reference

Standard	Title	Edition
DIN EN 60529	Degrees of protection provided by enclosures (IP-Code)	2000-09

Note:
^① Mounting depth "C" is only available with nominal capacitance "02M4" or "01M0"
Mounting depth "D" is only available with nominal capacitance "05M0"

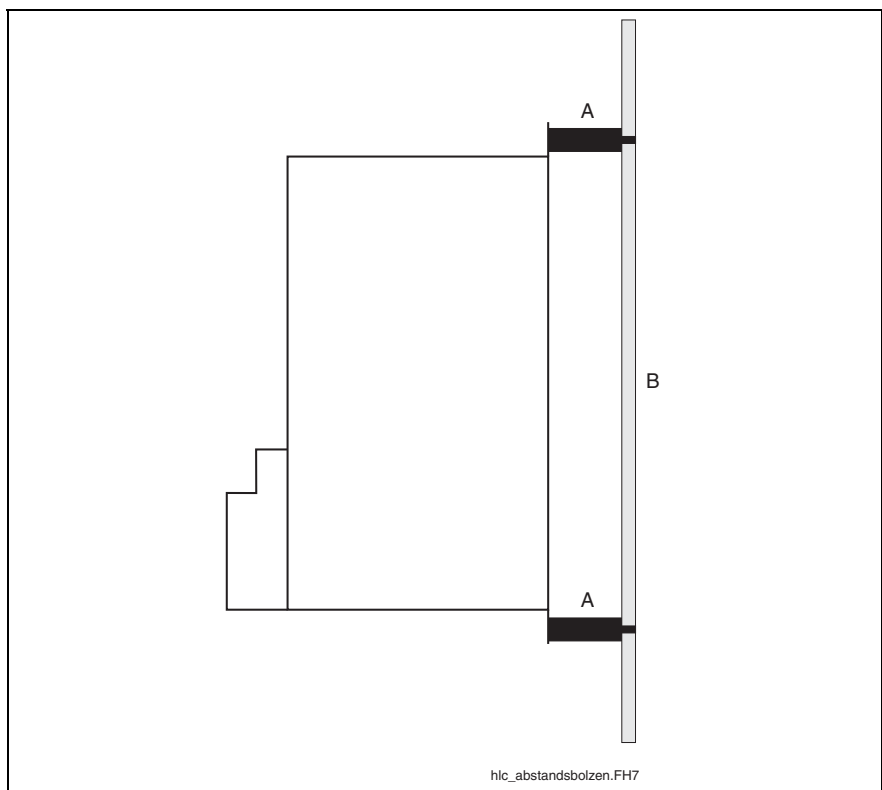
DT000001v01_en.FH9

Fig. 7-1: Type code

7.3 Mounting

Details

- Mount the DC bus capacitor unit directly besides the drive controller with the highest DC bus continuous power.
- To compensate the different mounting depths of a DC bus capacitor unit **HLC01.1C** and a **IndraDrive M drive controller** we have suitable distance bolts (accessories HAS03.1-002-NNN-NN; contains 2 distance bolts). You are allowed to use the distance bolts only when the DC bus capacitor unit is mounted directly besides the drive controller. In this case you also have to connect the contact rails for DC bus and control voltage connection to the DC bus capacitor unit.
- Mount the distance bolts on the bright mounting surface of the control cabinet.

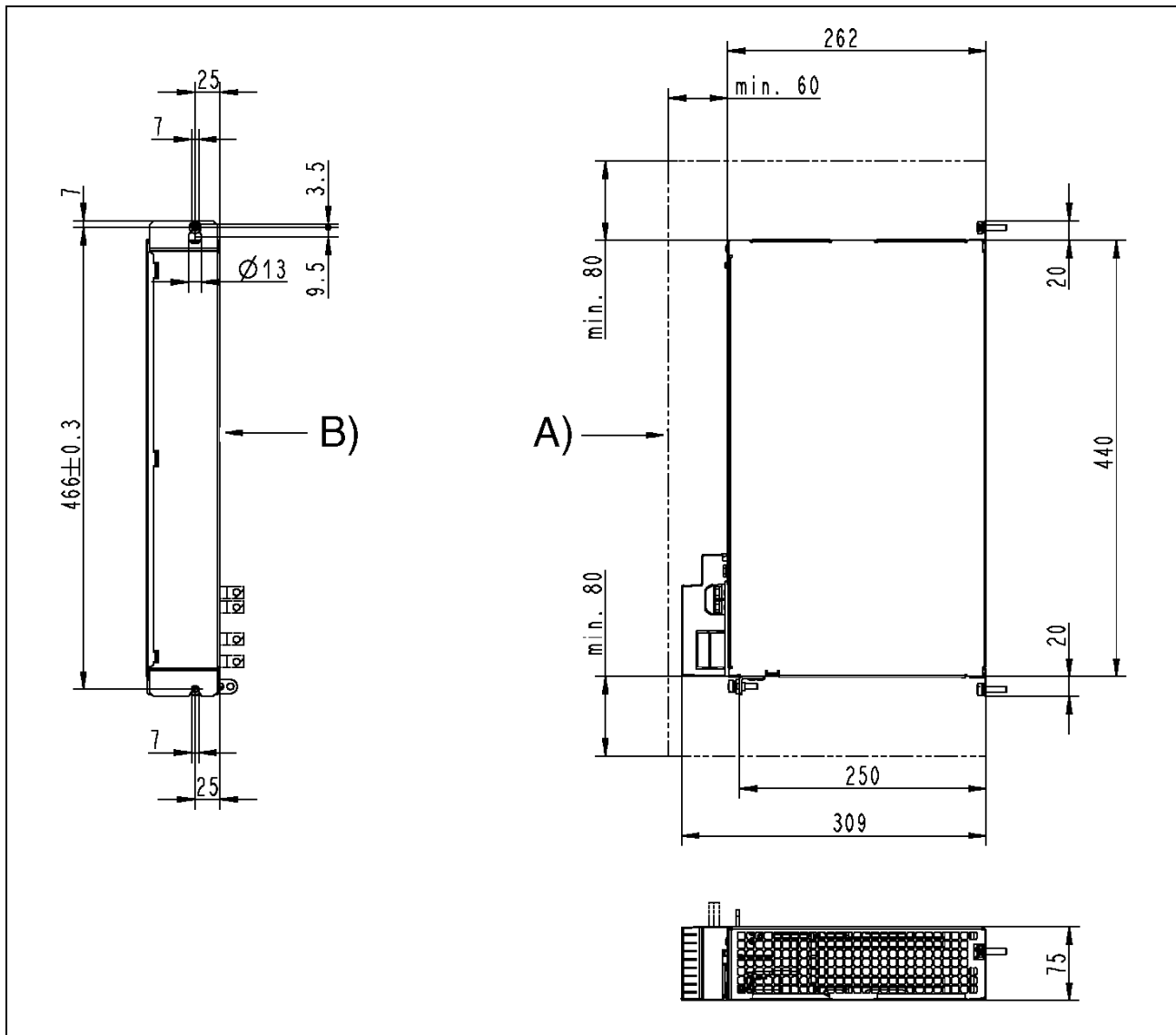


A: distance bolt

B: bright mounting surface in control cabinet

Fig. 7-2: How to use the distance bolts

Dimensions HLC01.1D



A) minimum mounting clearance

B) view from the rear!

Fig. 7-4: Dimensions HLC01.1D

Weight

Type	Weight [kg]
HLC01.1C-01M0-A-007-NNNN	3,2
HLC01.1C-02M4-A-007-NNNN	4,3
HLC01.1D-05M0-A-007-NNNN	8,6

Fig. 7-5: Weight

7.4 Installation

Electrical Data

Designation	Symbol	Unit	Type		
			HLC01.1C-01M0	HLC01.1 C-02M4	HLC01.1D-05M0
Allowed input voltage	U_{DC}	V	DC (254...750) V		
Nominal DC bus capacitance	C_{DC}	mF	1	2,4	5
Maximum discharge time from 820 V to 50 V	T	s	175	200	760
Power dissipation (at 750 V DC bus voltage)	P_v	W	12	24	13
Insulation resistance	R_{is}	MOhm	> 25		
Maximum current (at 55 °C ambient temperature)	I_{ceff}	A	20	50	60
Cooling			natural convection		

Fig. 7-6: Electrical Data

Note: The higher the input voltage the lower the energy that can be stored in the DC bus as the differential voltage between bleeder threshold and DC bus voltage (peak value of input voltage) decreases.

Connections



WARNING

High electrical voltage! Danger to life, severe bodily harm by electric shock!

Before working with electrical parts:

- ⇒ Wait thirty (30) minutes after switching off power to allow capacitor units to discharge.
- ⇒ Measure the voltage on the capacitor unit before beginning to work to make sure that the voltage is below 50 V.

Note: If you use a supply unit Rexroth IndraDrive HMV01 you can shorten the discharging time by using the DC bus short circuit function ("ZKS").

Overview

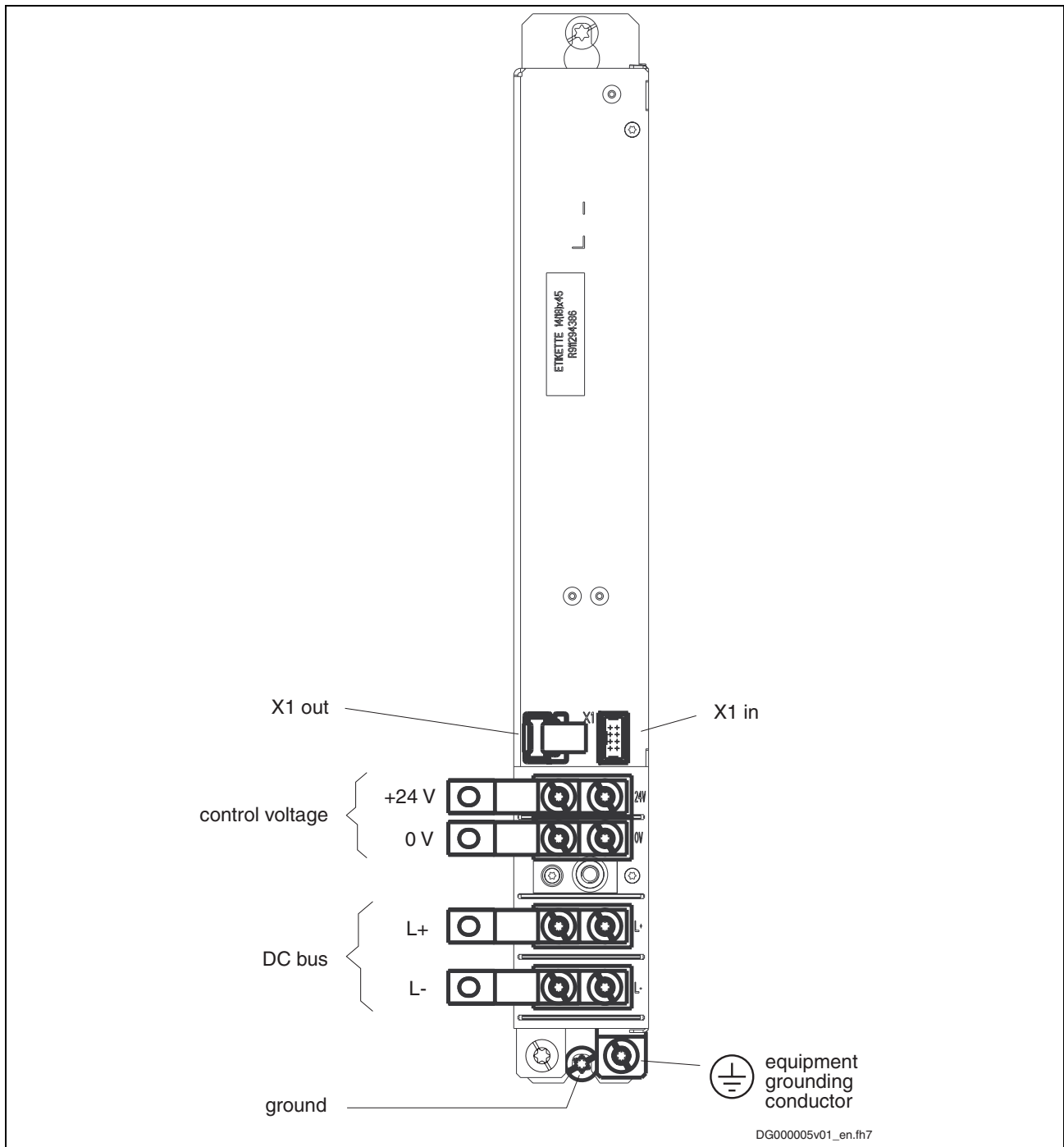


Fig. 7-7: Connections - overview

X1, Bus Module

The bus module permits data exchange between the devices. At the DC bus capacitor unit the bus module is looped through to other devices without influencing the data.

Note: Abandon on connecting the bus module, if the DC bus capacitor unit is the last device in the row of devices.

Graphic Representation

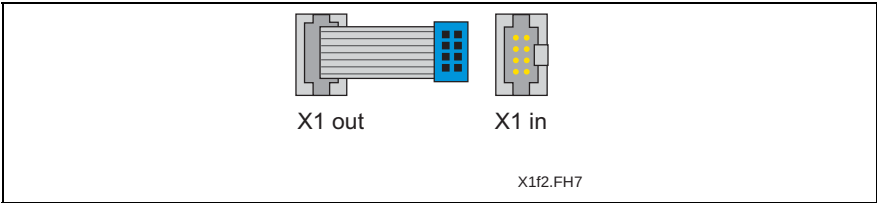


Fig. 7-8: X1

Design

Type	Number of poles	Type of design
Flat line connector	8	Connector on device
Flat line bushing	8	Bushing on flat line

Fig. 7-9: Design

Control Voltage (+24 V, 0 V)

Design

The control voltage supply is connected by contact rails and screws (M6) to the front of the drive. At the DC bus capacitor unit the control voltage is looped through to other devices. The DC bus capacitor unit itself needs no control voltage. The contact rails are parts of our accessories HAS01.1-xxx-072-MN.

Note: Abandon on connecting the control voltage, if the DC bus capacitor unit is the last device in the row of devices.

Tightening Torque 6 Nm

DC Bus (L+, L-)

Design

The DC bus is connected by contact rails and screws (M6) to the front of the device. The contact rails are parts of our accessories HAS01.1-xxx-072-MN.

Tightening Torque 6 Nm

DC Bus Wiring

If in special cases it is not possible to use the DC bus rails provided to make the connection, the connection must be made using the shortest possible twisted cables.

**CAUTION**

Risk of damage due to polarity reversal the DC bus connections L+ and L-!

⇒ Pay attention to correct polarity.

Length of the twisted cables	max. 2 m
Wire cross-section	min. 10 mm ² , however, no less than the cross-section of the power input line
Wire protection	by means of fuses in the mains supply
Voltage stability of a single strand against ground	≥ 750 V (e.g., strand type - H07)

Equipment Grounding Conductor Terminal

Design	Screw (M6 x 25)
Tightening Torque	6 Nm

Note: The equipment grounding conductor must have at least the cross section of the mains supply cable (but not smaller than 10 mm²).

Connect the equipment grounding conductor to the protective circuit within the control cabinet.

If the neighboring device is a drive controller IndraDrive M use the grounding bracket to interconnect the equipment grounding conductor connections of the two devices. The grounding bracket is part of our accessories HAS01.1-xxx-072-MN.

Ground Connection

Design	Screws (M6 x 25; screws for mounting the device)
Tightening Torque	6 Nm

Using the mounting screws provided connect the bright rear panel of the device electrically conductive to the mounting surface in the control cabinet. Connect the mounting surface electrically conductive to the protective circuit.

Notes

8 DC Bus Resistor Unit HLB01.1C and HLB01.1D

8.1 Description

The DC bus resistor unit is useful:

- for increasing the available continuous regenerated power
- for increasing the available peak regenerated power
- for DC bus dynamic braking

8.2 Identification

Type Code HLB01.1

Abbrev. Column →		1	2	3	4	5	6	7	8	9	1	0	1	2	3	4	5	6	7	8	9	2	0	1	2	3	4	5	6	7	8	9	3	0	1	2	3	4	5	6	7	8	9	4	0
Example:		H	L	B	0	1	.	1	C	-	0	2	K	0	-	N	0	6	R	0	-	A	-	0	0	7	-	N	N	N	N														

1. Product

1.1 HLB = HLB

2. Line

2.1 1 = 01

3. Design

3.1 1 = 1

4. Mounting depth

4.1 300 mm = C

4.2 400 mm = D

5. Nominal power

5.1 e.g., 2 kW = 02K0

6. Additional option

6.1 none = N

7. Resistance

7.1 e.g., 6 Ohm = 06R0

8. Degree of protection

8.1 IP 20 = A

9. DC bus nominal voltage

9.1 DC 700 V = 007

10. Other design

10.1 none = NNNN

10.2 Safe stopping process = NNNS

11. Standard reference

Standard

DIN EN 60529

Title

Degrees of protection provided by enclosures (IP-Code)

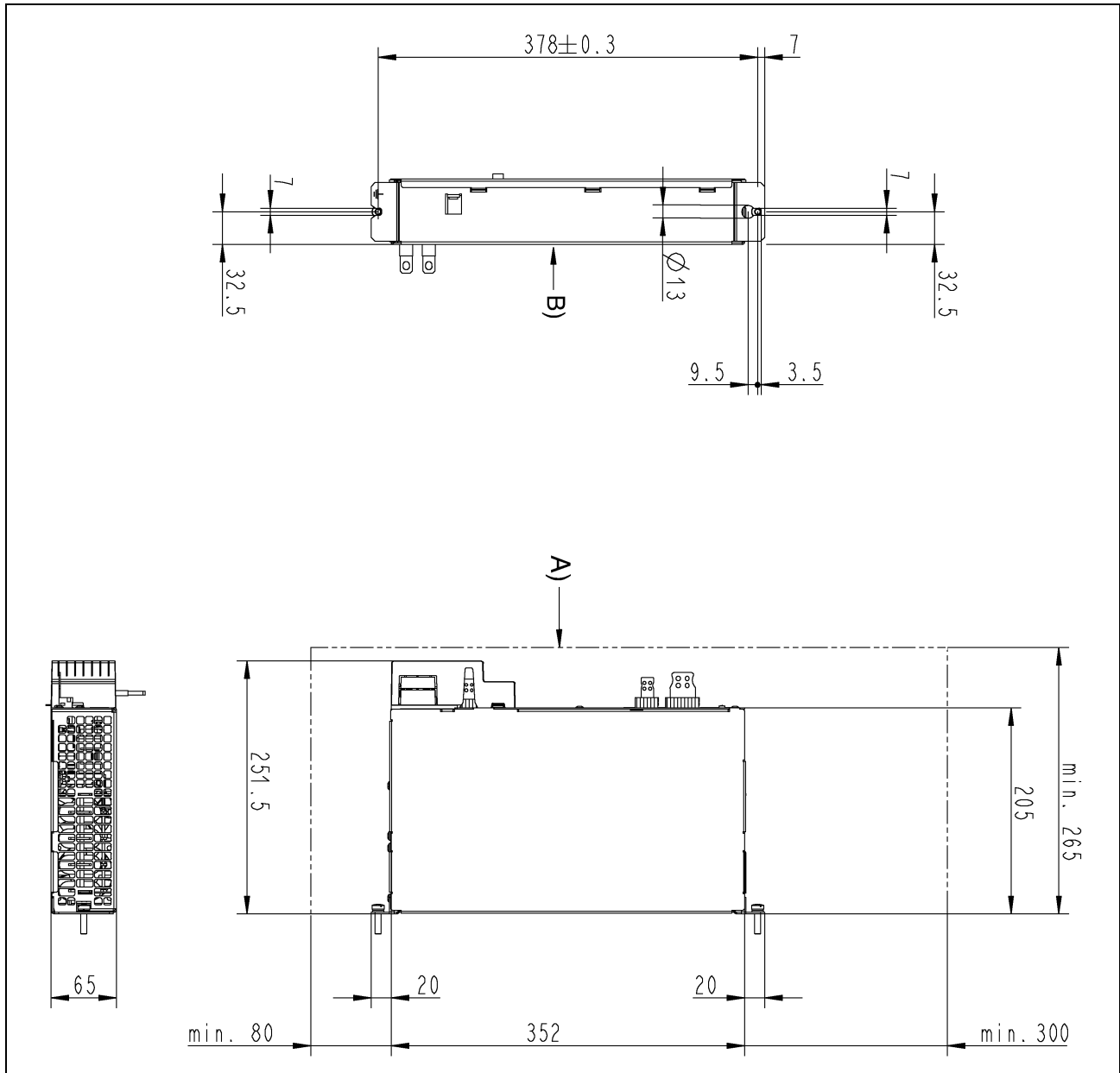
Edition

2000-09

8.3 Mounting HLB01.1

Mechanical Data HLB01.1

Dimensions HLB01.1C

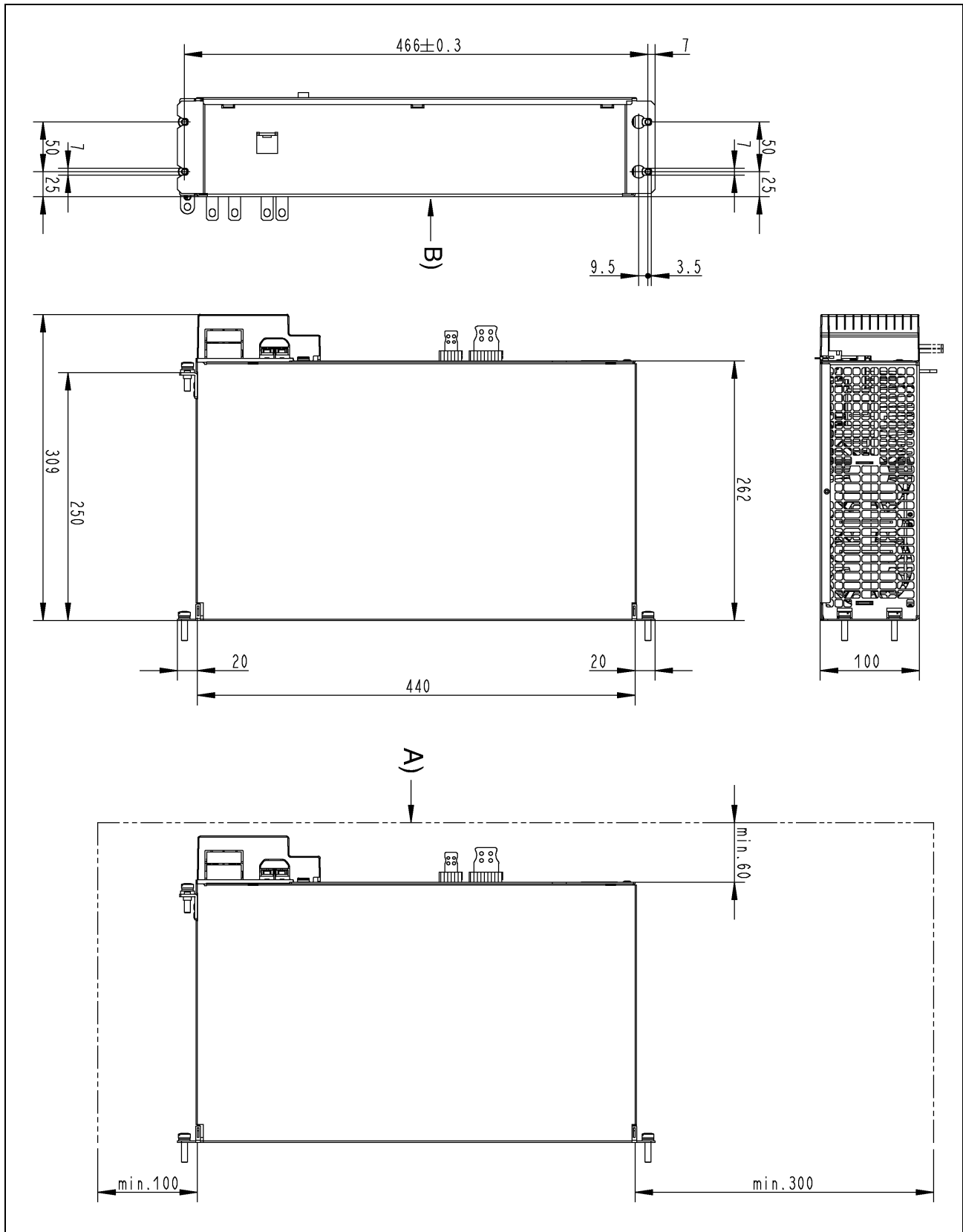


A) minimum mounting clearance

B) rear view!

Fig. 8-2: Dimensions HLB01.1C

Dimensions HLB01.1D



A) minimum mounting clearance
 B) rear view!

Fig. 8-3: Dimensions HLB01.1D

Weight

Type	Weight [kg]
HLB01.1C	5,8
HLB01.1D	12,2

Fig. 8-4: Weight

8.4 Installation HLB01.1

Electrical Data HLB01.1

Designation	Symbol	Unit	Type	
			HLB01.1C	HLB01.1D
mains input voltage	U_{ZK}	V	DC (254...750)	DC (254...750)
braking resistor switch-on threshold	U_{BI}	V	820	820
continuous power	P_{BD}	kW	1	2
peak power	P_{BS}	kW	100	100
maximum regenerated power	W_{MAX}	kWs	100	500
derating if devices are connected in parallel ¹⁾			0,8	0,8
control voltage:				
control voltage	U_{ST}	V	$24 \pm 20\%$	$24 \pm 5\%$
power consumption	P_{ST}	W	14	14
max. inrush current	$I_{EIN ST}$	A	3 (for 50 ms)	3 (for 50 ms)
power dissipation:				
basic power dissipation	P_{VG}	W	28	28

1) The factor of derating results from the balancing behavior. The balancing behavior is determined by tolerance afflicted voltage measurements in the single devices. Derating applies as well for the continuous power as for the maximum allowed regenerated power.

Fig. 8-5: Electrical Data

Connections

Overview HLB01.1C

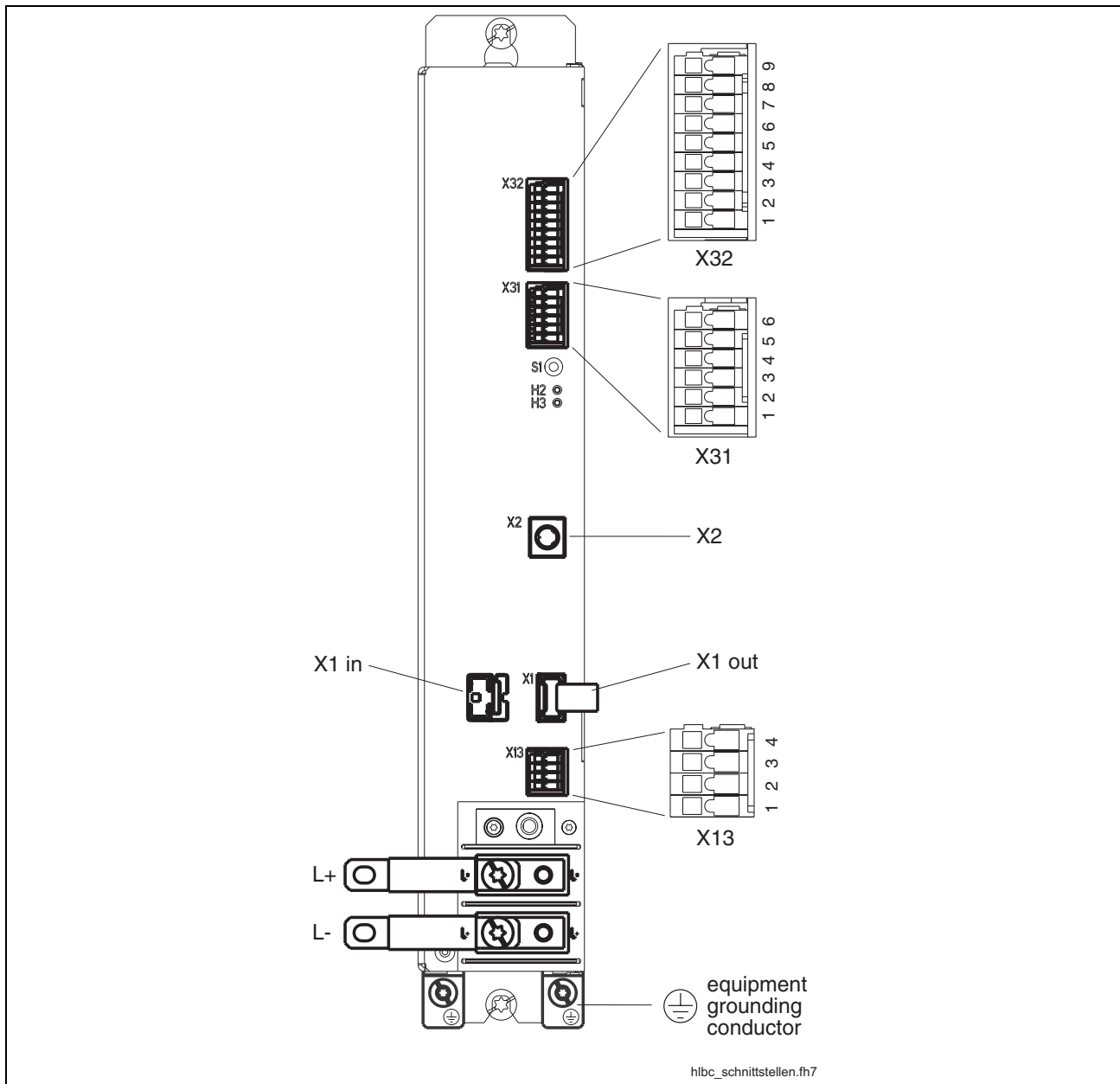


Fig. 8-6: Connections HLB01.1C

Connection	see page
L+, L- (DC bus)	8-8
Equipment grounding conductor	8-8
X1 in, X1 out (bus module)	8-10
X2 (RS232 interface)	8-12
X13 (control voltage)	8-9
X31 (ready for operation, and advance warning contact)	8-11
X32 (DC bus short circuit control, clearing errors, braking resistor threshold)	8-12

Overview HLB01.1D

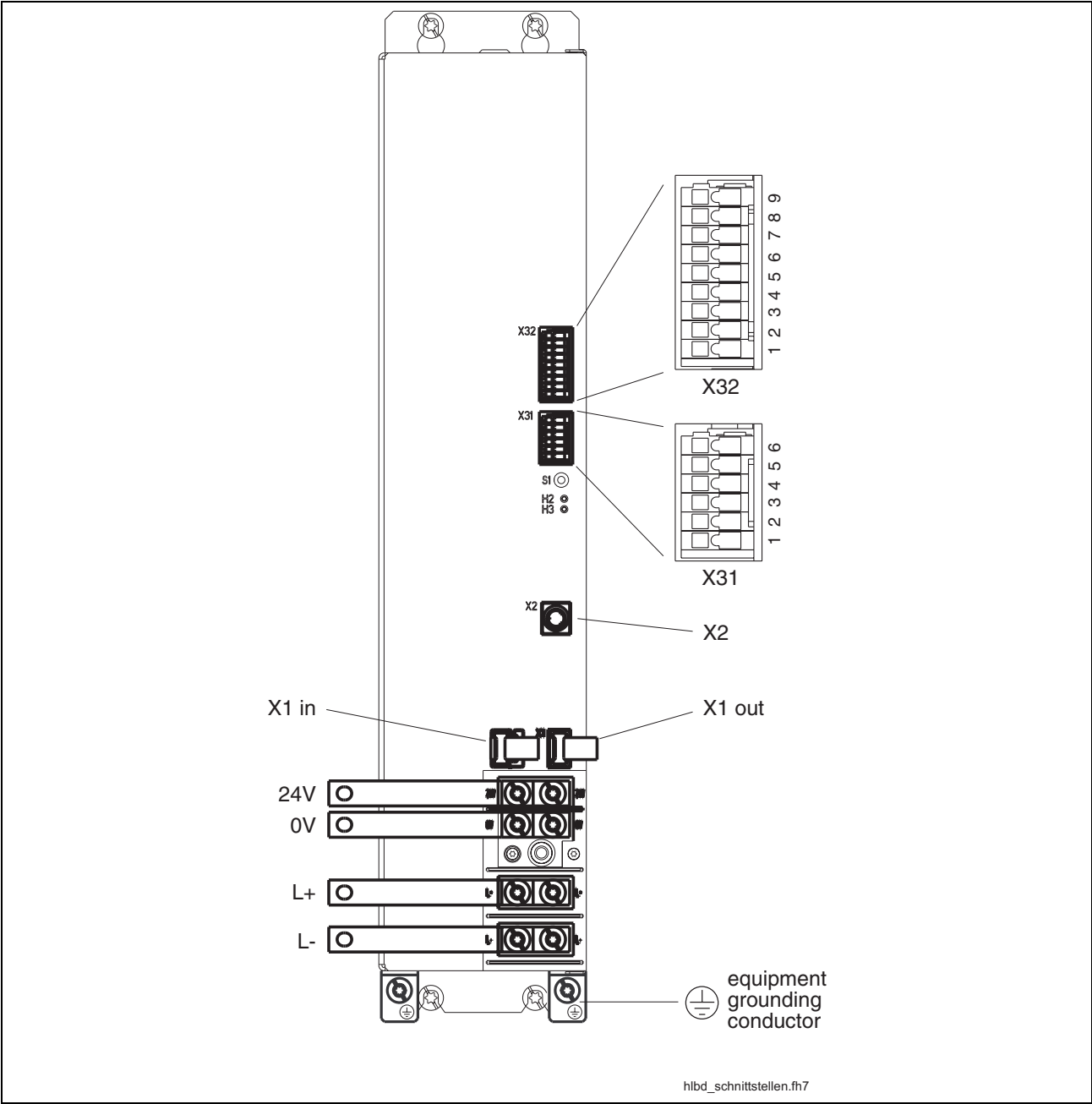


Fig. 8-7: Connections HLB01.1D

Connection	see page
24V, 0V (control voltage)	8-9
L+, L- (DC bus)	8-8
Equipment grounding conductor	8-8
X1 in, X1 out (bus module)	8-10
X2 (RS232 interface)	8-12
X31 (ready for operation, and prewarning contact)	8-11
X32 (DC bus short circuit control, clearing errors, braking resistor threshold)	8-12

DC Bus

Designation L+, L-

Design The DC bus is connected by contact rails and screws (M6) to the front of the device.

Cross section of a contact rail:

- HLB01.1D: 72 mm² (6 mm x 12 mm)
- HLB01.1C: 48 mm² (4 mm x 12 mm)

Tightening Torque 6 Nm

Equipment Grounding Conductor Terminal

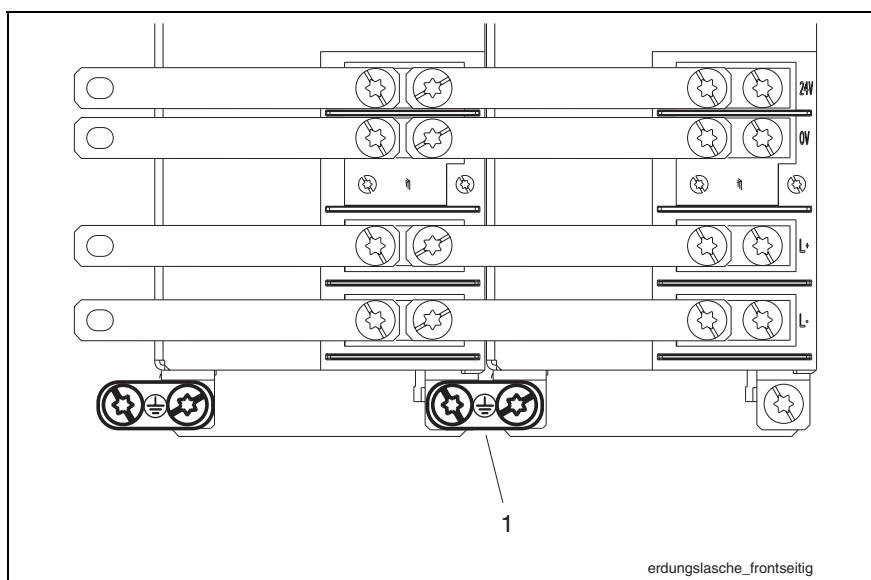
Design Screw (M6 x 25)

Tightening Torque 6 Nm

Note: The equipment grounding conductor must have at least the cross section of the mains supply cable (but not smaller than 10 mm²).

For HLB01.1C: Connect the equipment grounding conductor terminal to the protective circuit in the control cabinet.

For HLB01.1D: Using a grounding bracket connect equipment grounding conductor terminal to the equipment grounding conductor terminal of the neighboring device. The grounding bracket is part of our accessories HAS01.1-xxx-072-MN.



1: grounding bracket

Fig. 8-8: Grounding bracket

Ground Connection

Design Screws (M6 x 25; screws for mounting the device)

Tightening Torque 6 Nm

Using the mounting screws of the device connect the bare rear panel of the device electrically conductive to the mounting surface in the control cabinet. Connect the mounting surface in the control cabinet electrically conductive to the protective circuit.

Control Voltage HLB01.1C

Designation X13

Design

Type	Number of poles	Type of design
Spring power	4	female (connector)

Fig. 8-9: Design

Connection Cross Section

Cross section single-wire [mm ²]	Cross section multi-wire [mm ²]	Cross section in AWG Gauge-Nr.:
0,08-1,5	0,08-1,5	28 - 14

Fig. 8-10: Connection Cross Section

Assignment

4	+24 V
3	+24 V
2	0 V
1	0 V

Lines +24V and 0V

line cross section	min. 1 mm ²
line routing	preferably in parallel
max. allowed current if control voltage is looped through	DC 6 A

Control Voltage HLB01.1D

Designation 24V, 0V

Design

The control voltage is connected by contact rails and screws (M6) to the front of the drive controller.

Cross section of a contact rail: 72 mm² (6 mm x 12 mm)

Tightening Torque 6 Nm

Module Bus

Designation X1 in, X1 out

Function The module bus permits data exchange between the devices.

Design	Type	Number of poles	Type of design
	Flat line connector	8	Connector on device
	Flat line bushing	8	Bushing on flat line

Fig. 8-11: Design

X31, Connection for Messages

Design

Type	Number of poles	Type of design
Spring power	6	female (connector)

Fig. 8-12: Design

Connection Cross Section

Cross section single-wire [mm ²]	Cross section multi-wire [mm ²]	Cross section in AWG Gauge-Nr.:
0,08-1,5	0,08-1,5	28 - 14

Fig. 8-13: Connection Cross Section

Loadability

Max. switching voltage	DC 30V
Max. switched current	DC 1A
Max. continuous current	DC 1A
minimum load of contacts	10 mA

Assignment

6	ready for operation contact
5	ready for operation contact
4	n.c.
3	n.c.
2	prewarning contact
1	prewarning contact

Circuit states

Ready for operation contact:

Contact is **closed**, if **all** of following conditions are met:

- control voltage present
- heat sink temperature OK
- load resistor temperature OK
- continuous power is smaller than specified (see Electrical Data)
- maximum regenerated power is smaller than specified (see Electrical Data)

Prewarning contact:

Contact is **open**, if **one** of the following conditions is met:

- heat sink temperature is too high
- continuous power > 90%
- regenerated power > 90%

Shield connection

Connect the shields of the control lines to shield connection XS1 of the neighboring drive controller.

X32, DC Bus Short Circuit, Clearing Errors, Braking Resistor Threshold

Design	Type	Number of Poles	Type of Design
	Spring power	9	female (connector)

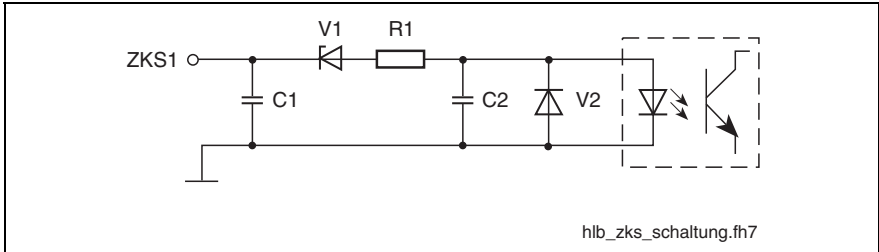
Fig. 8-14: Design

Connection Cross Section	Cross section single-wire [mm²]	Cross section multi-wire [mm²]	Cross section in AWG Gauge-Nr.:
	0,08-1,5	0,08-1,5	28 - 14

Fig. 8-15: Connection Cross Section

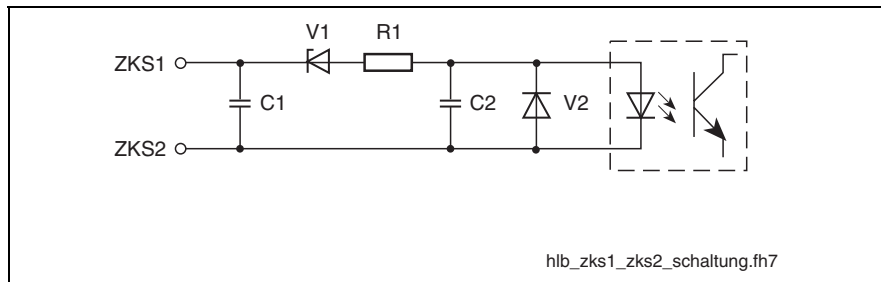
Assignment		HLB01.1C	HLB01.1D
	9	n.c.	n.c.
	8	ZKS1 (DC bus short circuit)	ZKS1 (DC bus short circuit)
	7	n.c.	ZKS1 (DC bus short circuit)
	6	n.c.	ZKS1 (DC bus short circuit)
	5	Clear errors	Clear errors
	4	GND	GND
	3	GND	GND
	2	Braking resistor threshold	Braking resistor threshold
	1	n.c.	ZKS2 (DC bus short circuit)

Input Circuit ZKS1 (HLB01.1C)



- R1: 2k2
- C1: 200 nF
- C2: 100 nF
- V1: 10 V
- V2: 0,7 V

Fig. 8-16: Circuit ZKS (HLB01.1C)

ZKS1, ZKS2 (HLB01.1D)

R1: 2k2
 C1: 200 nF
 C2: 100 nF
 V1: 10 V
 V2: 0,7 V

Fig. 8-17: Circuit ZKS (HLB01.1D)

ZKS1 Input

Input voltage:	Min.	Max.
High	19,2 V	28,8 V
Low	0 V	5 V
Operate delay t _d	approx. 160 ms	
Input resistance	2 kOhm ± 10%	
Reverse voltage protection within the allowed input voltage range		

ZKS1 Usage

- For supplying the ZKS input with current.
- For wiring further to the supply unit resp. to other DC bus resistor units HLB01.1D.

ZKS2 Usage

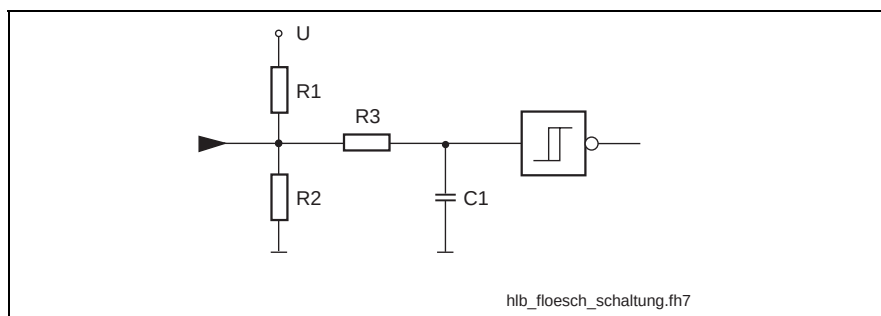
Has to be connected to GND. Bridge X32 Pin 1 and X32 Pin 4.

Release behavior of ZKS Equipment

ZKS Input	ZKS Equipment
not supplied with current	active
supplied with current	not active

Protection of ZKS equipment with mains supply voltage present

The ZKS input has to be activated by the NO contact of the mains contactor (auxiliary switch).

Input Circuit**Clearing Errors (X32/5)**

U: supply voltage
 R1: 10k
 R2: 2k3
 R3: 10k
 C1: 100 nF

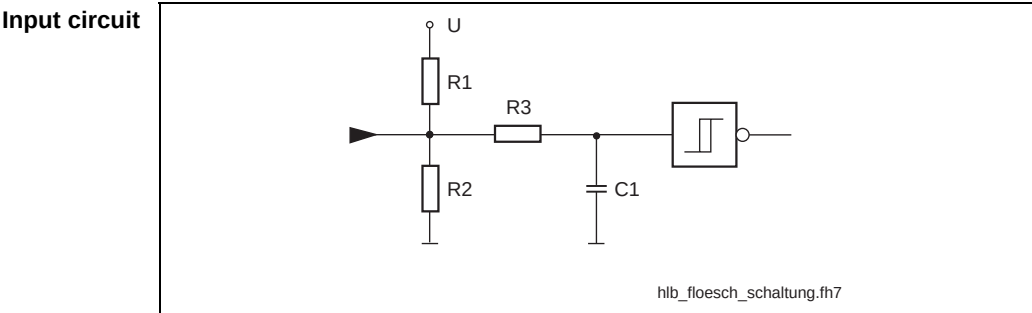
Fig. 8-18: Input circuit Clearing Errors

Clearing Errors Input	Input circuit:	
	High	input open
	Low	0 V (bridge Pin 4 / Pin 5)

Clearing Errors Usage A negative edge on the input effects a clearing of all pending and clearable errors.

Braking Resistor Threshold (X32/2)

Usage Determines the mode of the braking resistor threshold.



- U: supply voltage
- R1: 10k
- R2: 2k49
- R3: 10k
- C1: 100 nF

Fig. 8-19: Input circuit

Settings	Input X32/2	Effective Mode	Braking Resistor Threshold
	0 V (bridge Pin 2 / Pin 3)	independent from mains voltage	dependent on utilization within the range of DC (820..850) V
	open	independent from mains voltage	dependent on utilization within the range of DC (820..850) V

Shield connection Connect the shields of the control lines to shield connection XS1 of the neighboring drive controller.

X2, RS232 Interface

Note: For internal usage only!

8.5 Diagnostic LEDs and Reset-Pushbutton

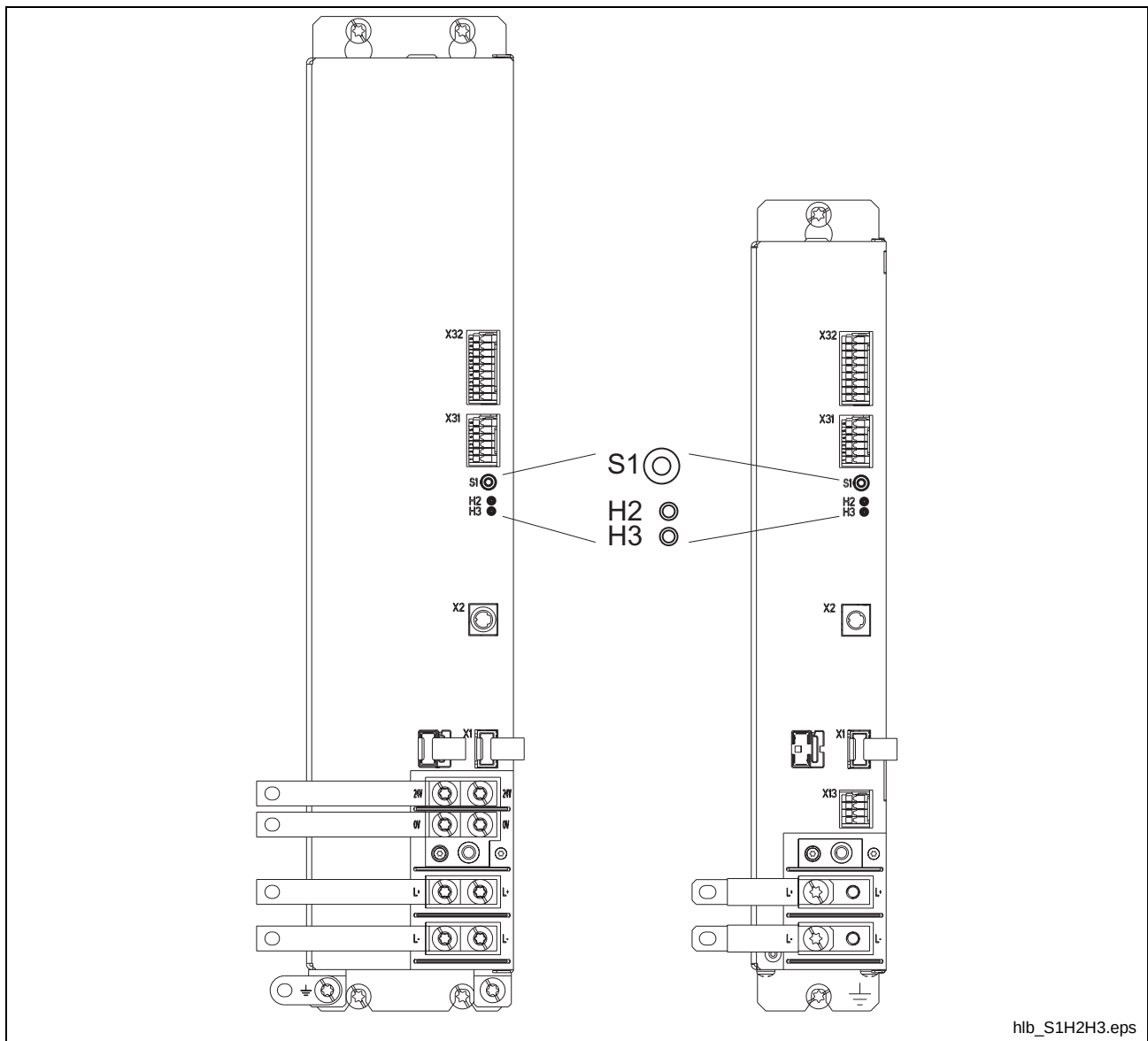


Fig. 8-20: Diagnostic LEDs and Reset-Pushbutton

Reset-Pushbutton S1 Is used for resetting errors.

Diagnostic LED H2 (green)

State	Meaning
blinking	device is ready for operation and $U_{ZK} \leq 50 \text{ V}$
steady light	device is ready for operation and $U_{ZK} > 50 \text{ V}$

Diagnostic LED H3 (red)

State	Meaning
blinking	overload prewarning at 90% utilization overtemperature prewarning
steady light	overload, overtemperature, internal error

Notes

9 Braking Resistors HLR01.1

9.1 General Information

For the drive controllers HCS02 and HCS03 there are external standard braking resistors HLR01.1N-xxxx-Nxxx-A-007-NNNN available. The drive controller HCS monitors the external braking resistor by means of the firmware which calculates a thermal image of the current braking power. When the limit values for the thermal image are exceeded, the converter switches off with error "F8820 braking resistor overload" to protect the braking resistor from overload.

Note: Choose drive controllers HCS03 in design -NNBN (braking transistor integrated) or drive controllers HCS02 with type current W0054 or W0070 so that you can operate braking resistors HLR01.1.

Standard and Reinforced Design HLR01.1

We distinguish two designs of braking resistors HLR01.1:

- The standard design is intended for assembly above the drive controller HCS03.1E.
- The **reinforced** design is **not** intended for assembly at the drive controller HCS03.1E.

For regenerative braking at higher power, occurring for example in drives with large centrifugal masses (e.g. centrifuges), long braking processes (e.g. in crane lifting gears), reinforced braking resistors should be connected to the converters with integrated brake chopper.



CAUTION

Damage to the drive controller HCS by braking resistor HLR01 incorrectly selected and parameterized!

- ⇒ When selecting the braking resistor HLR01, take the allowed minimum resistance value into account. For the minimum resistance value please see the electrical data of the drive controllers HCS02 and HCS03.
 - ⇒ At the drive controllers HCS02 and HCS03 parameterize the performance data of the selected braking resistor in parameter "P-0-0858, Data of external braking resistor".
 - ⇒ Activate the selected braking resistor by setting bit 8 = 1 in parameter "P-0-0860, Control word of power section".
-

Assigning Standard Braking Resistors to Drive Controllers HCS02 and HCS03

Braking resistor	Drive controller
tbd	HCS02.1E-W0054
tbd	HCS02.1E-W0070
HLR01.1N-0300-N17R5-A-007-NNBN	HCS03.1E-W0070
HLR01.1N-0470-N11R7-A-007-NNBN	HCS03.1E-W0100
HLR01.1N-0780-N07R0-A-007-NNBN	HCS03.1E-W0150
HLR01.1N-1K08-N05R0-A-007-NNBN	HCS03.1E-W0210

Fig. 9-1: Assignment of standard braking resistors HLR

Assigning Reinforced Braking Resistors to Drive Controllers HCS03

The table below shows the allowed combinations of reinforced braking resistors HLR01.1 with drive controllers HCS03.1.

Braking resistor	Drive controller
HLR01.1N-01K6-N18R0-A-007-NNNN	HCS03.1E-W0070-A-05-NNBx
HLR01.1N-02K0-N15R0-A-007-NNNN	HCS03.1E-W0100-A-05-NNBx
HLR01.1N-04K5-N07R4-A-007-NNNN	HCS03.1E-W0150-A-05-NNBx
HLR01.1N-06K5-N06R1-A-007-NNNN	HCS03.1E-W0210-A-05-NNBx
HLR01.1N-03K5-N19R0-A-007-NNNN	HCS03.1E-W0070-A-05-NNBx
HLR01.1N-05K0-N15R0-A-007-NNNN	HCS03.1E-W0100-A-05-NNBx
HLR01.1N-08K5-N08R0-A-007-NNNN	HCS03.1E-W0150-A-05-NNBx
HLR01.1N-12K5-N05R5-A-007-NNNN	HCS03.1E-W0210-A-05-NNBx
HLR01.1N-04K5-N18R0-A-007-NNNN	HCS03.1E-W0070-A-05-NNBx
HLR01.1N-07K0-N14R0-A-007-NNNN	HCS03.1E-W0100-A-05-NNBx
HLR01.1N-11K0-N07R3-A-007-NNNN	HCS03.1E-W0150-A-05-NNBx
HLR01.1N-17K0-N05R1-A-007-NNNN	HCS03.1E-W0210-A-05-NNBx
HLR01.1N-06K5-N18R0-A-007-NNNN	HCS03.1E-W0070-A-05-NNBx
HLR01.1N-09K5-N13R0-A-007-NNNN	HCS03.1E-W0100-A-05-NNBx
HLR01.1N-15K0-N08R1-A-007-NNNN	HCS03.1E-W0150-A-05-NNBx
HLR01.1N-23K0-N05R5-A-007-NNNN	HCS03.1E-W0210-A-05-NNBx
HLR01.1N-10K0-N18R0-A-007-NNNN	HCS03.1E-W0070-A-05-NNBx
HLR01.1N-14K5-N13R0-A-007-NNNN	HCS03.1E-W0100-A-05-NNBx
HLR01.1N-24K0-N07R2-A-007-NNNN	HCS03.1E-W0150-A-05-NNBx
HLR01.1N-36K0-N05R4-A-007-NNNN	HCS03.1E-W0210-A-05-NNBx

Fig. 9-2: Assignment of reinforced braking resistors HLR

9.2 Identification

Type Code HLR01.1

Abbrev. Column		→																																									
		1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0		
Example:		H	L	R	0	1	.	1	N	-	1	0	K	0	-	N	0	5	R	4	-	A	-	0	0	7	-	N	N	N	N												

1.

Product

1.1

HLR = HLR

2.

Line

2.1

1..... = 01

3.

Design

3.1

1 = 1

4.

Mountin style

4.1

Device mounting solution = A

4.2

Free mounting solution = N

5.

Continous output

5.1

e.g., 10 kW = 10K0

6.

Additional option

6.1

none = N

7.

Resistance

7.1

e.g., 5.4 Ohm..... = 05R4

8.

Degree of protection

8.1

IP20 = A

9.

DC bus nominal voltage

9.1

DC 700 V..... = 007

10.

Other design

10.1

none = NNNN

11.

Standard reference

Standard

DIN EN 60529

Title

Degrees of protection provided by enclosures (IP-Code)

Edition

2000-09

Fig. 9-3: Type code braking resistor HLR01.1

Type Plate HLR01.1

Each braking resistor is identified by a type designation. There is a type plate attached to all components.

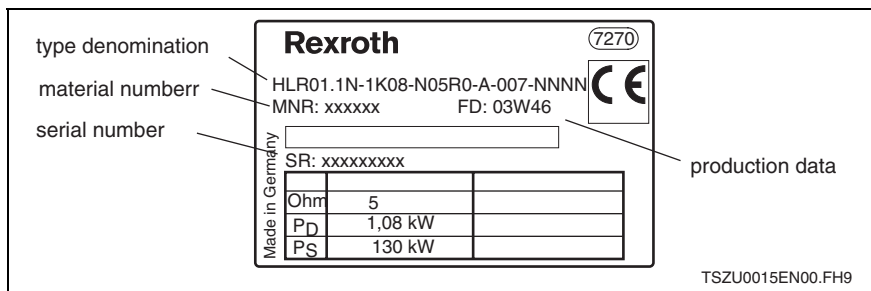


Fig. 9-4: Type Plate HLR01.1

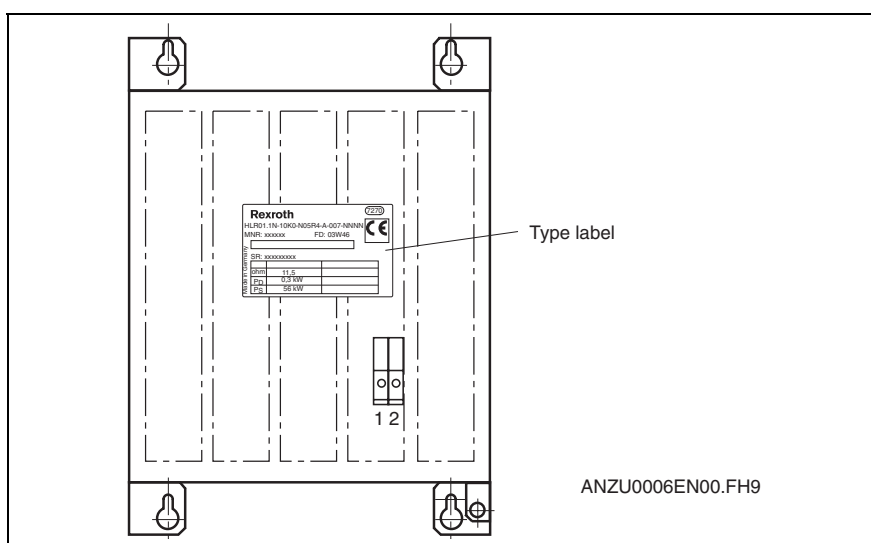


Fig. 9-5: Type plate position

9.3 Technical Characteristics of the Standard Braking Resistors HLR01.1

The resistors of these braking units consist of a heavy-duty metal alloy. The resistor element is embedded within an aluminum housing, laying in an insulating ceramics material of high thermal conductivity. Complete metal cladding ensures protection from damaging ambient conditions and unintentional touching of live parts. They have the following properties:

- compact design and low weight
- high power and impulse loading
- robust construction and high reliability
- high dielectric strength
- the resistor element is non-flammable

9.4 Mechanical Data of the Standard Braking Resistors HLR01.1

Dimensioned Drawings

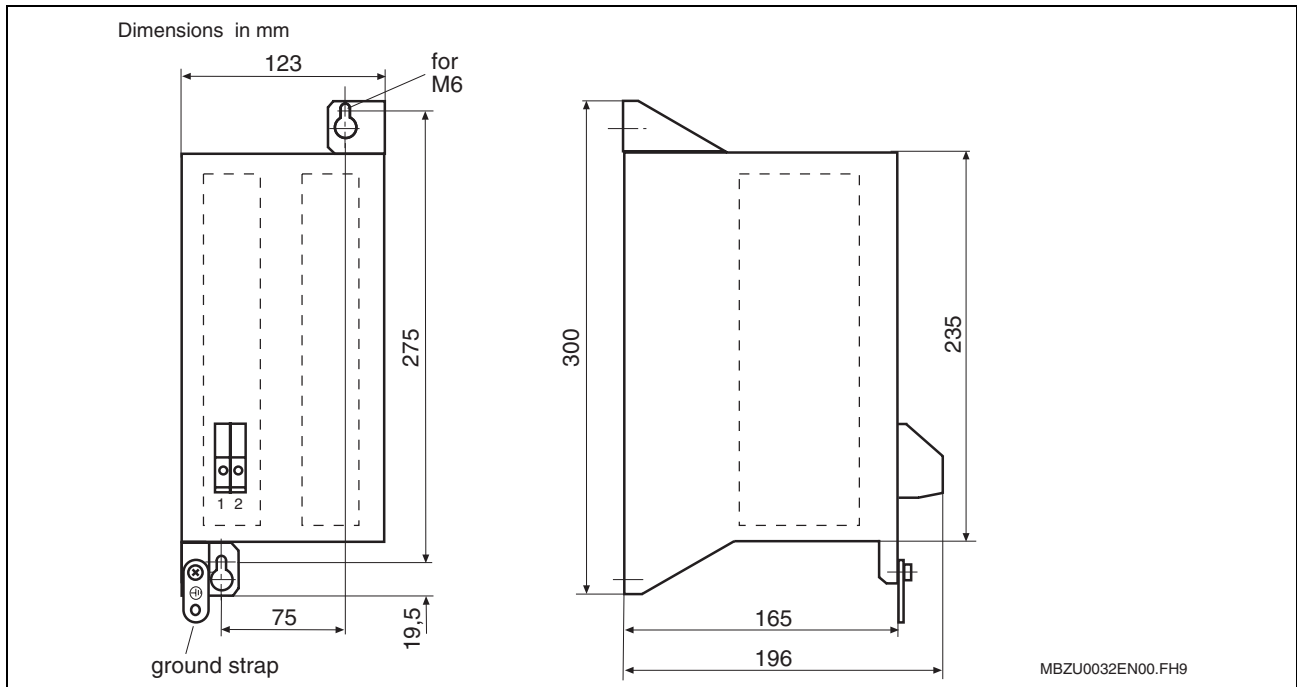


Fig. 9-6: Dimension sheet braking resistor HLR01.1N-0300-N17R5-A-007-NNNN

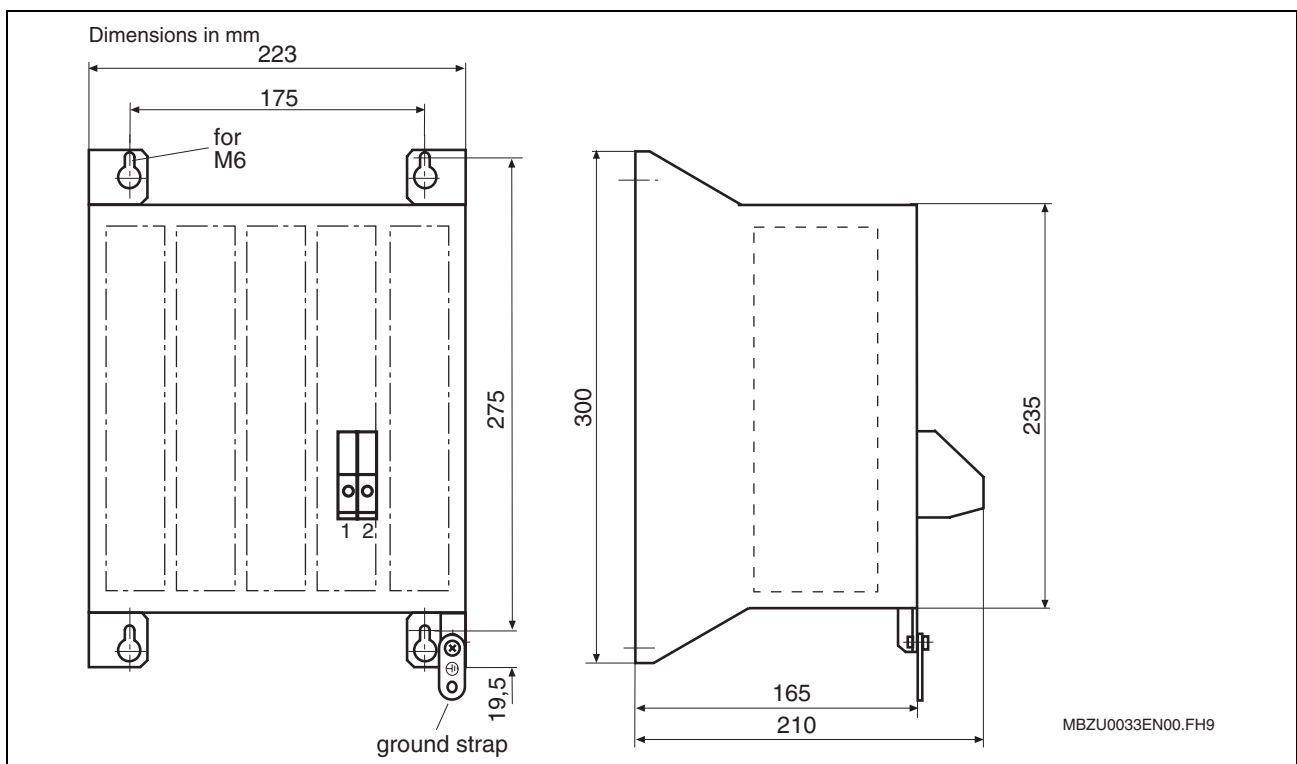


Fig. 9-7: Dimension sheet braking resistor
HLR01.1N-0470-N11R7-A-007-NNNN and
HLR01.1N-0780-N07R0-A-007-NNNN

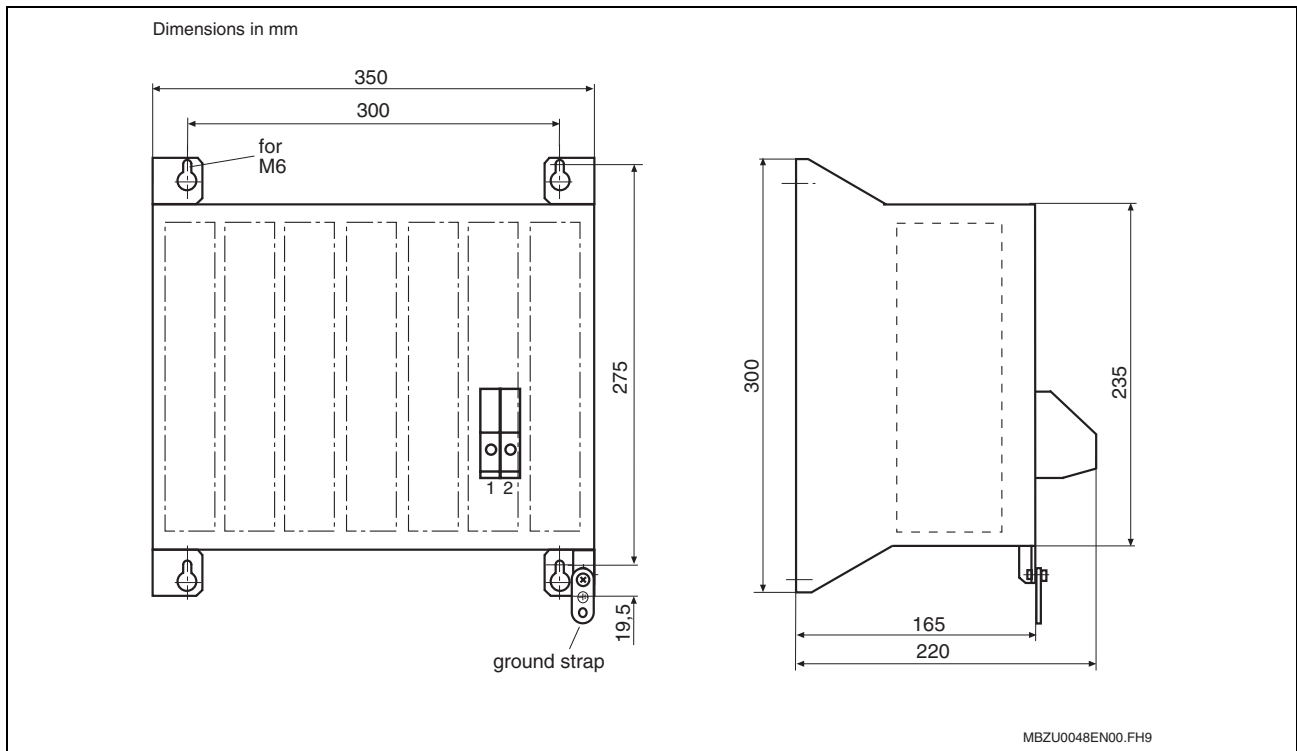


Fig. 9-8: Dimension sheet braking resistor HLR01.1N-1K08-N05R0-A-007-NNNN

Arrangement of the Components Device / Braking Resistor HLR01.1N-0300-N17R-A-007-NNNN

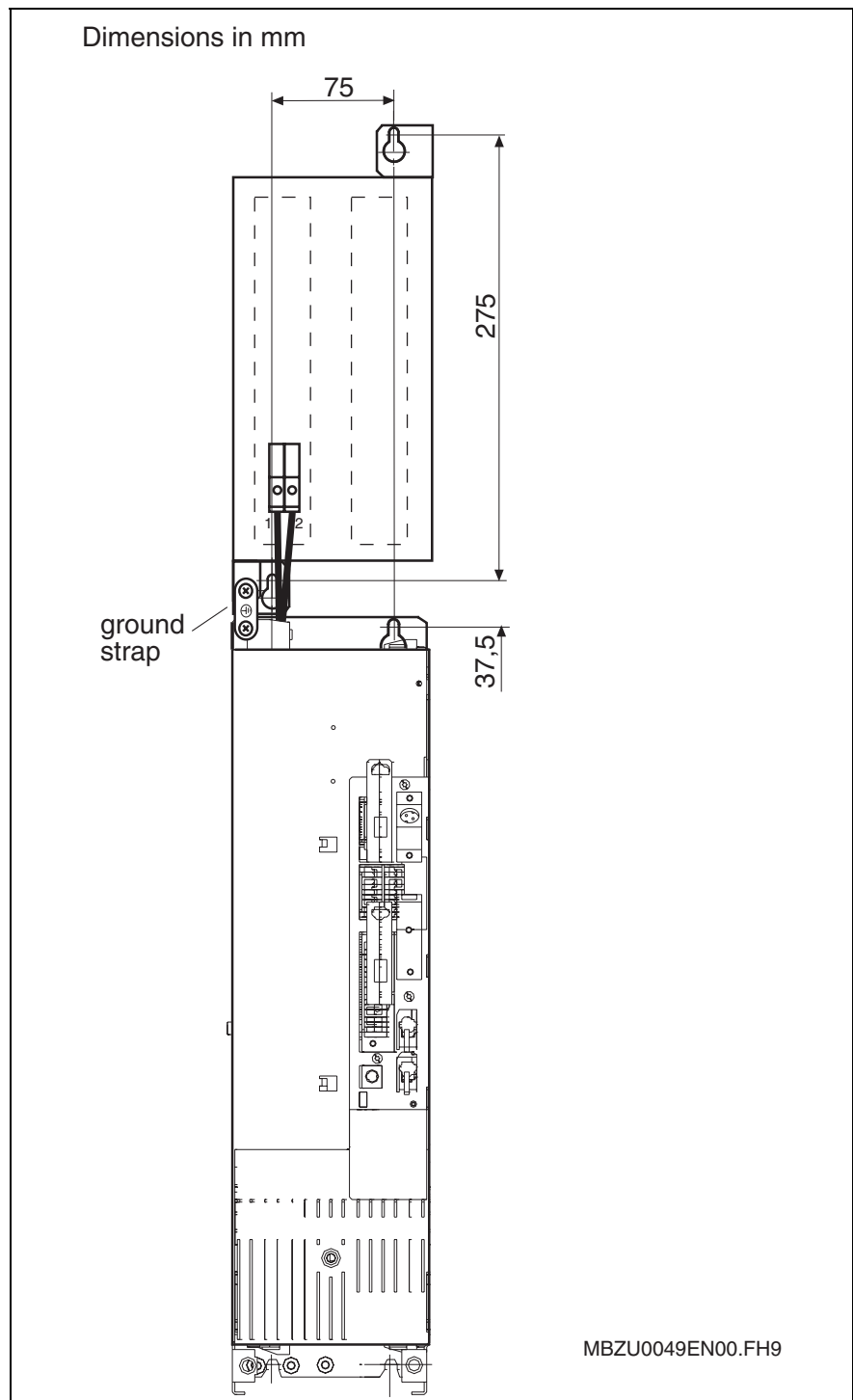


Fig. 9-9: Sample mounting device / braking resistor
HLR01.1N-0300-N17R5-A-007-NNNN

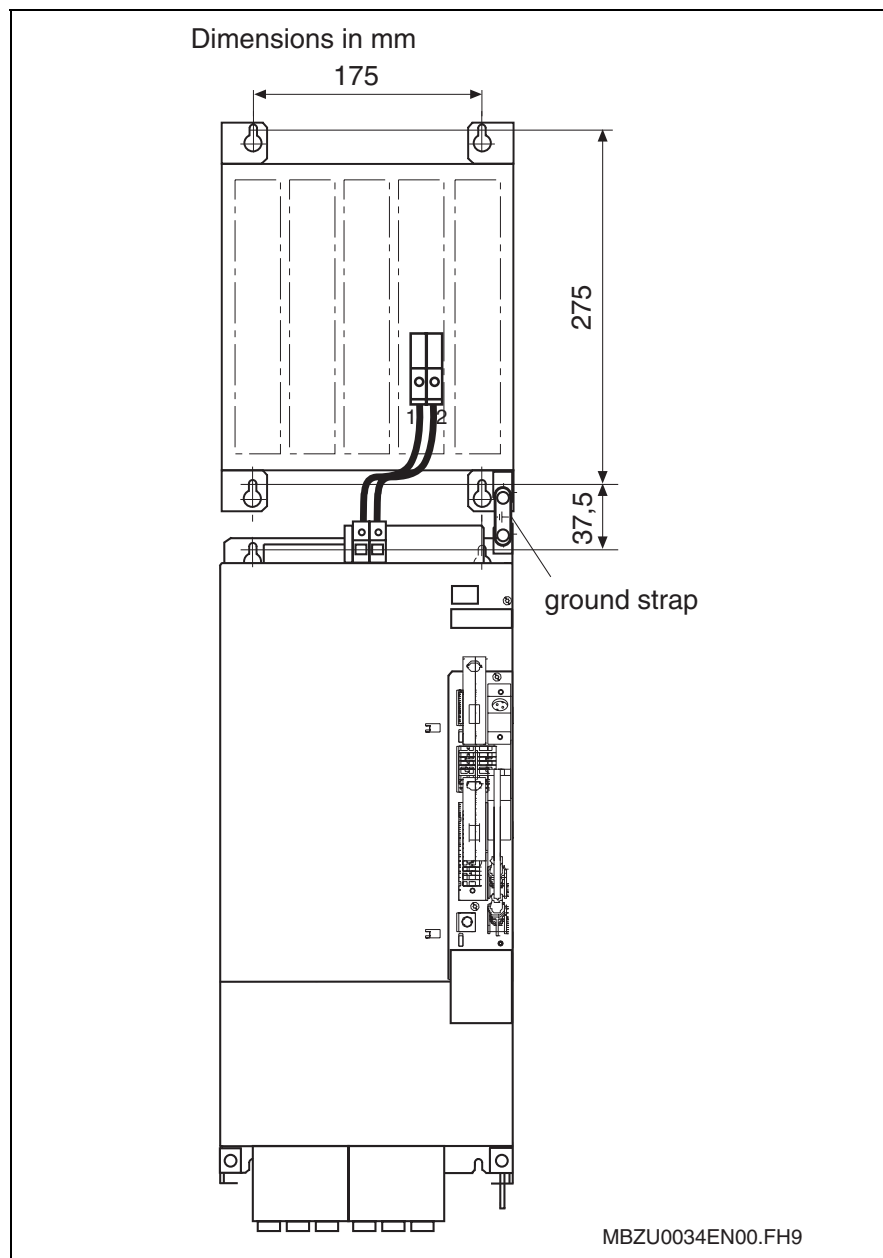
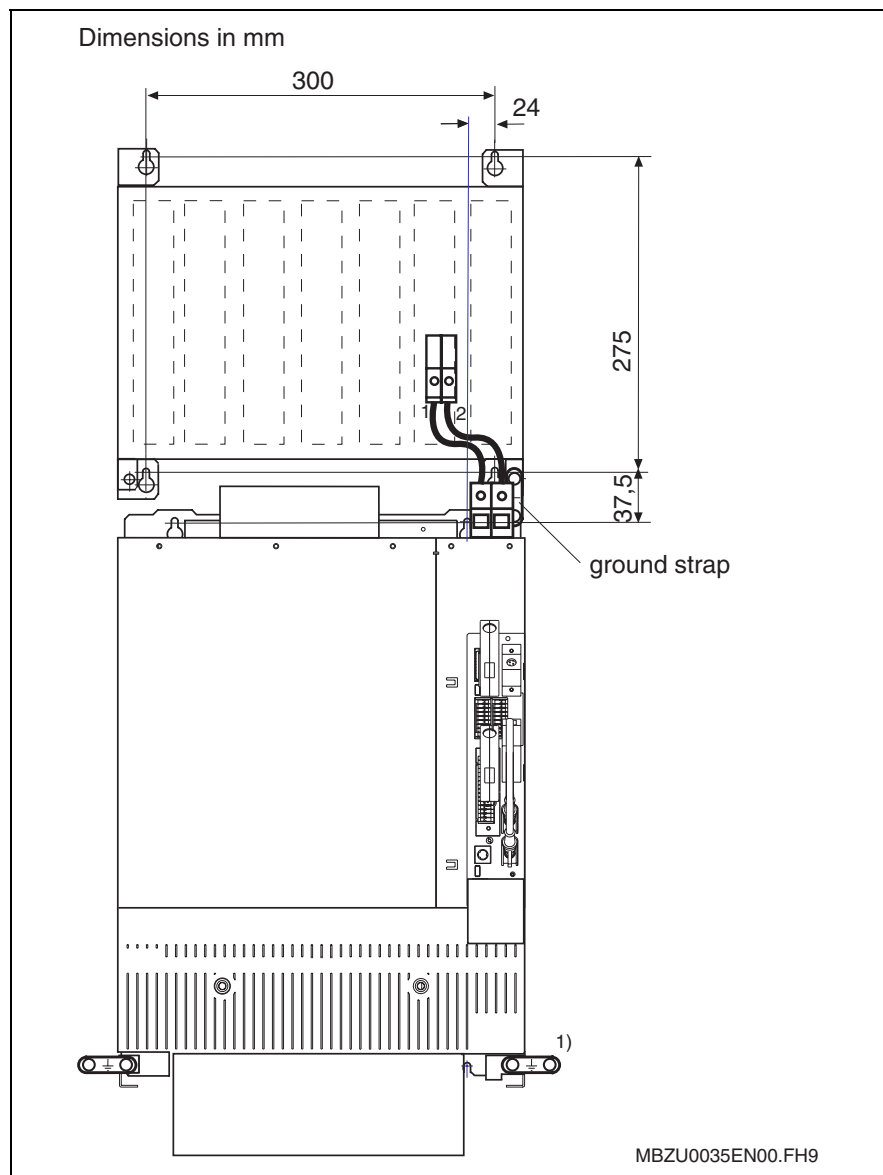


Fig. 9-10: Sample mounting device / braking resistor
HLR01.1N-0470-N11R7-A-007-NNNN and
HLR01.1N-0780-N07R7-A-007-NNNN



1): 1): Joint bar is part of accessories HAS01.
 Fig. 9-11: Sample mounting device / braking resistor
 HLR01.1N-1K08-N05R0-A-007-NNNN

Note: The standard braking resistors are intended to be mounted above the drive controllers HCS03; otherwise, forced convection (blower) with a volume stream of 200 m³/hour is to be provided.

9.5 Technical Data of the Standard Braking Resistors HLR01.1

Braking resistor HLR01.1N-xxxx-Nxxxx-A-007-NNNN	0300-N17R5	0470-N11R7	0780-N07R0	1K08-N05R0
peak power [kW]	37	56	93	130
peak power duty cycle load cycle T=120 s [ED %]	0,83	0,83	0,83	0,83
peak power t _{ein max} , T=120 s [s]	1	1	1	1
energy absorption [kWs]	37	56	93	130
continuous power [kW]	0,30	0,47	0,78	1,08
resistance value in operating status [Ω]	20,5	13,7	8,2	5,8
connection cross section [mm²]	10	25	25	50
connection cross section [AWG]	8	3	3	1/0
time constant [s]	780	780	780	780
weight kg	3	4,5	5,5	8

Fig. 9-12: Technical data standard braking resistors HLR01.1

9.6 Technical Features of the Reinforced Braking Resistors HLR01.1

Degree of protection IP 20

- Type A

Cement-coated, wire-wound, tube-type fixed resistors in screwed-on condition, with side walls and perforated cover, fixed parallel to the mounting surface, with two connections wired to porcelain terminals in the mounted terminal box with conduit thread connection.

- Type B and C

Fixed resistor in steel grid version. Depending on the power, with 2 flat terminals, 2 stud terminals M6 or M8.

Dimensioning

The specified braking power is valid for the mentioned relative duty cycle (ED) of 6 %, 15 %, 25 %, 40 %, and 60 %, relative to a load cycle of 120 s.

Obviously, other duty cycles are allowed as well. The required clearing time is reduced in the same ratio.

$$t_{aus} = \frac{t_{ein}}{t_{ein\max}} \times t_{aus\min}$$

t_{aus}: required clearing time in [s]

t_{ein}: duty cycle in [s]

t_{einmax}: maximum duty cycle in [s]

t_{ausmin}: minimum clearing time in [s]

Fig. 9-13: Calculation of the required clearing time

Example: With a duty cycle of 40%, the resistor may be activated for a maximum of 48 s (t_{einmax}), followed by a clearing time of 72 s. Obviously, shorter duty cycles are allowed as well. The required clearing time is

reduced in the same ratio. For example with a duty cycle of 10 s, the clearing time is as follows:

$$\text{example: } \frac{10s}{48s} \times 72s = 15s$$

Fig. 9-14: Sample calculation of the required clearing time

Duty Cycles

ED	t _{einmax} in s	t _{ausmin} in s
60%	72	48
40%	48	72
25%	30	90
15%	18	102
6 %	7,2	112,8

Fig. 9-15: Duty cycles

9.7 Technical Data of the Reinforced Braking Resistors HLR01.1

Braking Resistors for HCS03.1E-W0070

HLR01.1N-xxxx-xxxxx-A-007-NNNN		-01K6-N18R0	-03K5-N19R0	-04K5-N18R0	-06K5-N18R0	-10K0-N18R0
continuous power	kW	1,6	3,5	4,5	6,5	10
resistance value in operating status	Ω	20,0	21,3	20,2	20,2	20,2
type		A5	B1	B2	B2	B3
connection cross section	mm ²	2,5	4	6	10	10
connection cross section	AWG	14	12	10	8	8
Energy absorption at peak power, duty cycle t=x sec and load cycle T=120 sec						
energy absorption at peak power	kWs	109	252	432	686	1080
peak power	kW	34	31	33	33	33
t	sec	3,3	8	13	21	32
duty cycle ED	%	2,7	6,7	11	17	27
Braking power at x % ED and a load cycle T=120 sec						
braking power	kW	15	14	14	14	15
t	sec	7,2	18	30	48	72
ED	%	6	15	25	40	60

Fig. 9-16: Technical data for braking resistors at HCS03.1E-W0070

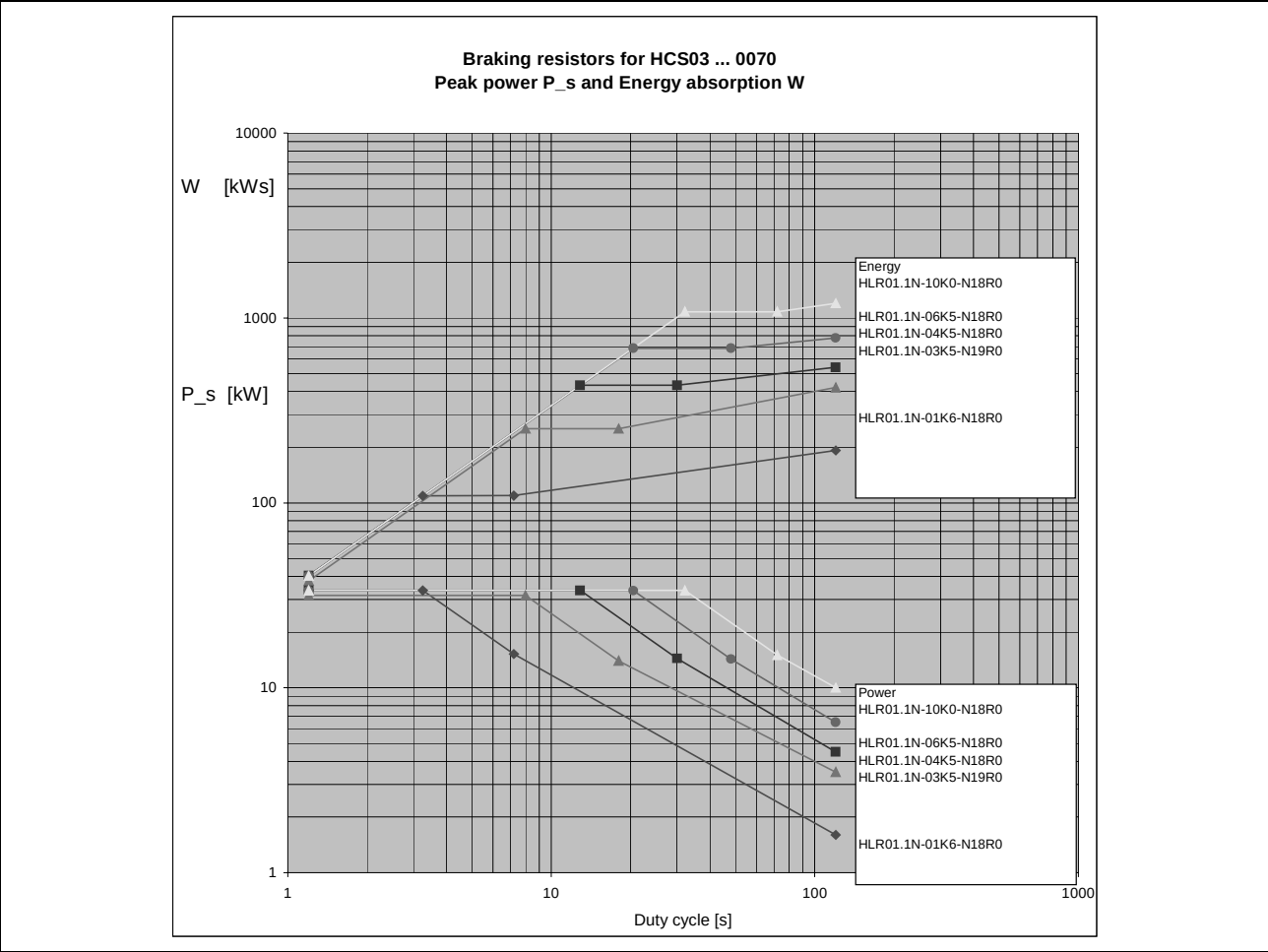


Fig. 9-17: Selection aid for braking resistors for HCS03.1E-W0070

Braking Resistors for HCS03.1E-W0100

HLR01.1N-xxxx-xxxxx-A-007-NNNN		-02K0-N15R0	-05K0-N15R0	-07K0-N14R0	-09K5-N13R0	-14K5-N13R0
continuous power	kW	2	5	7	9,5	14,5
resistance value in operating status	Ω	16,7	16,9	15,7	14,6	14,6
type		A6	B2	B3	B3	B4
connection cross section	mm ²	4	6	10	10	16
connection cross section	AWG	12	10	8	8	6
Energy absorption at peak power, duty cycle t=x sec and load cycle T=120 sec						
energy absorption at peak power	kWs	137	360	672	1003	1566
peak power	kW	40	40	43	46	46
t	sec	3,4	9	16	22	34
duty cycle ED	%	2,8	7,5	13	18	28
Braking power at x % ED and a load cycle T=120 sec						
braking power	kW	19	20	22	21	22
t	sec	7,2	18	30	48	72
ED	%	6	15	25	40	60

Fig. 9-18: Technical data for braking resistors at HCS03.1E-W0100

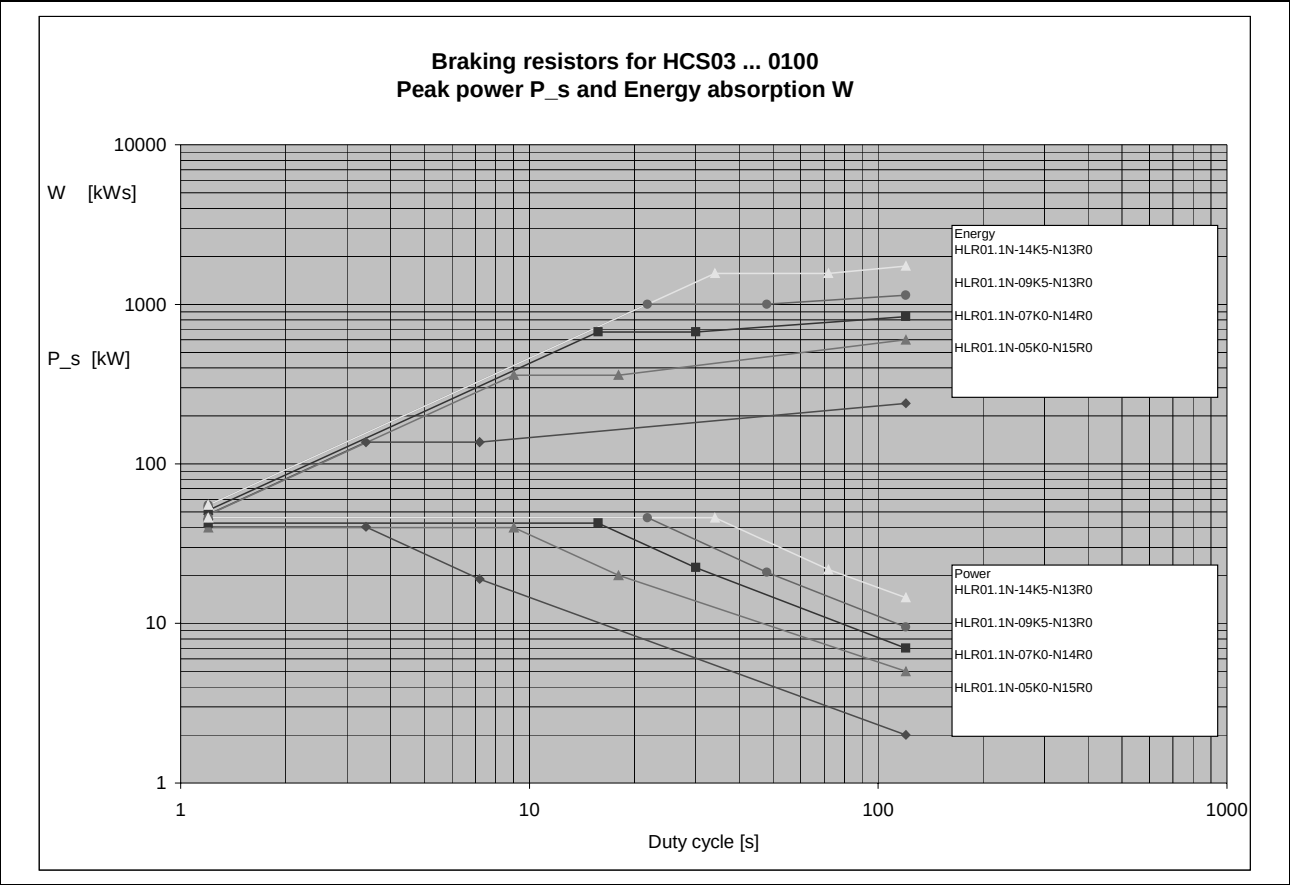


Fig. 9-19: Selection aid for braking resistors for HCS03.1E-W0100

Braking Resistors for HCS03.1E-W0150

HLR01.1N-xxxx-xxxxx-A-007-NNNN		-04K5-N07R4	-08K5-N08R0	-11K0-N07R3	-15K0-N08R1	-24K0-N07R2
continuous power	kW	4,5	8,5	11	15	24
resistance value in operating status	Ω	8,3	9,0	8,2	9,1	8,1
type		B1	B3	B3	B4	C3
connection cross section	mm ²	10	16	25	25	35
connection cross section	AWG	8	6	3	3	2
Energy absorption at peak power, duty cycle t=x sec and load cycle T=120 sec						
energy absorption at peak power	kWs	246	612	1056	1584	2592
peak power	kW	81	75	82	74	83
t	sec	3	8,2	13	21	31
duty cycle ED	%	2,5	6,8	11	18	26
Braking power at x % ED and a load cycle T=120 sec						
braking power	kW	34	34	35	33	36
t	sec	7,2	18	30	48	72
ED	%	6	15	25	40	60

Fig. 9-20: Technical data for braking resistors at HCS03.1E-W0150

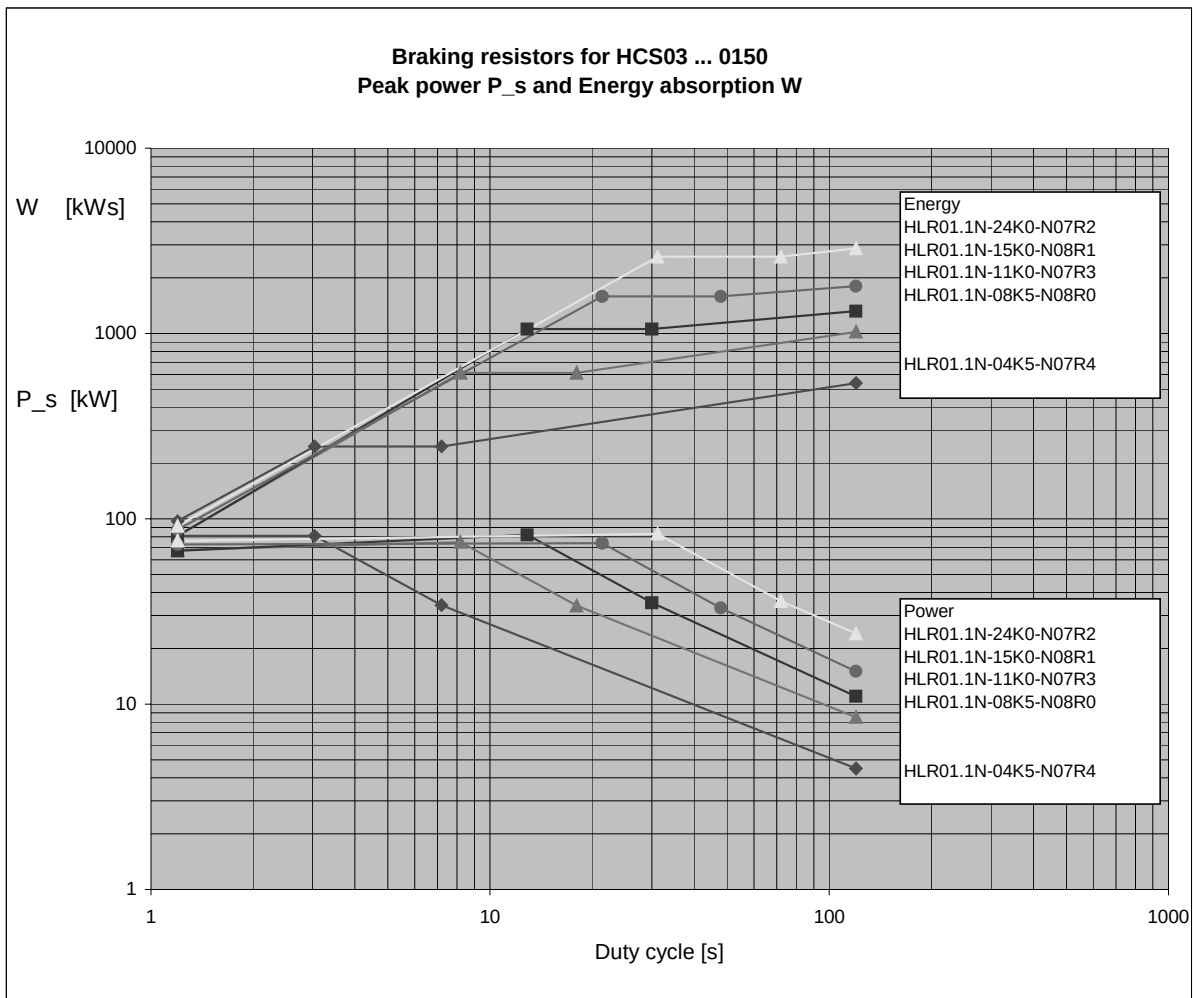


Fig. 9-21: Selection aid for braking resistors for HCS03.1-W0150

Braking Resistors for HCS03.1E-W0210

HLR01.1N-xxxx-xxxxx-A-007-NNNN		-06K5-N06R1	-12K5-N05R5	-17K0-N05R1	-23K0-N05R5	-36K0-N05R4
continuous power	kW	6,5	12,5	17	23	36
resistance value in operating status	Ω	7	6	6	6	6
type		B2	B4	B5	C2	C4
connection cross section	mm ²	16	35	35	50	50
connection cross section	AWG	6	2	2	1/0	1/0
Energy absorption at peak power, duty cycle t=x sec and load cycle T=120 sec						
energy absorption at peak power	kWs	356	900	1632	2429	3888
peak power	kW	98	109	117	109	111
t	sec	3,6	8,3	14	22	35
duty cycle ED	%	3	6,9	12	18	29
Braking power at x % ED and a load cycle T=120 sec						
braking power	kW	49	50	54	51	54
t	sec	7,2	18	30	48	72
ED	%	6	15	25	40	60

Fig. 9-22: Technical data for braking resistors at HCS03.1E-W0210

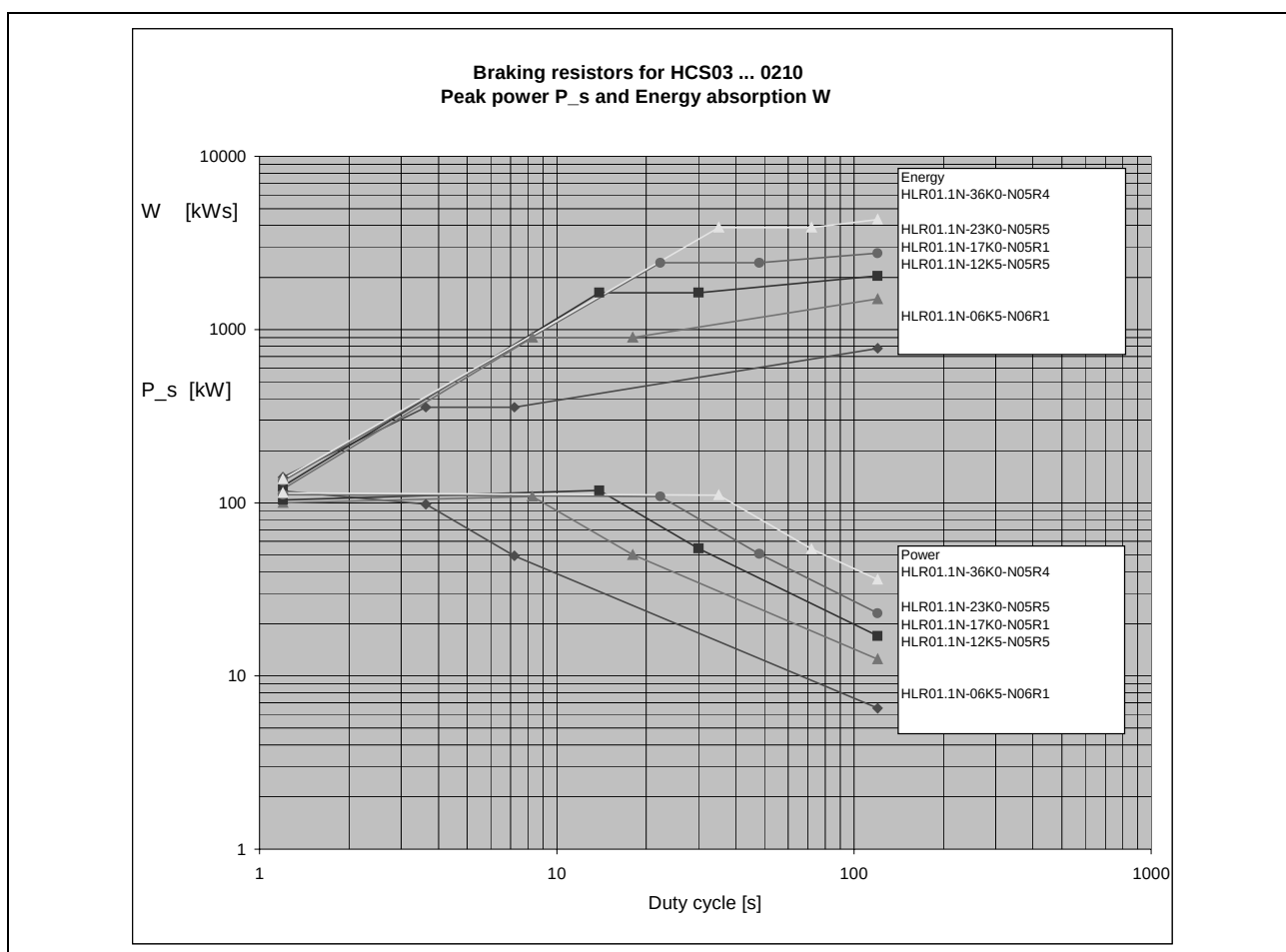


Fig. 9-23: Selection aid for braking resistors for HCS03.1E-W0210

9.8 Line Cross Sections for Reinforced Braking Resistors HLR01.1

According to the standards "Electronic equipment for use in power installations" (EN 50178, chapter 5.3.2.1) and "Adjustable speed electrical power drive systems" (EN 61800-5-1, chapter 4.2.5.4.2), a stationary connection of the equipment grounding conductor is required. The following requirement has to be complied with: cross section of equipment grounding conductor at least 10 mm^2 (mechanical stability).

Note: For the equipment grounding conductor run lines with a cross section corresponding to that of the lines for mains connection at the drive controller HCS and having at least 10 mm^2 .

Requirements for Line Routing

- The cross sections are valid for one phase each with multi-wire conductors and are specified according to the regulations of VDE0298.
- Up to 50 mm^2 , single strands in the cable duct.

9.9 Mechanical Data of the Reinforced Braking Resistors HLR01.1

Fixed Resistors IP 20 Type A

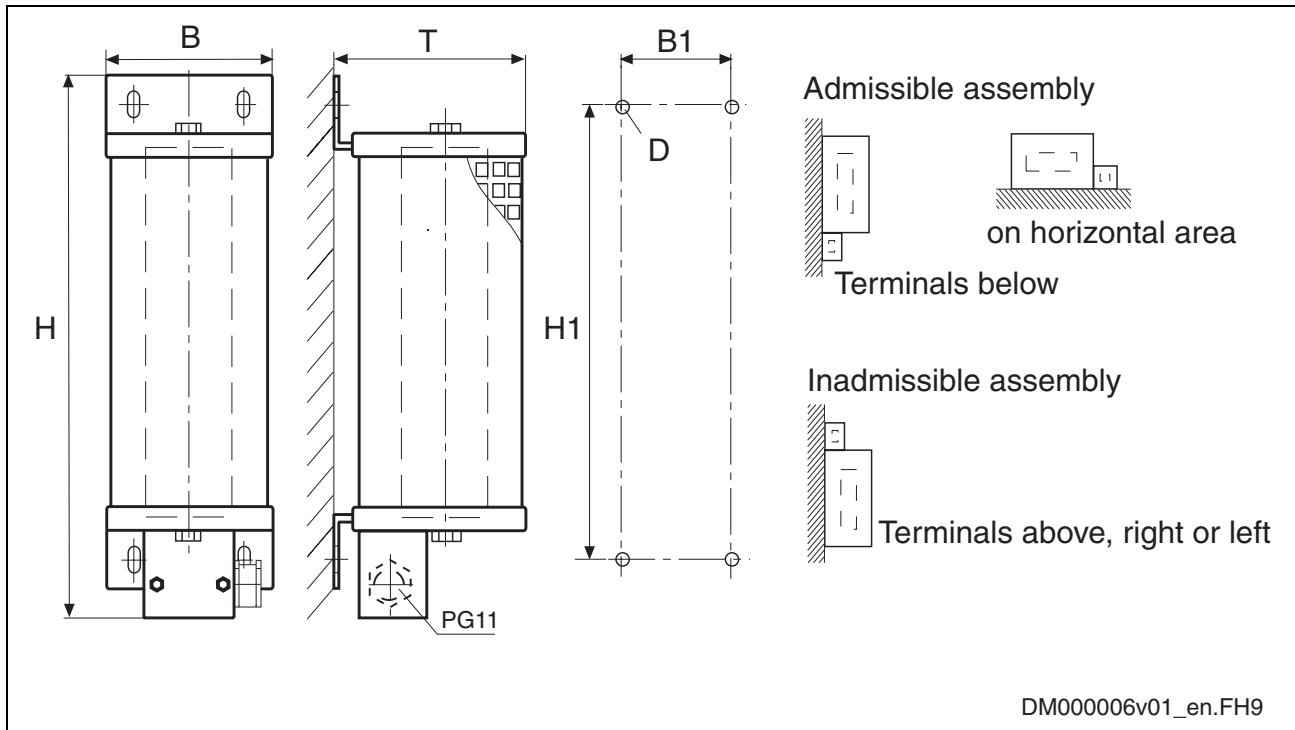


Fig. 9-24: Type A5 – A6

Type	Dimensions in mm						Weight in kg
	H	B	T	H1	B1	D	
A5	586	185	120	526	150	M6	5,2
A6	686	185	120	626	150	M6	6,2

Fig. 9-25: Dimensions table braking resistor type A

Steel Grid Fixed Resistors IP 20 Type B

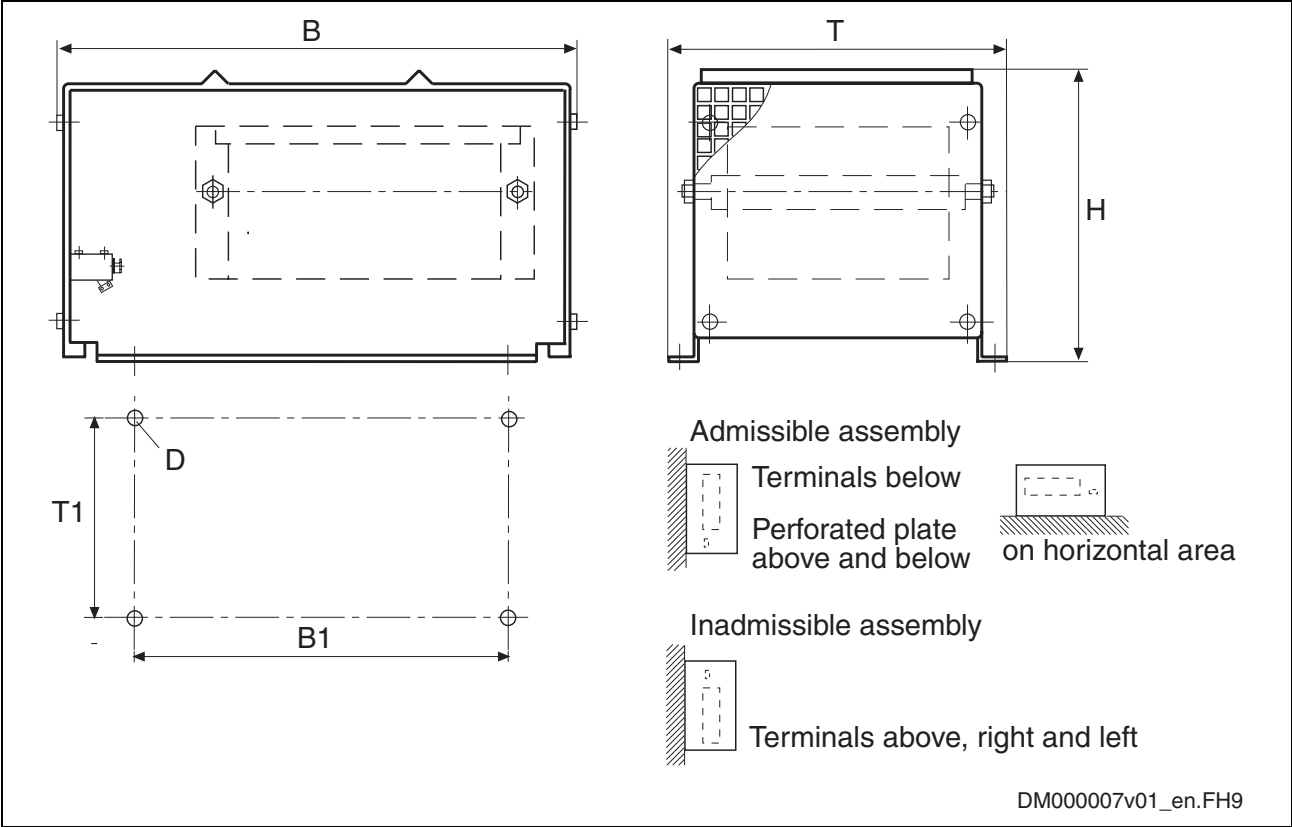


Fig. 9-26: Type B1 – B4

Type	Dimensions in mm						Weight in kg
	B	T	H	B1	T1	D	
B1	490	300	270	380	270	M10	9,5
B2	490	400	270	380	370	M10	13
B3	490	600	270	380	570	M10	22
B4	490	800	270	380	770	M10	33
B5	490	1000	270	380	970	M10	44

Fig. 9-27: Dimensions table braking resistor type B

Steel Grid Fixed Resistors IP 20 Type C

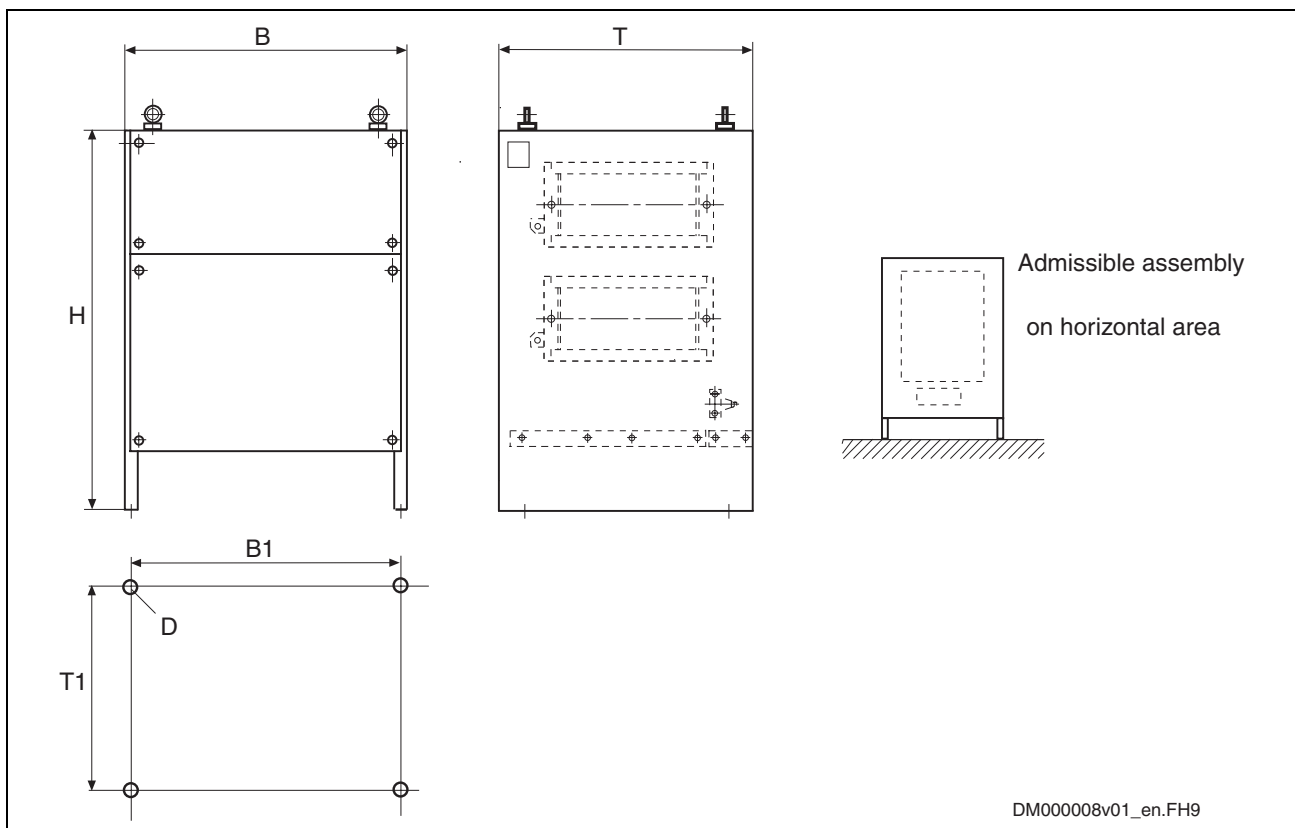


Fig. 9-28: Type C1 – C7

Type	Dimensions in mm						Weight in kg
	B	T	H	B1	T1	D	
C2	595	490	710	570	380	M10	56
C3	795	490	710	770	380	M10	80
C4	995	490	710	970	380	M10	93

Fig. 9-29: Dimensions table braking resistor type C

Note: Under consideration of the technical data for the minimum braking resistance and the brake chopper, other braking resistors can be used alternatively.

Notes

10 Motor Filter HMF01.1

10.1 Type Code

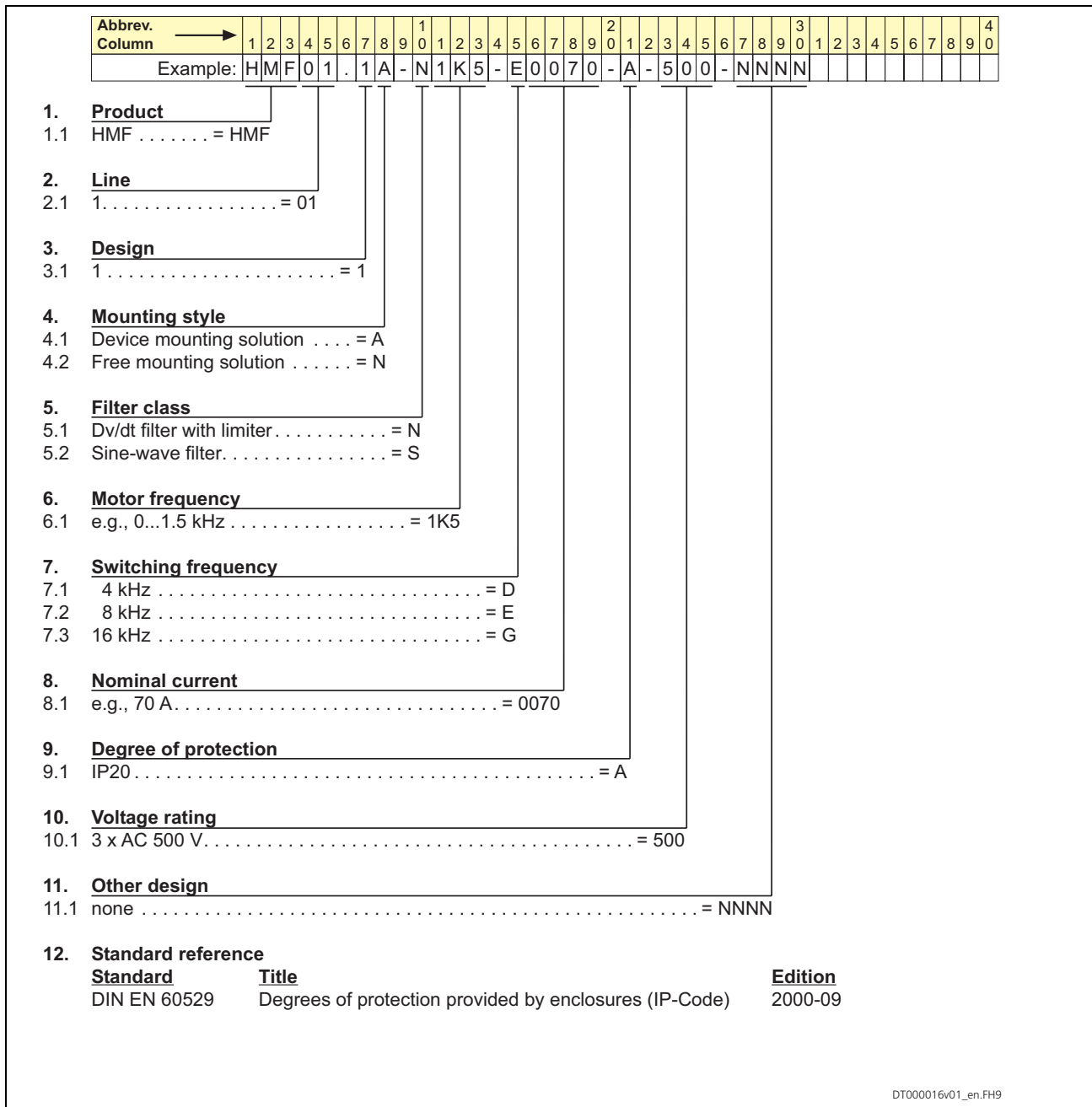


Fig. 10-1: Type code of motor filter HMF01.1

10.2 Type Plate

Each motor filter is identified by a type designation. There is a type plate attached to all components.

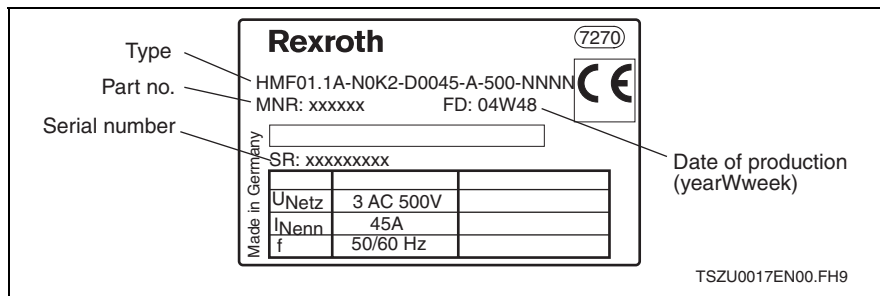


Fig. 10-2: Type plate

10.3 Electrical Data

Motor filter	I _N [A]	P _V [W]	L _M μH	f _{out_max} Hz	f _{p_max} kHz	du/dt kV / us	U _{out_max} kV
HMF01.1A-N4K0-A0045-A-007-NNNN	45	120	3 x 160	200	4	< 4	1,2
HMF01.1A-N4K0-A0072-A-007-NNNN	72	160	3 x 100	200	4	< 4	1,2
HMF01.1A-N4K0-A0095-A-007-NNNN	95	190	3 x 78	200	4	< 4	1,2
HMF01.1A-N4K0-A0145-A-007-NNNN	145	220	3 x 50	200	4	< 4	1,2

10.4 Mechanical Data

Dimensional Drawings

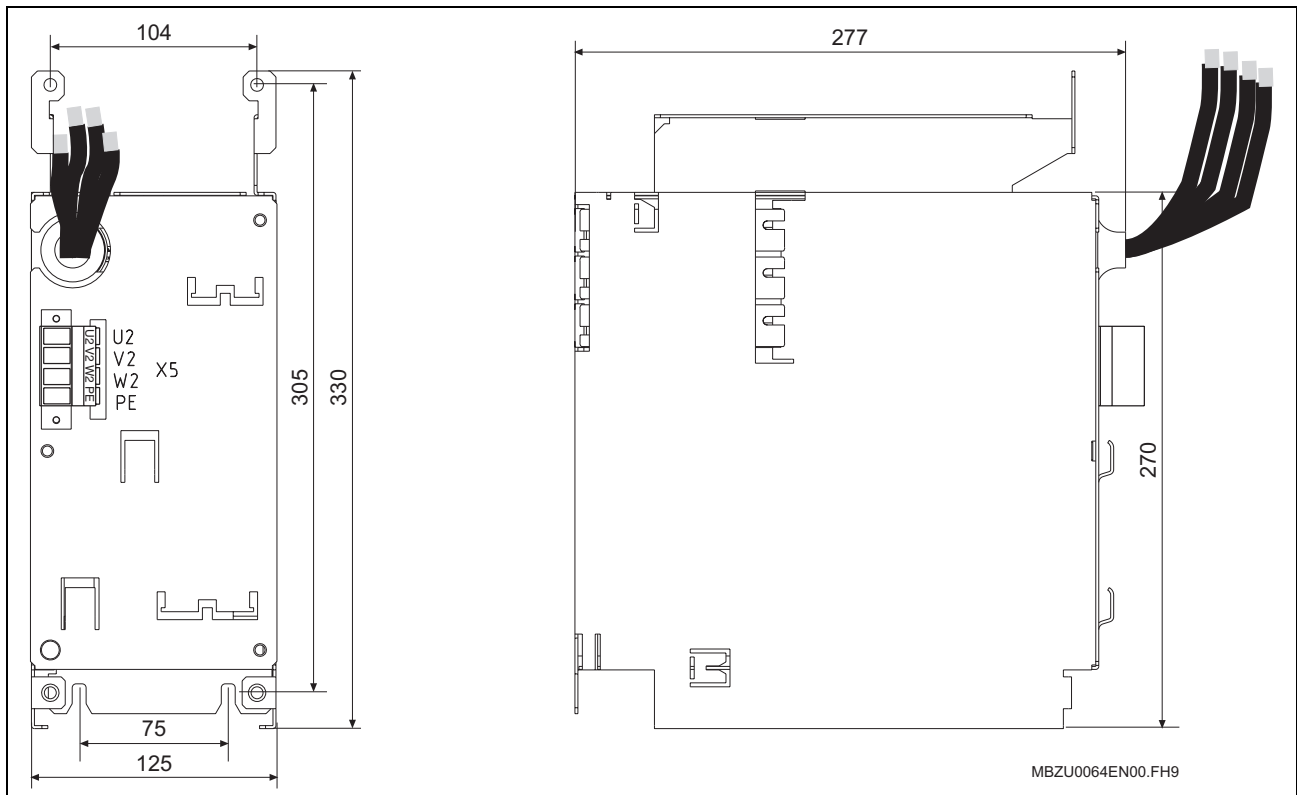


Fig. 10-3: Dimensional drawing motor filter HMF01.1A-D0K2-D0045-A-500-NNNN

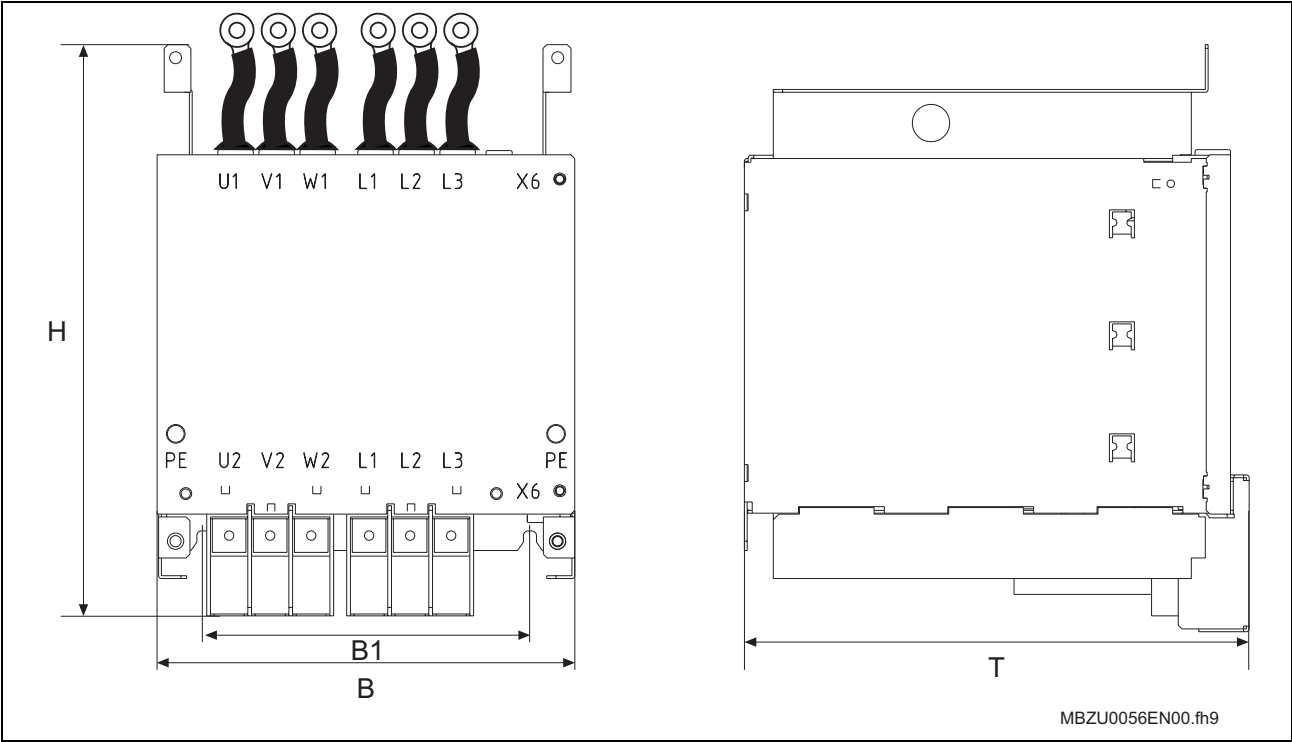


Fig. 10-4: Dimensional drawing motor filter
HMF01.1A-N0K2-D0073-A-500-NNNN,
HMF01.1A-N0K2-D0095-A-500-NNNN and
HMF01.1A-N0K2-D0145-A-500-NNNN

Type	Dimensions in mm				
	B	H	T	B1	H1
0073	225	315	270	175	257
0095	225	315	270	175	257
0145	350	400	260	250	310

Fig. 10-5: Dimensional drawing motor filter

10.5 Arrangement of the Components HCS03.1 / Motor Filter HMF

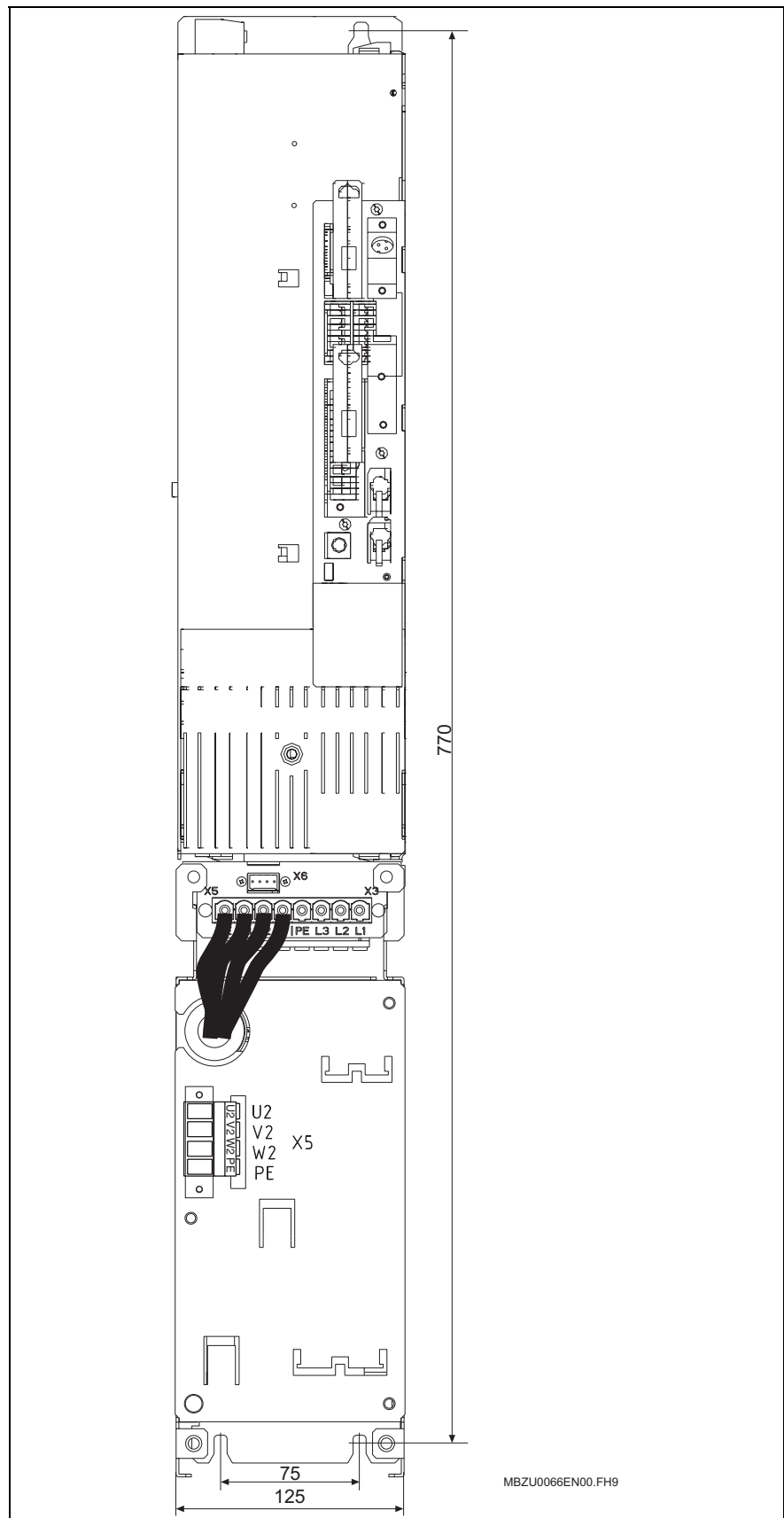


Fig. 10-6: Sample mounting HCS03.1E-W0070 / HMF01.1A-D0K2-D0045

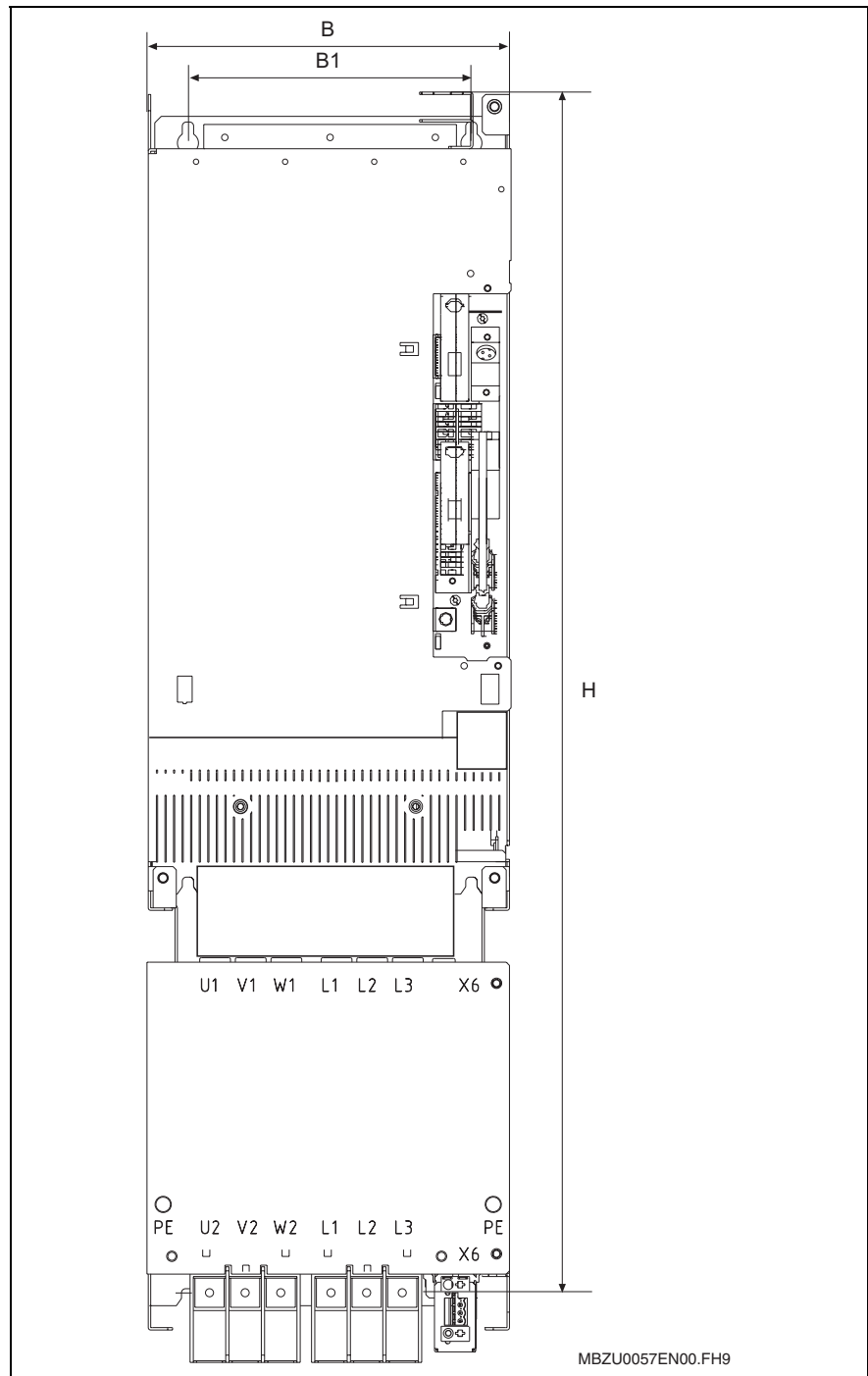


Fig. 10-7: Sample mounting HCS03.1E-W0100 0150 0210 / motor filter

HCS03.1E-	H	B	B1
W0100	720	225	175
W0150	720	225	175
W0210	780	350	250

Fig. 10-8: Dimensions table for sample mounting HCS03.1E-.. / motor filter HMF

10.6 Arrangement of the Components HCS03.1 / Motor Filter and Mains Filter

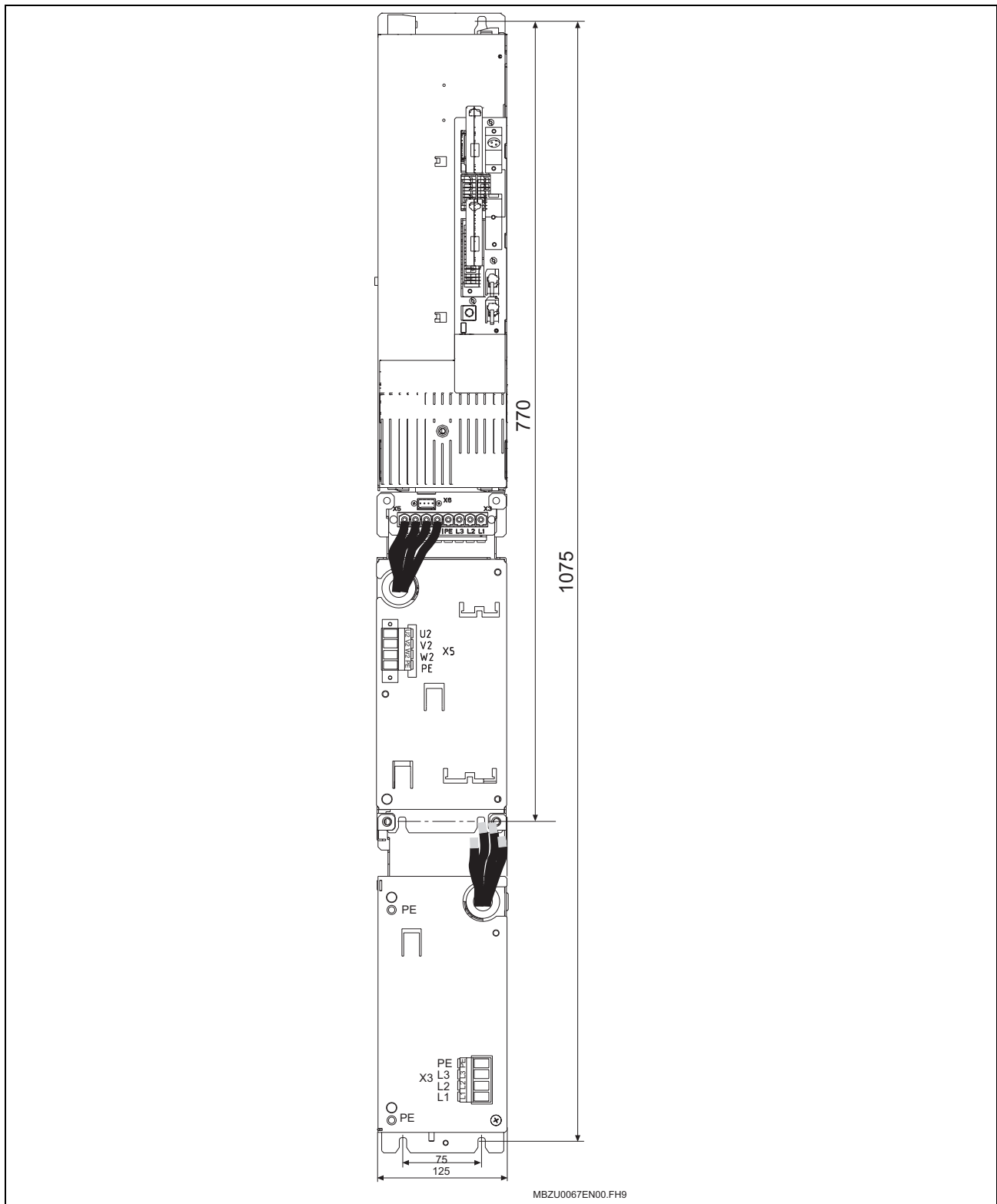


Fig. 10-9: Arrangement HCS03.1E-W0070 / motor filter and mains filter + HAS05.1-001 + HAS05.1-002

Note: For information on the HAS accessories see Project Planning Manual on the drive system Rexroth IndraDrive.

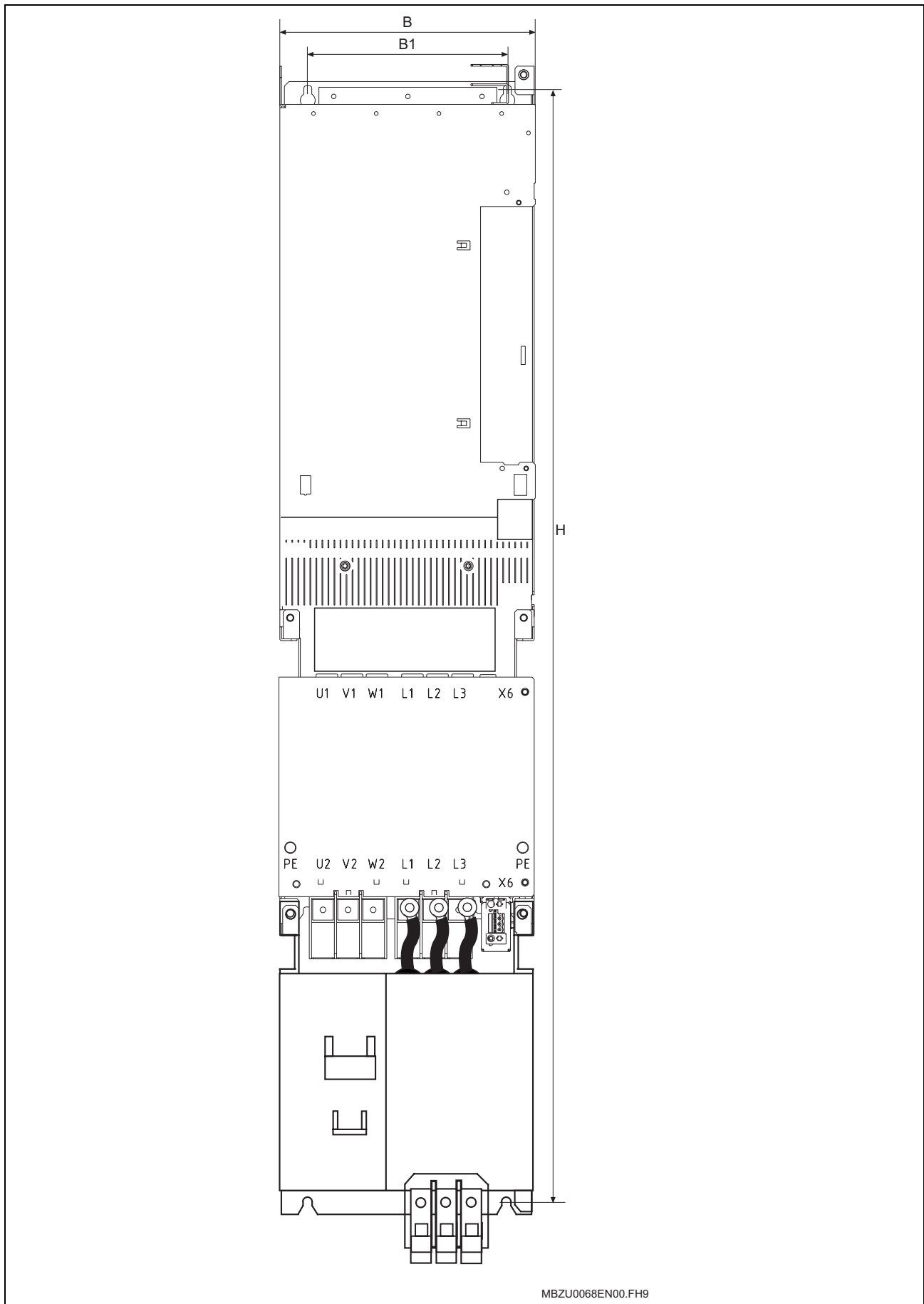


Fig. 10-10: Arrangement of device 0100, 0150, 0210 / motor filter and mains filter

HCS03.1E-	H	B1	B
W0100	980	175	225
W0150	980	175	225
W0210	1090	250	350

Fig. 10-11: Dimensions table for sample mounting HCS03.1E-.. / mains filter / motor filter

Notes

11 Service & Support

11.1 Helpdesk

Unser Kundendienst-Helpdesk im Hauptwerk Lohr am Main steht Ihnen mit Rat und Tat zur Seite. Sie erreichen uns

Our service helpdesk at our headquarters in Lohr am Main, Germany can assist you in all kinds of inquiries. Contact us

- | | |
|--|---|
| <ul style="list-style-type: none"> - telefonisch - by phone:
über Service Call Entry Center
- via Service Call Entry Center - per Fax - by fax: - per e-Mail - by e-mail: <u>service.svc@boschrexroth.de</u> | <p>+49 (0) 9352 40 50 60
Mo-Fr 07:00-18:00
Mo-Fr 7:00 am - 6:00 pm</p> <p>+49 (0) 9352 40 49 41</p> |
|--|---|

11.2 Service-Hotline

Außerhalb der Helpdesk-Zeiten ist der Service direkt ansprechbar unter

After helpdesk hours, contact our service department directly at

+49 (0) 171 333 88 26
oder - or **+49 (0) 172 660 04 06**

11.3 Internet

Unter **www.boschrexroth.com** finden Sie ergänzende Hinweise zu Service, Reparatur und Training sowie die **aktuellen** Adressen *) unserer auf den folgenden Seiten aufgeführten Vertriebs- und Servicebüros.



Verkaufsniederlassungen



Niederlassungen mit Kundendienst

Außerhalb Deutschlands nehmen Sie bitte zuerst Kontakt mit unserem für Sie nächstgelegenen Ansprechpartner auf.

*) Die Angaben in der vorliegenden Dokumentation können seit Drucklegung überholt sein.

At **www.boschrexroth.com** you may find additional notes about service, repairs and training in the Internet, as well as the **actual** addresses *) of our sales- and service facilities figuring on the following pages.



sales agencies



offices providing service

Please contact our sales / service office in your area first.

*) Data in the present documentation may have become obsolete since printing.

11.4 Vor der Kontaktaufnahme... - Before contacting us...

Wir können Ihnen schnell und effizient helfen wenn Sie folgende Informationen bereithalten:

1. detaillierte Beschreibung der Störung und der Umstände.
2. Angaben auf dem Typenschild der betreffenden Produkte, insbesondere Typenschlüssel und Seriennummern.
3. Tel./Faxnummern und e-Mail-Adresse, unter denen Sie für Rückfragen zu erreichen sind.

For quick and efficient help, please have the following information ready:

1. Detailed description of the failure and circumstances.
2. Information on the type plate of the affected products, especially type codes and serial numbers.
3. Your phone/fax numbers and e-mail address, so we can contact you in case of questions.

11.5 Kundenbetreuungsstellen - Sales & Service Facilities

Deutschland – Germany

vom Ausland:

(0) nach Landeskennziffer weglassen!

from abroad:

don't dial (0) after country code!

Vertriebsgebiet Mitte Germany Centre	SERVICE AUTOMATION	SERVICE AUTOMATION	SERVICE AUTOMATION
Rexroth Indramat GmbH Bgm.-Dr.-Nebel-Str. 2 / Postf. 1357 97816 Lohr am Main / 97803 Lohr Kompetenz-Zentrum Europa Tel.: +49 (0)9352 40-0 Fax: +49 (0)9352 40-4885	CALL ENTRY CENTER Helpdesk MO – FR von 07:00 - 18:00 Uhr from 7 am – 6 pm Tel. +49 (0) 9352 40 50 60 Fax +49 (0) 9352 40 49 41 service.svc@boschrexroth.de	HOTLINE 24 / 7 / 365 außerhalb der Helpdesk-Zeit out of helpdesk hours Tel.: +49 (0)172 660 04 06 oder / or Tel.: +49 (0)171 333 88 26	ERSATZTEILE / SPARES verlängerte Ansprechzeit - extended office time - ♦ nur an Werktagen - only on working days - ♦ von 07:00 - 18:00 Uhr - from 7 am - 6 pm - Tel. +49 (0) 9352 40 42 22
Vertriebsgebiet Süd Germany South	Vertriebsgebiet West Germany West	Gebiet Südwest Germany South-West	
Bosch Rexroth AG Landshuter Allee 8-10 80637 München Tel.: +49 (0)89 127 14-0 Fax: +49 (0)89 127 14-490	Bosch Rexroth AG Regionalzentrum West Borsigstrasse 15 40880 Ratingen Tel.: +49 (0)2102 409-0 Fax: +49 (0)2102 409-406 +49 (0)2102 409-430	Bosch Rexroth AG Service-Regionalzentrum Süd-West Siemensstr. 1 70736 Fellbach Tel.: +49 (0)711 51046-0 Fax: +49 (0)711 51046-248	
Vertriebsgebiet Nord Germany North	Vertriebsgebiet Mitte Germany Centre	Vertriebsgebiet Ost Germany East	Vertriebsgebiet Ost Germany East
Bosch Rexroth AG Walsroder Str. 93 30853 Langenhagen Tel.: +49 (0) 511 72 66 57-0 Service: +49 (0) 511 72 66 57-256 Fax: +49 (0) 511 72 66 57-93 Service: +49 (0) 511 72 66 57-783	Bosch Rexroth AG Regionalzentrum Mitte Waldecker Straße 13 64546 Mörfelden-Walldorf Tel.: +49 (0) 61 05 702-3 Fax: +49 (0) 61 05 702-444	Bosch Rexroth AG Beckerstraße 31 09120 Chemnitz Tel.: +49 (0)371 35 55-0 Fax: +49 (0)371 35 55-333	Bosch Rexroth AG Regionalzentrum Ost Walter-Köhn-Str. 4d 04356 Leipzig Tel.: +49 (0)341 25 61-0 Fax: +49 (0)341 25 61-111

Europa (West) - Europe (West)

vom Ausland: (0) nach Landeskennziffer weglassen,
from abroad: don't dial (0) after country code,

Italien: 0 nach Landeskennziffer mitwählen
Italy: dial 0 after country code

Austria – Österreich Bosch Rexroth GmbH Electric Drives & Controls Stachegasse 13 1120 Wien Tel.: +43 (0)1 985 25 40 Fax: +43 (0)1 985 25 40-93	Austria – Österreich Bosch Rexroth GmbH Electric Drives & Controls Industriepark 18 4061 Pasching Tel.: +43 (0)7221 605-0 Fax: +43 (0)7221 605-21	Belgium - Belgien Bosch Rexroth NV/SA Henri Genessestraat 1 1070 Bruxelles Tel: +32 (0) 2 451 26 08 Fax: +32 (0) 2 451 27 90 info@boschrexroth.be service@boschrexroth.be	Denmark - Dänemark BEC A/S Zinkvej 6 8900 Randers Tel.: +45 (0)87 11 90 60 Fax: +45 (0)87 11 90 61
Great Britain – Großbritannien Bosch Rexroth Ltd. Electric Drives & Controls Broadway Lane, South Cerney Cirencester, Glos GL7 5UH Tel.: +44 (0)1285 863000 Fax: +44 (0)1285 863030 sales@boschrexroth.co.uk service@boschrexroth.co.uk	Finland - Finnland Bosch Rexroth Oy Electric Drives & Controls Ansatie 6 017 40 Vantaa Tel.: +358 (0)9 84 91-11 Fax: +358 (0)9 84 91-13 60	France - Frankreich Bosch Rexroth SAS Electric Drives & Controls Avenue de la Trentaine (BP. 74) 77503 Chelles Cedex Tel.: +33 (0)164 72-63 22 Fax: +33 (0)164 72-63 20 Hotline: +33 (0)608 33 43 28	France - Frankreich Bosch Rexroth SAS Electric Drives & Controls ZI de Thibaud, 20 bd. Thibaud (BP. 1751) 31084 Toulouse Tel.: +33 (0)5 61 43 61 87 Fax: +33 (0)5 61 43 94 12
France – Frankreich Bosch Rexroth SAS Electric Drives & Controls 91, Bd. Irène Joliot-Curie 69634 Vénissieux – Cedex Tel.: +33 (0)4 78 78 53 65 Fax: +33 (0)4 78 78 53 62	Italy - Italien Bosch Rexroth S.p.A. Via G. Di Vittorio, 1 20063 Cernusco S/N.MI Hotline: +39 02 92 365 563 Tel.: +39 02 92 365 1 Service: +39 02 92 365 300 Fax: +39 02 92 365 500 Service: +39 02 92 365 516	Italy - Italien Bosch Rexroth S.p.A. Via Paolo Veronesi, 250 10148 Torino Tel.: +39 011 224 88 11 Fax: +39 011 224 88 30	Italy - Italien Bosch Rexroth S.p.A. Via Mascia, 1 80053 Castellammare di Stabia NA Tel.: +39 081 8 71 57 00 Fax: +39 081 8 71 68 85
Italy - Italien Bosch Rexroth S.p.A. Via del Progresso, 16 (Zona Ind.) 35020 Padova Tel.: +39 049 8 70 13 70 Fax: +39 049 8 70 13 77	Italy - Italien Bosch Rexroth S.p.A. Via Isonzo, 61 40033 Casalecchio di Reno (Bo) Tel.: +39 051 29 86 430 Fax: +39 051 29 86 490	Netherlands - Niederlande/Holland Bosch Rexroth Services B.V. Technical Services Kruisbroeksestraat 1 (P.O. Box 32) 5281 RV Boxtel Tel.: +31 (0) 411 65 19 51 Fax: +31 (0) 411 67 78 14 Hotline: +31 (0) 411 65 19 51 services@boschrexroth.nl	Netherlands – Niederlande/Holland Bosch Rexroth B.V. Kruisbroeksestraat 1 (P.O. Box 32) 5281 RV Boxtel Tel.: +31 (0) 411 65 16 40 Fax: +31 (0) 411 65 14 83 www.boschrexroth.nl
Norway - Norwegen Bosch Rexroth AS Electric Drives & Controls Berghagan 1 or: Box 3007 1405 Ski-Langhus 1402 Ski Tel.: +47 (0) 64 86 41 00 Fax: +47 (0) 64 86 90 62 Hotline: +47 (0)64 86 94 82 jul.ruud@rexroth.no	Spain – Spanien Goimendi S.A. Electric Drives & Controls Parque Empresarial Zuatzu C/ Francisco Grandmontagne no.2 20018 San Sebastian Tel.: +34 9 43 31 84 21 - service: +34 9 43 31 84 56 Fax: +34 9 43 31 84 27 - service: +34 9 43 31 84 60 sat.indramat@goimendi.es	Spain - Spanien Bosch Rexroth S.A. Electric Drives & Controls Centro Industrial Santiga Obradors s/n 08130 Santa Perpetua de Mogoda Barcelona Tel.: +34 9 37 47 94 00 Fax: +34 9 37 47 94 01	Spain - Spanien Bosch Rexroth S.A. Electric Drives & Controls c/ Almazara, 9 28760 Tres Cantos (Madrid) Tel.: +34 91 806 24 79 Fax: +34 91 806 24 72 fernando.bariego@boschrexroth.es
Sweden - Schweden Bosch Rexroth AB Electric Drives & Controls - Varuvägen 7 (Service: Konsumentvägen 4, Älfsjö) 125 81 Stockholm Tel.: +46 (0)8 727 92 00 Fax: +46 (0)8 647 32 77	Sweden - Schweden Bosch Rexroth AB Electric Drives & Controls Ekvändan 7 254 67 Helsingborg Tel.: +46 (0) 42 38 88 -50 Fax: +46 (0) 42 38 88 -74	Switzerland East - Schweiz Ost Bosch Rexroth Schweiz AG Electric Drives & Controls Hemrietstrasse 2 8863 Buttikon Tel. +41 (0) 55 46 46 111 Fax +41 (0) 55 46 46 222	Switzerland West - Schweiz West Bosch Rexroth Suisse SA Av. Général Guisan 26 1800 Vevey 1 Tel.: +41 (0)21 632 84 20 Fax: +41 (0)21 632 84 21

Europa (Ost) - Europe (East)

vom Ausland: (0) nach Landeskennziffer weglassen
from abroad: don't dial (0) after country code

Czech Republic - Tschechien Bosch -Rexroth, spol.s.r.o. Hviezdoslavova 5 627 00 Brno Tel.: +420 (0)5 48 126 358 Fax: +420 (0)5 48 126 112	Czech Republic - Tschechien DEL a.s. Strojirenská 38 591 01 Zdar nad Sázavou Tel.: +420 566 64 3144 Fax: +420 566 62 1657	Hungary - Ungarn Bosch Rexroth Kft. Angol utca 34 1149 Budapest Tel.: +36 (1) 422 3200 Fax: +36 (1) 422 3201	Poland – Polen Bosch Rexroth Sp.zo.o. ul. Staszica 1 05-800 Pruszków Tel.: +48 22 738 18 00 – service: +48 22 738 18 46 Fax: +48 22 758 87 35 – service: +48 22 738 18 42
Poland – Polen Bosch Rexroth Sp.zo.o. Biuro Poznan ul. Dabrowskiego 81/85 60-529 Poznan Tel.: +48 061 847 64 62 /-63 Fax: +48 061 847 64 02	Romania - Rumänien East Electric S.R.L. Bdul Basarabia no.250, sector 3 73429 Bucuresti Tel./Fax.: +40 (0)21 255 35 07 +40 (0)21 255 77 13 Fax: +40 (0)21 725 61 21 eastel@rdsnet.ro	Romania - Rumänien Bosch Rexroth Sp.zo.o. Str. Drobety nr. 4-10, app. 14 70258 Bucuresti, Sector 2 Tel.: +40 (0)1 210 48 25 +40 (0)1 210 29 50 Fax: +40 (0)1 210 29 52	Russia - Russland Bosch Rexroth OOO Wjatskaja ul. 27/15 127015 Moskau Tel.: +7-095-785 74 78 +7-095 785 74 79 Fax: +7 095 785 74 77 laura.kanina@boschrexroth.ru
Russia Belarus - Weissrussland ELMIS 10, Internationalnaya 246640 Gomel, Belarus Tel.: +375/ 232 53 42 70 +375/ 232 53 21 69 Fax: +375/ 232 53 37 69 elmis_ltd@yahoo.com	Turkey - Türkei Bosch Rexroth Otomasyon San & Tic. A..S. Fevzi Cakmak Cad No. 3 34630 Sefaköy Istanbul Tel.: +90 212 413 34 00 Fax: +90 212 413 34 17 www.boschrexroth.com.tr	Turkey - Türkei Servo Kontrol Ltd. Sti. Perpa Ticaret Merkezi B Blok Kat: 11 No: 1609 80270 Okmeydani-Istanbul Tel: +90 212 320 30 80 Fax: +90 212 320 30 81 remzi.sali@servokontrol.com www.servokontrol.com	Slowenia - Slowenien DOMEL Otoki 21 64 228 Zelezniki Tel.: +386 5 5117 152 Fax: +386 5 5117 225 brane.ozebek@domel.si

Africa, Asia, Australia – incl. Pacific Rim

Australia - Australien AIMS - Australian Industrial Machinery Services Pty. Ltd. 28 Westside Drive Laverton North Vic 3026 Melbourne Tel.: +61 3 93 14 3321 Fax: +61 3 93 14 3329 Hotlines: +61 3 93 14 3321 +61 4 19 369 195 enquires@aimservices.com.au	Australia - Australien Bosch Rexroth Pty. Ltd. No. 7, Endeavour Way Braeside Victoria, 31 95 Melbourne Tel.: +61 3 95 80 39 33 Fax: +61 3 95 80 17 33 mel@rexroth.com.au	China Shanghai Bosch Rexroth Hydraulics & Automation Ltd. Waigaoqiao, Free Trade Zone No.122, Fu Te Dong Yi Road Shanghai 200131 - P.R.China Tel.: +86 21 58 66 30 30 Fax: +86 21 58 66 55 23 richard.yang_sh@boschrexroth.com.cn gf.zhu_sh@boschrexroth.com.cn	China Shanghai Bosch Rexroth Hydraulics & Automation Ltd. 4/f, Marine Tower No.1, Pudong Avenue Shanghai 200120 - P.R.China Tel.: +86 21 68 86 15 88 Fax: +86 21 58 40 65 77
China Bosch Rexroth China Ltd. 15/F China World Trade Center 1, Jianguomenwai Avenue Beijing 100004, P.R.China Tel.: +86 10 65 05 03 80 Fax: +86 10 65 05 03 79	China Bosch Rexroth China Ltd. Guangzhou Repres. Office Room 1014-1016, Metro Plaza, Tian He District, 183 Tian He Bei Rd Guangzhou 510075, P.R.China Tel.: +86 20 8755-0030 +86 20 8755-0011 Fax: +86 20 8755-2387	China Bosch Rexroth (China) Ltd. A-5F., 123 Lian Shan Street Sha He Kou District Dalian 116 023, P.R.China Tel.: +86 411 46 78 930 Fax: +86 411 46 78 932	China Melchers GmbH BRC-SE, Tightening & Press-fit 13 Floor Est Ocean Centre No.588 Yanan Rd. East 65 Yanan Rd. West Shanghai 200001 Tel.: +86 21 6352 8848 Fax: +86 21 6351 3138
Hongkong Bosch Rexroth (China) Ltd. 6 th Floor, Yeung Yiu Chung No.6 Ind Bldg. 19 Cheung Shun Street Cheung Sha Wan, Kowloon, Hongkong Tel.: +852 22 62 51 00 Fax: +852 27 41 33 44 alexis.siu@boschrexroth.com.hk	India - Indien Bosch Rexroth (India) Ltd. Electric Drives & Controls Plot. No.96, Phase III Peenya Industrial Area Bangalore – 560058 Tel.: +91 80 51 17 0-211...-218 Fax: +91 80 83 94 345 +91 80 83 97 374 mohanvelu.t@boschrexroth.co.in	India - Indien Bosch Rexroth (India) Ltd. Electric Drives & Controls Advance House, II Floor Ark Industrial Compound Narol Naka, Makwana Road Andheri (East), Mumbai - 400 059 Tel.: +91 22 28 56 32 90 +91 22 28 56 33 18 Fax: +91 22 28 56 32 93 singh.op@boschrexroth.co.in	India - Indien Bosch Rexroth (India) Ltd. S-10, Green Park Extension New Delhi – 110016 Tel.: +91 11 26 56 65 25 +91 11 26 56 65 27 Fax: +91 11 26 56 68 87 koul.rp@boschrexroth.co.in
Indonesia - Indonesien PT. Bosch Rexroth Building # 202, Cilandak Commercial Estate Jl. Cilandak KKO, Jakarta 12560 Tel.: +62 21 7891169 (5 lines) Fax: +62 21 7891170 - 71 rudny.karimun@boschrexroth.co.id	Japan Bosch Rexroth Automation Corp. Service Center Japan Yutakagaoka 1810, Meito-ku, NAGOYA 465-0035, Japan Tel.: +81 52 777 88 41 +81 52 777 88 53 +81 52 777 88 79 Fax: +81 52 777 89 01	Japan Bosch Rexroth Automation Corp. Electric Drives & Controls 2F, I.R. Building Nakamachidai 4-26-44, Tsuzuki-ku YOKOHAMA 224-0041, Japan Tel.: +81 45 942 72 10 Fax: +81 45 942 03 41	Korea Bosch Rexroth-Korea Ltd. Electric Drives and Controls Bongwoo Bldg. 7FL, 31-7, 1Ga Jangchoong-dong, Jung-gu Seoul, 100-391 Tel.: +82 234 061 813 Fax: +82 222 641 295
Korea Bosch Rexroth-Korea Ltd. 1515-14 Dadae-Dong, Saha-gu Electric Drives & Controls Pusan Metropolitan City, 604-050 Tel.: +82 51 26 00 741 Fax: +82 51 26 00 747 eunkyoung.kim@boschrexroth.co.kr	Malaysia Bosch Rexroth Sdn.Bhd. 11, Jalan U8/82, Seksyen U8 40150 Shah Alam Selangor, Malaysia Tel.: +60 3 78 44 80 00 Fax: +60 3 78 45 48 00 hockhwa@hotmail.com rexroth1@tm.net.my	Singapore - Singapur Bosch Rexroth Pte Ltd 15D Tuas Road Singapore 638520 Tel.: +65 68 61 87 33 Fax: +65 68 61 18 25 sanjay.nemade@boschrexroth.com.sg	South Africa - Südafrika TECTRA Automation (Pty) Ltd. 71 Watt Street, Meadowdale Edenvale 1609 Tel.: +27 11 971 94 00 Fax: +27 11 971 94 40 Hotline: +27 82 903 29 23 georgv@tectra.co.za
Taiwan Bosch Rexroth Co., Ltd. Taichung Industrial Area No.19, 38 Road Taichung, Taiwan 407, R.O.C. Tel.: +886 - 4 -235 08 383 Fax: +886 - 4 -235 08 586 jim.lin@boschrexroth.com.tw david.lai@boschrexroth.com.tw	Taiwan Bosch Rexroth Co., Ltd. Tainan Branch No. 17, Alley 24, Lane 737 Chung Cheng N.Rd. Yungkang Tainan Hsien, Taiwan, R.O.C. Tel.: +886 - 6 -253 6565 Fax: +886 - 6 -253 4754 charlie.chen@boschrexroth.com.tw	Thailand NC Advance Technology Co. Ltd. 59/76 Moo 9 Ramintra road 34 Tharang, Bangkok, Bangkok Bangkok 10230 Tel.: +66 2 943 70 62 +66 2 943 71 21 Fax: +66 2 509 23 62 Hotline +66 1 984 61 52 sonkawin@hotmail.com	

Nordamerika – North America

USA Headquarters - Hauptniederlassung Bosch Rexroth Corporation Electric Drives & Controls 5150 Prairie Stone Parkway Hoffman Estates, IL 60192-3707 Tel.: +1 847 6 45 36 00 Fax: +1 847 6 45 62 01 servicebrc@boschrexroth-us.com repairbrc@boschrexroth-us.com	USA Central Region - Mitte Bosch Rexroth Corporation Electric Drives & Controls Central Region Technical Center 1701 Harmon Road Auburn Hills, MI 48326 Tel.: +1 248 3 93 33 30 Fax: +1 248 3 93 29 06	USA Southeast Region - Südwest Bosch Rexroth Corporation Electric Drives & Controls Southeastern Technical Center 3625 Swiftwater Park Drive Suwanee, Georgia 30124 Tel.: +1 770 9 32 32 00 Fax: +1 770 9 32 19 03	USA SERVICE-HOTLINE - 7 days x 24hrs - +1-800-REX-ROTH +1 800 739 7684
USA East Region – Ost Bosch Rexroth Corporation Electric Drives & Controls Charlotte Regional Sales Office 14001 South Lakes Drive Charlotte, North Carolina 28273 Tel.: +1 704 5 83 97 62 +1 704 5 83 14 86	USA Northeast Region – Nordost Bosch Rexroth Corporation Electric Drives & Controls Northeastern Technical Center 99 Rainbow Road East Granby, Connecticut 06026 Tel.: +1 860 8 44 83 77 Fax: +1 860 8 44 85 95	USA West Region – West Bosch Rexroth Corporation 7901 Stoneridge Drive, Suite 220 Pleasant Hill, California 94588 Tel.: +1 925 227 10 84 Fax: +1 925 227 10 81	
Canada East - Kanada Ost Bosch Rexroth Canada Corporation Burlington Division 3426 Mainway Drive Burlington, Ontario Canada L7M 1A8 Tel.: +1 905 335 5511 Fax: +1 905 335 4184 Hotline: +1 905 335 5511 michael.moro@boschrexroth.ca	Canada West - Kanada West Bosch Rexroth Canada Corporation 5345 Goring St. Burnaby, British Columbia Canada V7J 1R1 Tel.: +1 604 205 5777 Fax: +1 604 205 6944 Hotline: +1 604 205 5777 david.gunby@boschrexroth.ca	Mexico Bosch Rexroth Mexico S.A. de C.V. Calle Neptuno 72 Unidad Ind. Vallejo 07700 Mexico, D.F. Tel.: +52 55 57 54 17 11 Fax: +52 55 57 54 50 73 mariofelipe.hernandez@boschrexroth.com.mx	Mexico Bosch Rexroth S.A. de C.V. Calle Argentina No 3913 Fracc. las Torres 64930 Monterrey, N.L. Tel.: +52 81 83 65 22 53 +52 81 83 65 89 11 +52 81 83 49 80 91 Fax: +52 81 83 65 52 80

Südamerika – South America

Argentina - Argentinien Bosch Rexroth S.A.I.C. "The Drive & Control Company" Rosario 2302 B1606DLD Carapachay Provincia de Buenos Aires Tel.: +54 11 4756 01 40 +54 11 4756 02 40 +54 11 4756 03 40 +54 11 4756 04 40 Fax: +54 11 4756 01 36 +54 11 4721 91 53 victor.jabif@boschrexroth.com.ar	Argentina - Argentinien NAKASE Servicio Tecnico CNC Calle 49, No. 5764/66 B1653AOX Villa Balester Provincia de Buenos Aires Tel.: +54 11 4768 36 43 Fax: +54 11 4768 24 13 Hotline: +54 11 155 307 6781 nakase@usa.net nakase@nakase.com gerencia@nakase.com (Service)	Brazil - Brasilien Bosch Rexroth Ltda. Av. Tégula, 888 Ponte Alta, Atibaia SP CEP 12942-440 Tel.: +55 11 4414 56 92 +55 11 4414 56 84 Fax sales: +55 11 4414 57 07 Fax serv.: +55 11 4414 56 86 alexandre.wittwer@rexroth.com.br	Brazil - Brasilien Bosch Rexroth Ltda. R. Dr.Humberto Pinheiro Vieira, 100 Distrito Industrial [Caixa Postal 1273] 89220-390 Joinville - SC Tel./Fax: +55 47 473 58 33 Mobil: +55 47 9974 6645 prochnow@zaz.com.br
Columbia - Kolumbien Reflutech de Colombia Ltda. Calle 37 No. 22-31 Santafé de Bogotá, D.C. Colombia Tel.: +57 1 368 82 67 +57 1 368 02 59 Fax: +57 1 268 97 37 reflutech@etb.net.co			

12 Index

A

Additional components
 mains choke 6-1
Appropriate use 2-1
autotransformers 4-2

B

Battery safety 3-11
bleeder 8-1
Braking resistor 9-1
Bus module 7-8

C

Capacitor Unit 7-1
Choke 6-1
Commutation choke 6-1
Connection
 bus module 7-8
 X1 7-8

D

DC Bus Capacitor Unit 7-1
DC bus resistor unit 8-1
Documentation
 overview 1-2
DST 4-1, 4-2

E

Electrical installation 1-3

F

filter
 mains 5-1

G

grounded power supply lines 4-1

H

Hazards by Improper Use 3-2
HCS03.1
 motor filter HMF01 10-1
HLB01.1C 8-1
HLB01.1D 8-1
HLC01.1 7-1
HLR01.1 9-1
HMF01 10-1
HNF 5-8
HNC01.1 5-12
HNL
 choke 6-1
 type code 6-1
HNL01.1E-
 0100-N0202-A-480-NNNN 6-3
 0170-N0146-A-500-NNNN 6-3
 0200-N0125-A-480-NNNN 6-3
 0240-N0106-A-500-NNNN 6-3

0362-N0080-A-500-NNNN 6-3
0400-N0051-A-480-NNNN 6-3
0571-N0050-A-500-NNNN 6-3
0600-N0032-A-500-NNNN 6-3
1000-N0012-A-500-NNNN 6-3
1000-N0020-A-500-NNNN 6-3
2800-S0125-A-480-NNNN 6-7
3400-S0202-A-480-NNNN 6-7
5700-S0051-A-480-NNNN 6-7
HNL01.1R-
0540-C0094-A-480-NNNN 6-6
0590-C0065-A-480-NNNN 6-6
0980-C0026-A-480-NNNN 6-6
3000-S0094-A-480-NNNN 6-7
4200-S0026-A-480-NNNN 6-7
6300-S0065-A-480-NNNN 6-7

I

Improper use
 hazards 3-2
Inappropriate use 2-2
 Consequences, Discharge of liability 2-1
IndraDrive
 documentation 1-2
Installation
 electrical 1-3

M

mains choke
 HNL01.1R (regenerating) 6-6
Mains choke 6-1
mains filter 5-1
 HNF 5-8
 HNK 5-12
 NFD 5-1
 NFE 5-1
motor filter
 HMF01 10-1

N

Netzdrossel
 HNL01.1E (feeding) 6-3
 HNL01.1E-****-S (stromkompensiert) 6-7

P

Protection
 against contact with electrical parts 3-5
 against contact with hot parts 3-10
 against dangerous movements 3-7
 against electric shock by protective low voltage (PELV) 3-6
 against magnetic and electromagnetic fields during operation and mounting 3-9
 against pressurized systems 3-11
 during handling and mounting 3-10

R

resistor unit 8-1

S

Safety Instructions for Electric Drives and Controls 3-1
single-phase filter NFE01.1-250-006 5-7

T

transformers 4-1

Type code

HNL 6-1

mains choke 6-1

U

ungrounded power supply lines 4-1

Use

directions for use 2-1

hazards by improper use 3-2

inappropriate 2-2

W

Warning symbols 3-1

X

X1 7-8

Bosch Rexroth AG
Electric Drives and Controls
P.O. Box 13 57
97803 Lohr, Germany
Bgm.-Dr.-Nebel-Str. 2
97816 Lohr, Germany
Phone +49 93 52-40-50 60
Fax +49 93 52-40-49 41
service.svc@boschrexroth.de
www.boschrexroth.com



R9111306140

Other ManualsLib Projects

 www.manualslib.com

 www.manualslib.de

 www.manualslib.es

 www.manualslib.fr

 www.manualslib.nl

 www.manualslib.mx

 www.manualslib.tech 30+ Languages